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QUALITY ASSESSMENT REPORT

LAKE WATER QUALITY

100M PRODUCT

VERSION 2.2

Issue I1.00

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GLWD	Global Lakes and Wetlands Database
GPT	Graph Processing Tool
LSR	Lake Surface Reflectance
NERC	Natural Environment Research Council (UK)
NIR	Near-Infrared, part of the electromagnetic spectrum
obs	Observation
OC	Ocean Colour
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAA	Quasi-Analytical Approach
QAR	Quality Assessment Report
rep	representative
R_w	Fully normalised water-leaving reflectance
SWIR	Shortwave Infrared, part of the electromagnetic spectrum
TS	Trophic State
TSM	Total Suspended Matter
TUR	Turbidity
WGS84	World Geodetic System 1984

1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) Lake Water Quality 100m (LWQ-100m) product provides an optical characterisation of inland waterbodies at 100m spatial resolution based on Sentinel-2 Multispectral Imager (MSI) data. This product was temporarily implemented between December 2018 and March 2020 in demonstration mode (v1.3), to complement the Sentinel-3 OLCI LWQ-300m product that includes more than 4000 lakes globally. The 100m service was reordered from July 2021 (v1.5), has since been further updated with new evolutions (v2.0, v2.1 and v2.2) and continues to be provided in near-real-time. Products are available for all resolvable water bodies within the outline of 225 Sentinel-2 MSI tiles over Europe and Africa. The 100m products contain six (sets) of variables: lake water-leaving reflectance (all wavebands available after atmospheric correction), turbidity, a trophic state index (derived from phytoplankton biomass by proxy of chlorophyll-*a*), total suspended matter concentration, chlorophyll-a concentration and a floating cyanobacteria risk index. Production and delivery cover up to 10-day intervals (starting the 1st, 11th and 21st day of each month), mapped onto a common global grid.

The current version (v2.2) is an update to the previous release v2.1 which introduced a new scene classification layer based on the CLMS Water and Ice Cover (WIC) Sentinel-1+Sentinel-2 merged product. Version 2.2 uses ECMWF (European Centre for Medium-Range Weather Forecasts) meteorological data as input to the atmospheric correction processor, compared to the previous versions that relied on (now discontinued) NASA (US National Aeronautics and Space Administration) data.

While Sentinel-2 MSI was primarily designed for land applications, some properties of waterbodies can also be observed, as demonstrated by the LWQ-100m product collection. The algorithms used to derive the input for the optical lake water products are implemented in the *Calimnos* processing chain, which also handles production of the S-3 OLCI LWQ-300m products. This Quality Assessment Report (QAR) describes the validation and intercomparison activities undertaken to demonstrate the quality and limitations of previous and current product versions.

1.2 SCOPE AND OBJECTIVES

The quality of Sentinel-2 MSI based LWQ-100m product collection is assessed by using examples of individual 10-day products. The analysis also gives a wider overview by providing time series and matchup analyses of lake surface reflectance (and improvements foreseen from upgrading the Polymer atmospheric correction) and water quality variables.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3 describes the methodology for quality assessment, the metrics and criteria for evaluation
- Chapter 4 presents the results of the analysis
- Chapter 5 summarizes the main conclusions of the study
- Chapter 6 makes recommendations based upon the results

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global Land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/2020/OP/3213 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2020)7715271, dated 17/12/2020

AD4: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service
CGLOPS2_SVP	Service Validation Plan of the Global Land Service
CGLOPS2_ATBD_LWQ100_S2_v2.2.0_I1.00	Description of algorithms for 100m lake water quality products demonstration products
CGLOPS2_PUM_LWQ100_S2_v2.2.0_I1.00	Product User Manual summarizing all information about the 100m lake water quality demonstration products

CGLOPS2_ATBD_Polymer_v1.0.0_I1.02	Description of Polymer Algorithm for different instruments

2 REVIEW OF USER REQUIREMENTS

According to applicable document [AD4], the user requirements relevant for Lake Water Quality products are:

- **Definition**:

The Lake Water Quality products v2.2 are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3, Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3 and lake surface reflectance spectra (LSR, given in terms of fully normalized water-leaving reflectance R_w; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of LSR is not physically accurate and would not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. LSR describes the apparent colour of the water body. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to monitoring eutrophication. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The baseline datasets described in this document shall be provided at a resolution of 100m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

- **Geographical coverage**:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data:

- geographic projection: lat lon,
- geodetical datum: WGS84
- accuracy: min 10 digits
- coordinate position: pixel centre
- global window coordinates: UL: 180W-75N, BR: 180E, 56S

The following output specifications are further optimised with respect to the requirements:

- pixel size at 100m: 0.1/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 100m: 400752 columns, 200376 lines

Continental windows:

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

- **Accuracy requirements**:

Lake Water Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and trophic state has an uncertainty threshold of 0.3 in log10 or 50% [AD4].

- **Temporal Definition**

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally ten and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

3 QUALITY ASSESSMENT METHOD

3.1 OVERALL PROCEDURE

Various complementary validation methods were applied to assess the quality of the lake water products. First, time series of selected lakes were extracted and plotted for selected variables to assess seasonal cycles, outliers and any unexpected patterns.

Visual inspection has been performed to assess realistic spatial behaviour of the products. The distribution of values (histograms) was inspected to assess the quality of the pixel identification. These results are shown in section 4.1.

For the validation of floating cyanobacteria, reports available via online platforms or social media of blooming events were used. Reports for macrophytes (sargassum, hyacinths) as well as cyanobacteria were used in order to verify that the differentiation works properly.

Further, validation against in situ data was carried out (section 4.1.3). Here, the assessment was performed on (a) time series (Level 2 products) at selected measurement stations and (b) matchup analyses. For the matchup analysis, the time difference between an in situ measurement and the day of acquisition is set to max. ±3 hours. For most of the comparison exercises with in situ data, a window of 3 x 3 pixels (macro-pixel) around the location of the measurement stations was extracted from the satellite products and filtered for invalid pixels before averaging.

Consistency tests have been performed that compare the outcome of different sensors (OLCI, MSI) at 300m product resolution, to highlight any bias and systematic error between 300 and 100m products. Here, time series have been plotted and, in the case of no temporal overlap, the seasonal trends were compared inner-annually. The results are presented in 4.3.2. Spatial consistency between the 300m and 100m products was further checked by analysing transects (sections 4.3.3 and 4.3.4), spectral consistency in reflectance is assessed in section 4.3.5. In all these cases, the 100m product is resampled and the nearest neighbour is selected to the 300m product grid.

When comparing consistency between satellite sensors, alignment of MSI to OLCI (co-occurring) is sought because the latter, with a larger set of narrower wavebands, is considered the more reliable for retrieval of water properties. Both MSI and OLCI validation is currently limited by scarce in situ observation data suitable for algorithm accuracy validation. OLCI performance has been, in turn, compared to MERIS performance because their radiometric performance and spectral configuration is considered very similar. Because these sensors do not overlap in time, consistency checks between MERIS and OLCI were based on multi-annual analysis (i.e., similar seasonal patterns) over multiple lakes. Results on this analysis are not included in this document; the reader is referred to the equivalent documentation for the 300m Lake Water Quality products.

It is worth noting that the comparison between satellite water quality products and in situ data aims to compare different measurement techniques. In situ observations are subject to measurement inaccuracies, some of which are unknown. This is particularly the case for global datasets collected by different laboratories with different qualifications, sampling and analysis protocols. The range in

optical properties of lakes globally is very wide. Consequently, sampling methods vary (single depths, integrated depth samples), analysis methods vary (HPLC, spectrophotometric and fluorometric for Chl-a, for example), and each sample typically represents a point in space rather than an area the size of a satellite pixel.

3.2 SATELLITE REFERENCE PRODUCTS

Analyses are performed on the L3 Copernicus Land user products, which are 10-day averages. Wherever necessary, L2 products processed with the *Calimnos* processing chain (not a Copernicus Land user product) are utilised specifically to validate atmospheric correction outputs, and turbidity and chlorophyll-*a* concentration retrieval algorithms underlying the Lake Water Quality products. Some of the results are made from analysis of v1.5 of the data products, some of v2.0. For the purposes of this validation the products are interchangeable. These analyses will be repeated for v3.0 once these products are available as v3.0 will not be interchangeable with v1.5 and v2.0 due to a significant change in the atmospheric correction and masking algorithms.

Detailed results are provided for a number of example lakes: Lake Mueritz, Lake Garda, Lake Turkana, Lake Michigan and Lake Balaton.

3.3 IN SITU REFERENCE PRODUCTS

To assess the atmospheric correction outputs, various in situ datasets were used.

Inland water measurements supplied by the H2020 EOMORES project were available for lakes in Italy, the Netherlands, Estonia, Spain and Scotland, and represent a range of optical conditions including clear and productive waters. Measurements from the Baltic Sea were obtained from the Alg@line network through the BONUS FerryScope project, with sensors mounted on two Baltic ferries. The Western English Channel region was sampled from the RV Quest. Full details of the lakes sampled and the sensors used are detailed in Warren et al. (2019).

From the H2020 MONOCLE project, WaterInsight's WISPCloud data were available for several European lakes; most stations are near to shore on fixed platforms. Plymouth Marine Laboratory (PML) SoRad radiometers were installed on ships and provide reflectance data for coastal regions and lakes.

Finally, the GLORIA dataset of Lehmann et al. (2023) was used: a global dataset of reflectance and water quality parameters collated for 450 inland water bodies by 59 institutions.

4 RESULTS

The results are presented starting with the visual inspection of pixel identification and water quality parameters. This is followed by the comparison with in situ data, for Lake Water Reflectance and for coastal waters (matchup analysis, 4.1.3). Chapter 4.3 provides different consistency tests between the 300m and 100m products. It shows results in time (time series, 4.3.2), in space (transects, 4.3.3 and 4.3.4) and spectrally (4.3.5).

4.1 VISUAL INSPECTION

4.1.1 Pre-processing

4.1.1.1 Cloud screening

The pixel identification software *IdePix* is used to generate masks for Sentinel-2 data. The elimination of non-valid pixels is crucial for the retrieval of non-disturbed 10-day averaged products. A visual qualitative analysis of the pixel identification performance follows.

The CLOUD flag consists of different elements, dedicated to different cloud types: CLOUD_SURE, CLOUD_AMBIGUOUS, CLOUD_BUFFER, CIRRUS_SURE and CIRRUS_AMBIGUOUS. Currently, the CLOUD_SURE, CLOUD_BUFFER and CLOUD_AMBIGUOUS flags are applied and pixel values for the different parameters are not provided for those pixels. The visual tests indicate that it will be necessary for the L3 aggregation to 10-day averaged products to apply not only the CLOUD_SURE, CLOUD_AMBIGUOUS and CLOUD_BUFFER flags but also the CIRRUS_SURE and probably even the CIRRUS_AMBIGUOUS flags. Figure 1 and Figure 2 show two examples of the different cloud flags and how they behave (i) in comparison with an RGB (clouds included) and (ii) on water retrieval (Figure 2). Artefacts are visible in the water products indicating that the cloud screening was not strong enough for this specific application. Adding the CIRRUS flags improves this result.

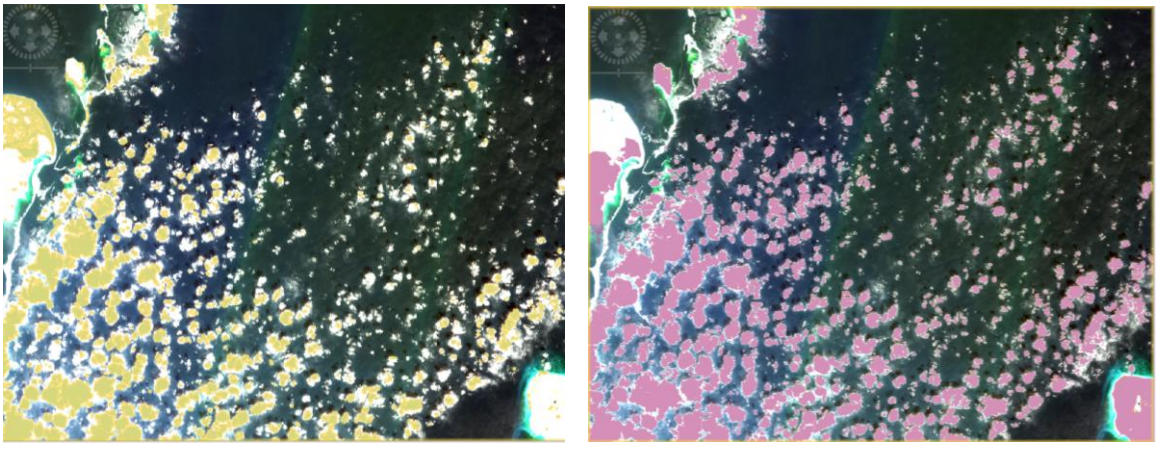


Figure 1: Lake Michigan (22.03.2017, T16TER) - cloud flag on small and thin cumulus clouds. Left: cloud and CLOUD BUFFER (2pixels); right: CLOUD and CIRRUS_SURE flag

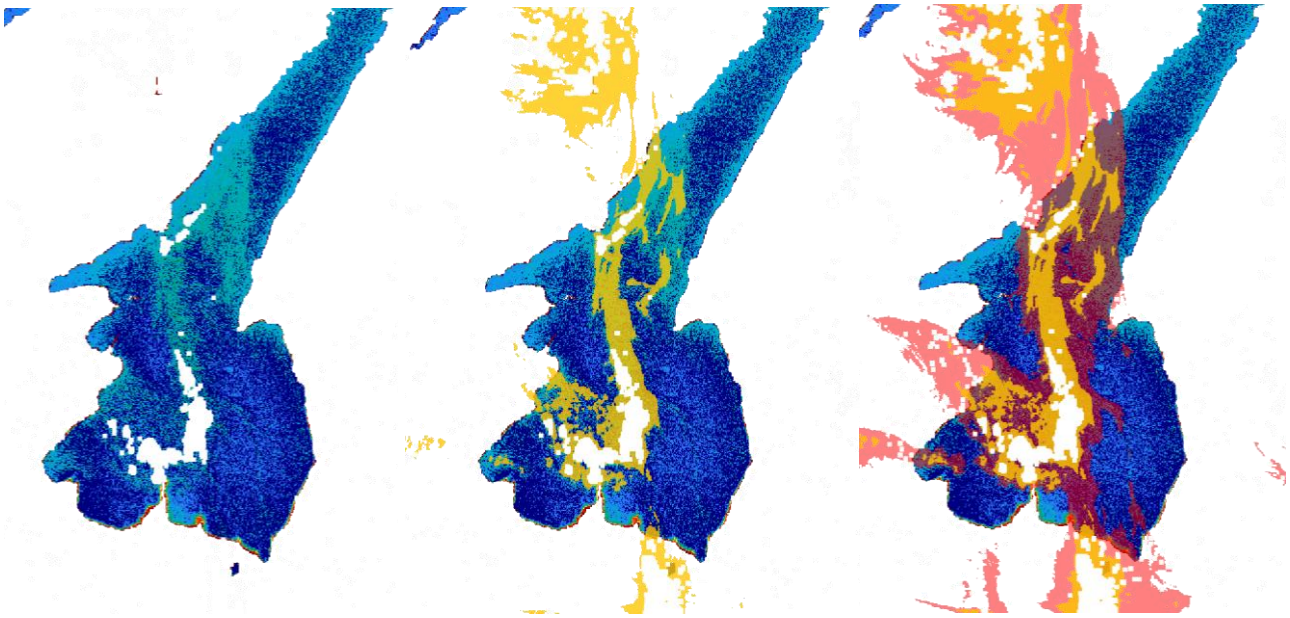


Figure 2: Blended TUR in Lake Garda left: all overlaid with CLOUD, CLOUD_BUFFER and CLOUD_AMBIGUOUS FLAGs; middle: in addition, the CIRRUS_SURE flag is raised; right: in addition the CIRRUS_AMBIGUOUS flag is raised

4.1.1.2 Cloud shadow

The cloud shadow is detected by geometry based on the identified cloud pixels and clustering of potential cloud shadow areas. Cloud shadow in high resolution data has a strong influence on the water products and should be removed from the final products. The examples shown in Figure 3 and Figure 4 present Lake Turkana, a turbid lake, showing clear influence of the cloud shadow on atmospheric correction. The combined CLOUD SHADOW flag detects the areas very well. However, in clear lakes the detection is less good because the low signal of dark water and even darker cloud shadow pixels are within the noise level of Sentinel-2. In clear water, the effect of cloud shadow is the strongest in R_w560 (followed by R_w at 490nm) and almost invisible in the other bands. Therefore, the cloud shadow has less impact on the retrieval of TSM or TUR using the Nechad single-band algorithms (Nechad et al., 2010; 2015; 2016).

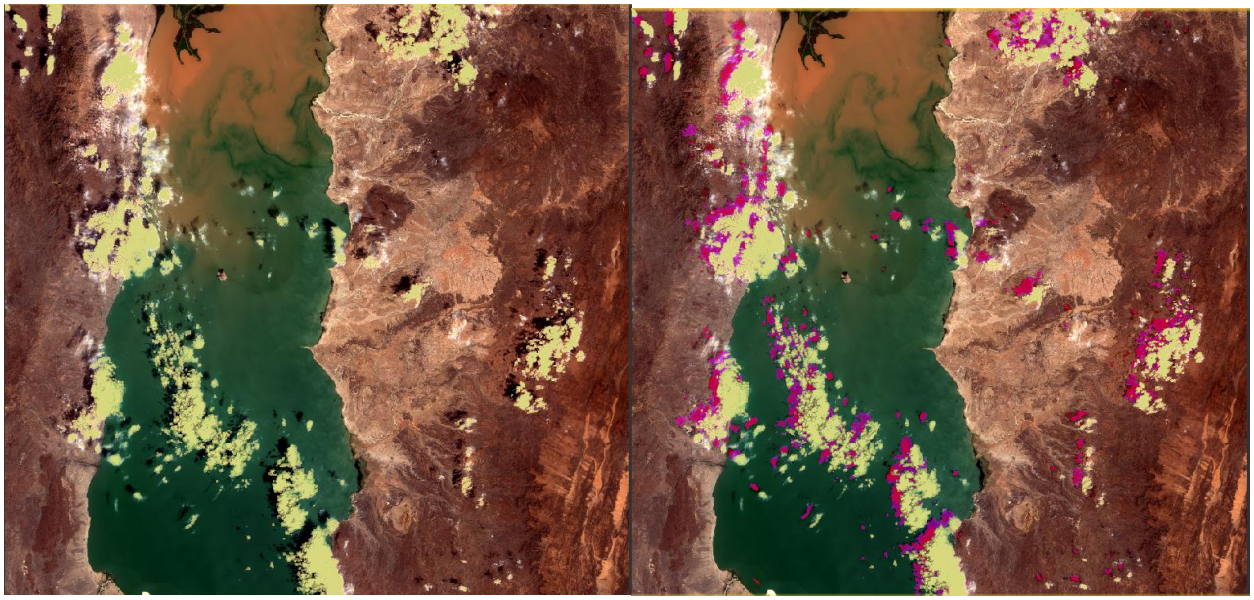


Figure 3: Lake Turkana on 27.01.2018, RGB, with (left) *IdePix* cloud mask flagging CLOUD and CLOUD_BUFFER and (right) cloud shadow flag added

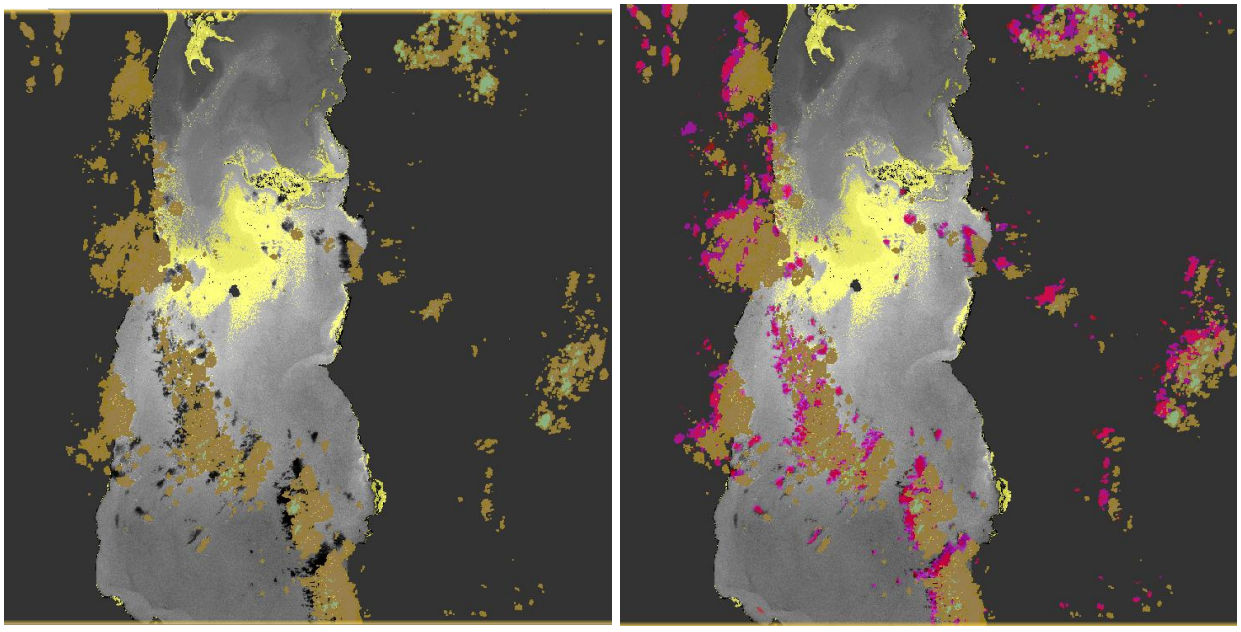


Figure 4: As previous but here showing the Polymer R_w560 band. (left) The cloud mask (IdePix CLOUD + CLOUD_BUFFER, brown) and the Polymer-bounds (yellow) flags are shown. In turbid water, the cloud shadow is visible as a strong contrast. (right) Pink pixels indicate the cloud shadow flag, which captures the dark cloud shadow pixels very well. Cloud shadows can be identified either by clustering or shifting the cloud mask

4.1.1.3 Shoreline detection

The standard (static) land flag for Sentinel-2 is not adequate for inland waters. Shoreline pixels remain and are included in the water mask (Figure 5). To mitigate this, the *IdePix* VEG_RISK flag is used to identify the land-water boundary and flag mixed pixels of land and water. Therefore, these pixels are excluded from the final product.

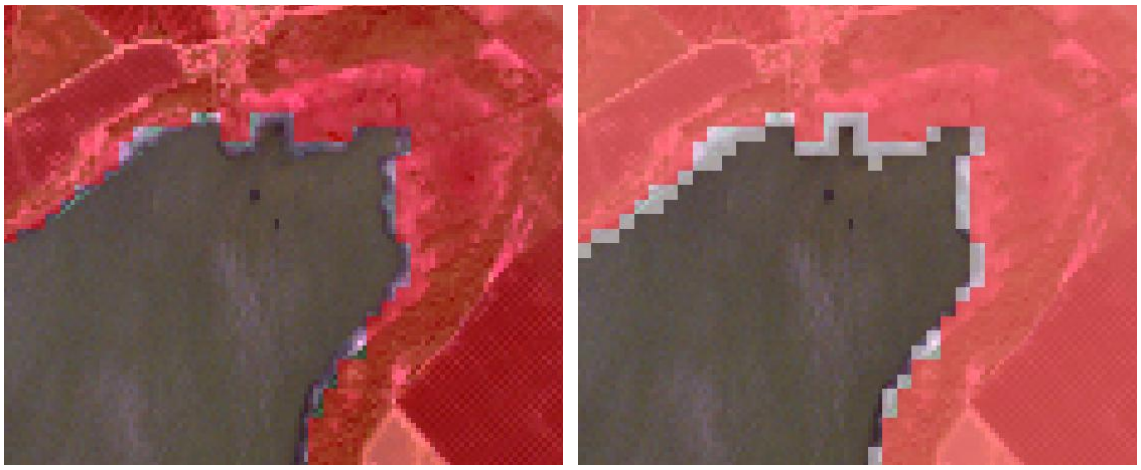


Figure 5: RGB image overlaid with IDEPIX_LAND (red) and on the right also with IDEPIX_VEG_RISK (white) flags. The VEG_RISK flag identifies the mixed pixels at the shoreline. The RGB image has 20m resolution; flags are overlaid in 60m resolution to show mixed pixels within 60m products

A second flag to detect shoreline pixels is the poly_OUT_OF_BOUNDS flag which is raised by Polymer when the optimised parameters of the bio-optical model (log chlorophyll a and f_b) depart from within a predefined range. Comparison between the IDEPIX_VEG_RISK and poly_OUT_OF_BOUNDS flags shows that the latter identifies more shoreline pixels but also flags some shallow waters (Figure 6).

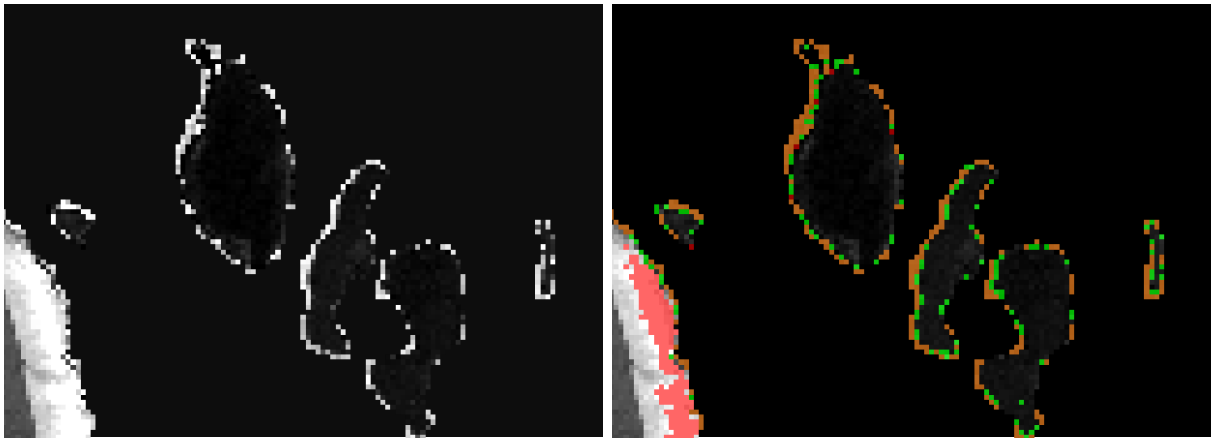


Figure 6: (left) Shoreline pixels (mixed pixels) not flagged by LAND flag show higher values in R_w560. (right) Flagged with poly_OUT_OF_BOUNDS (red) and *IdePix* (green)

The expected result of processing shoreline and mixed land/water pixels as water is an elevated R_w product. This shows clearly in histograms of the distribution of R_w560 where the pixels masked by the two added flags cluster around 0.10 (Figure 7), although it is likely that the VEG_RISK flag also affects some open water pixels. Combining all flags results in a realistic distribution of R_w560 (Figure 8).

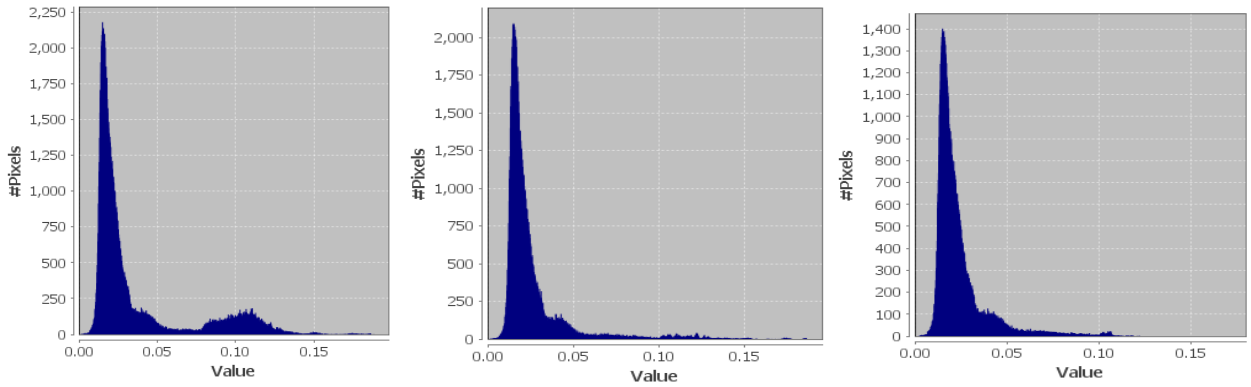


Figure 7: Histograms of water pixels for R_w560. (left) No flagging; the histogram shows a secondary mode around 0.10 which is expected to be caused by shoreline and mixed pixels. (centre) Only pixels flagged as *Idepix* VEG_RISK have been removed. (right) Only pixels flagged as poly_OUT_OF_BOUNDS have been removed

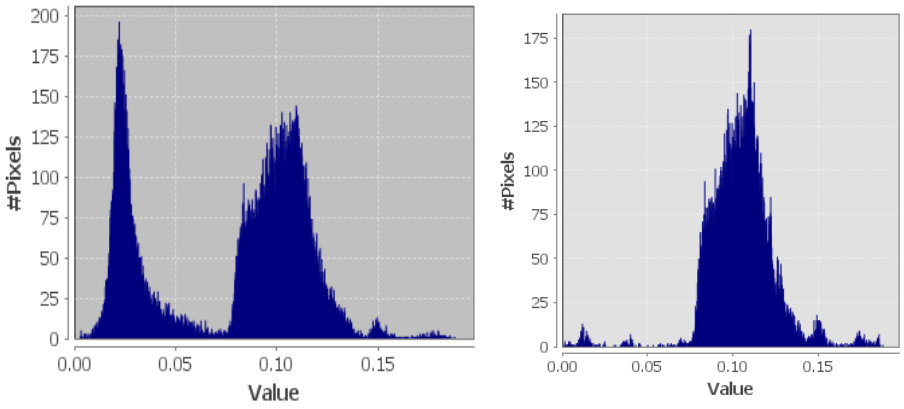


Figure 8: (left) Histogram of water pixels that are flagged with VEG_RISK flag; mixed pixels clearly shown, but also clear water pixels are covered by the VEG_RISK flags. (right) Water pixels flagged with poly_OUT_OF_BOUNDS flag. Mixed pixels are shown, water pixels are less affected

4.1.2 Water products

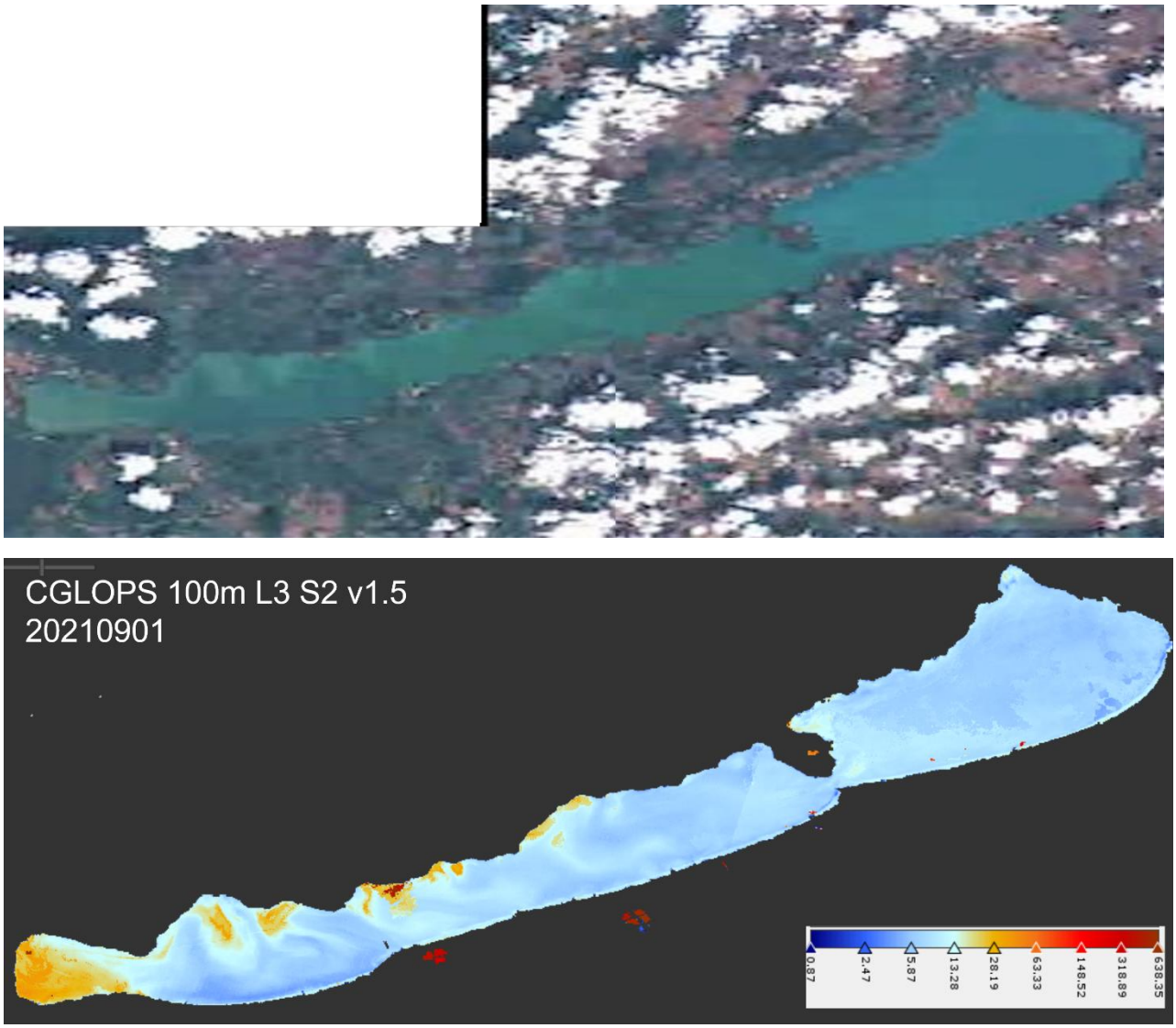


Figure 9: Lake Balaton as stretched RGB image (top) and blended_turbidity derived with *Calimnos* processing (bottom) on 01.09.2021

The visible structures that can be seen in RGB images are well captured by the CGLOPS water products (Figure 9). Very turbid water visible in the RGB image correspond to high values in the blended turbidity product. A visible line in the L3 product near the narrow part between lake basins originates from the aggregation of two data days, one of which lies at the swath edge (Figure 10). Also, the eastern part of the lake shows some spatial patterns which are introduced by the aggregation technique. Artefacts like these are unavoidable over a 10-day aggregation period with a relatively long revisit time between observations.

New Optical Water Types (OWT) have been introduced to Sentinel-2 MSI processing, which lead to reasonable turbidity values for the most part. However, turbidity values exceed 100 FNU in some parts, which appear like strong artefacts in the turbidity spatial distribution. This needs to be further investigated and validated against in situ measurements.

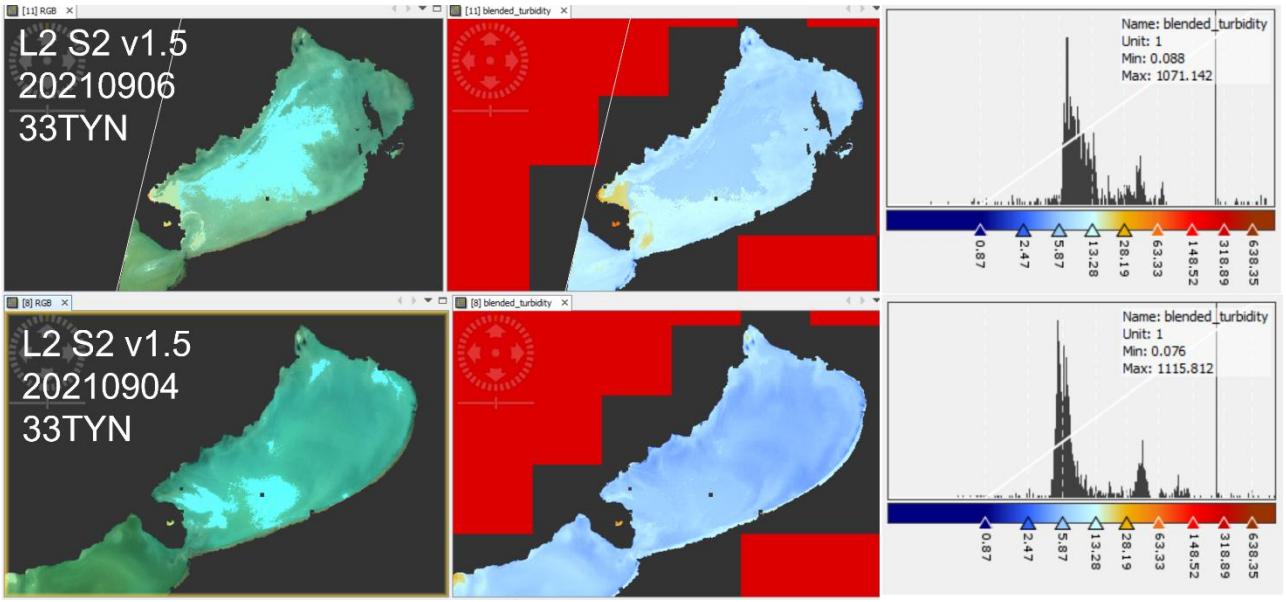


Figure 10: Examples of turbidity values in a Level 2 Sentinel-2 MSI product, which are combined in the L3 product shown in Figure 9. The north-eastern part is covered by overpasses on two different days, the edge of the swath remains visible after aggregation

An example with more extensive spatial artefacts of very high turbidity values is shown in Figure 11 (top). The cloud detection seems to be working. Still, in some areas strange spectra are retrieved by the atmospheric correction (for examples see the red lines in the bottom-right panel of Figure 11), which lead to unrealistically high turbidity values.

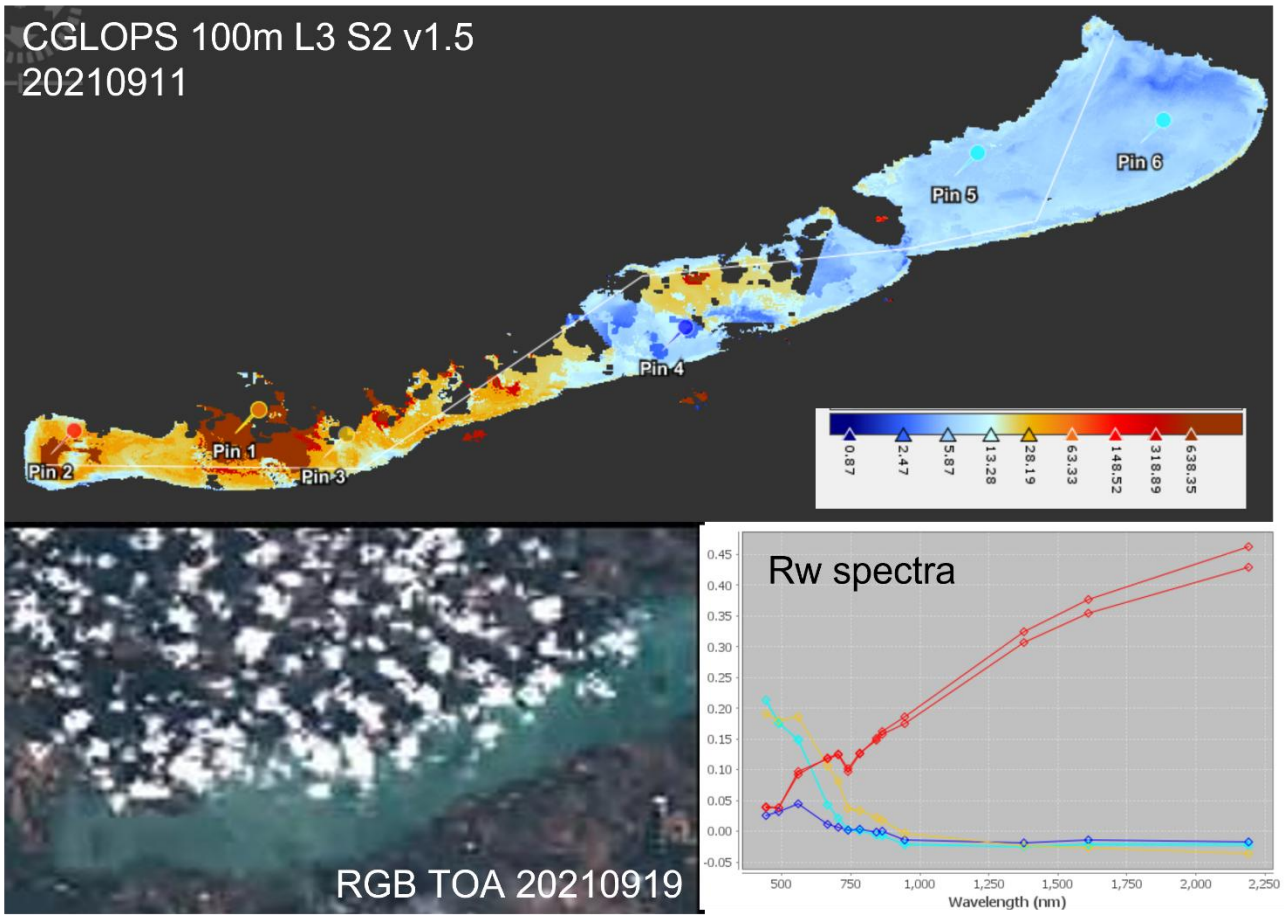


Figure 11: (top) Example of spatial artefacts with very high turbidity values. (bottom-left) The corresponding original data (RGB). (bottom-right) Example R_w spectra

4.1.3 Floating cyanobacteria

As a first assessment, a visual inspection was performed. Anecdotal records from articles and social media posts were used to identify occurrences of and assess the ability of the satellite products to represent surface blooms. Visual validation particularly helps to improve the quality of the methodology in terms of omission or commission errors. In addition, the social media channel CyanoTracker (https://twitter.com/CyanoTracker) was used to identify a number of reported cases of surface blooms.

Several CyanoTracker posts included blooms that were visually identified from satellite images across various lakes. These posts were used to locate bloom events and test the methodology described here. Similarly, a range of published media articles related to these bloom events were studied to specify the type of bloom that took place. Articles listed in Table 1 included information about the presence of floating macrophytes. Table 2 lists the remote sensing products used to identify and validate the cyanobacteria surface bloom products.

Using observations from CyanoTracker and media articles (Table 1 and Table 2), it was found that all events could be consistently identified. Nevertheless, some uncertainty remains regarding the accuracy of mapping the extent for each floating biomass class. To assess the sensitivity of the class assignments, in situ measurements would be needed for further validation.

Table 1: Articles containing information on the presence of floating macrophytes, used to visually validate the proposed methodology

Journal	Link	Accessed
Mercator Ocean International	https://www.mercator-ocean.eu/en/news/mercator-ocean-arcan36sargassesmte_en/	28-03-2023
The New Humanitarian	https://www.thenewhumanitarian.org/report/74270/kenya-livelihoods-hit-water-hyacinth-takes-over-lake-victoria	28-03-2023
The Scientific World Journal	https://www.researchgate.net/publication/257938835_Gichuki_et_al_Water_hyacinth_succession_in_Lake_Victoria	28-03-2023

Table 2: EO products used to test the surface bloom product

Date	Sensor	Lake
2016-09-13	Sentinel 2 MSI	Curonian lagoon
2018-06-03	Sentinel 2 MSI (T19QCA)	Caribbean Sea
2019-07-24	Sentinel 2 MSI	Lake Erie
2019-08-09	Sentinel 2 MSI (T34VCK)	Baltic Sea
2020-08-19	Sentinel 2 MSI	Lake Pepso
2022-04-11	Sentinel 2 MSI	Lake Suchitlán

In a second step, the spectral information within the detected bloom and in the surrounding water were compared and assessed. We assume that the water around surface blooms of cyanobacteria is affected by immersed cyanobacteria. Therefore, we should expect the typical light absorption features of cyanobacteria to be present in atmospherically corrected reflectance spectra. In severe bloom conditions, these features may also show clearly in TOA spectra. Examples of likely omission errors would be the detection of a macrophyte patch surrounded by water in which cyanobacteria are evident (which is possible, but less likely to occur), whereas a commission error example would be a floating cyanobacteria detection event without evidence of cyanobacteria in surrounding waters, which is very unlikely to occur. Omission and commission errors are more likely to occur with MSI

imagery due to the absence of several narrow diagnostic bands in the red part of the spectrum, compared to OLCI.

The image from Winam Gulf shown in Figure 12 highlights areas identified as having cyanobacteria (predominantly in the southwest) and macrophytes (in the centre of the image and further east). Spectra in the open water around these occurrences (plotted in blue) do not show distinct differences in shape, noting that phycocyanin detection is not possible due to absence of a band in the 620nm region. Spectra in the floating macrophyte mask show a sharper rise in the 740-780nm region but are otherwise similar to those associated with floating cyanobacteria. Compared to macrophytes, the cyanobacterial surface blooms show a higher absorption in the SWIR bands, indicating that the effects of water absorption on cyanobacterial blooms are stronger than those on macrophytes. Since the water absorption in the SWIR band is much stronger than that in the NIR bands, the SWIR band is considered more effective than the NIR band for distinguishing cyanobacterial blooms and aquatic macrophytes.

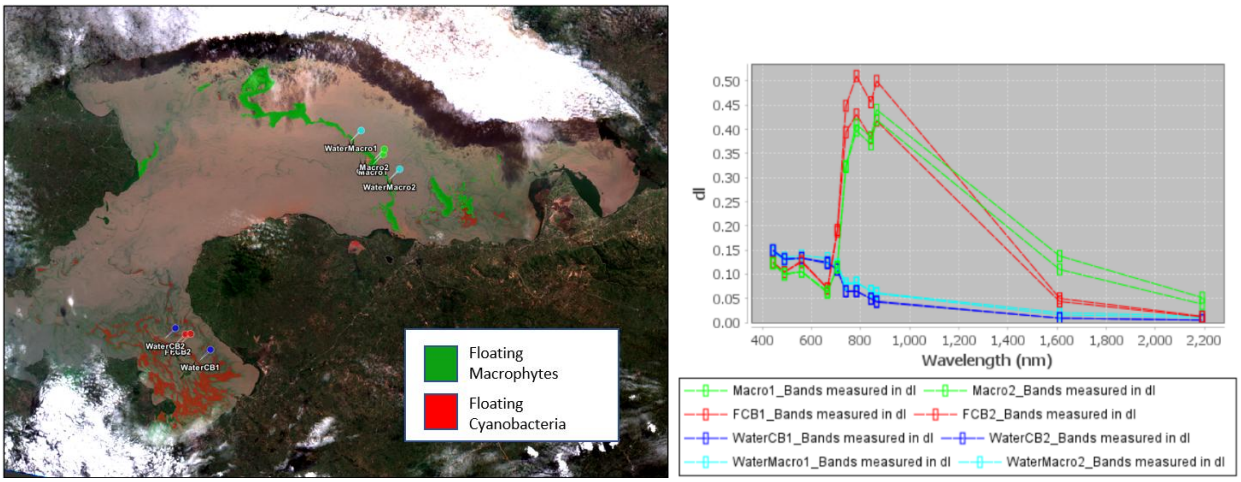


Figure 12: Spectra associated with areas identified as having floating cyanobacteria and macrophytes in Winam Gulf, Lake Victoria. Key to the spectra: Blue – open water in vicinity of floating cyanobacteria, Red – floating cyanobacteria identified, Light Blue – open water near macrophytes, Green – floating macrophytes

4.2 COMPARISON WITH IN SITU DATA: LAKE WATER REFLECTANCE

As part of a joint analysis with the H2020 EOMORES project, a Sentinel-2 atmospheric correction analysis was performed and published (Warren et al. 2019). Sentinel-2A data, resampled such that all bands are at the same spatial resolution of 60m, underwent a matchup analysis against in situ data of inland water bodies and coastal regions (Baltic Sea and Western Channel UK). In situ data within ± 3 hours of a Sentinel-2A overpass were used to generate the matchups. Table 3 shows the number of Sentinel-2A granules and matchups produced by Polymer v4.6 (an earlier version of

Polymer, released 23.11.2017) for each region. Figure 13 shows per-band scatterplots of the matchup analysis. The 444, 497 and 560nm wavebands show regression slopes close to 1 albeit with varying bias. For the 665 and 704nm wavebands, both essential in retrieval of chlorophyll-*a* in relatively turbid and productive inland waters, regression slopes < 1 suggest that Polymer overcorrects these bands and algorithm optimization for chlorophyll-a retrieval is essential. Figure 14 shows a plot of the coefficient of determination, root mean square (RMSE), mean absolute percentage difference (MAPD) and bias for Polymer v4.6 by region. The inland water region has a higher RMSE and bias than the coastal regions but also much lower number of matchups. In the Western Channel, UK, the MAPD is much higher than in the Baltic and inland bodies, in particular in the red and near-infrared (NIR) bands.

Table 3: Number of MSI granules with valid matchups, after removing duplicates, in the time windows of ±3 h for Western Channel, UK, and Baltic Sea datasets and ±24 h for the inland dataset

	Inland	Baltic	Western Channel
granules	35	30	5
matchups	230	986	452

© C-GLOPS Lot2 consortium

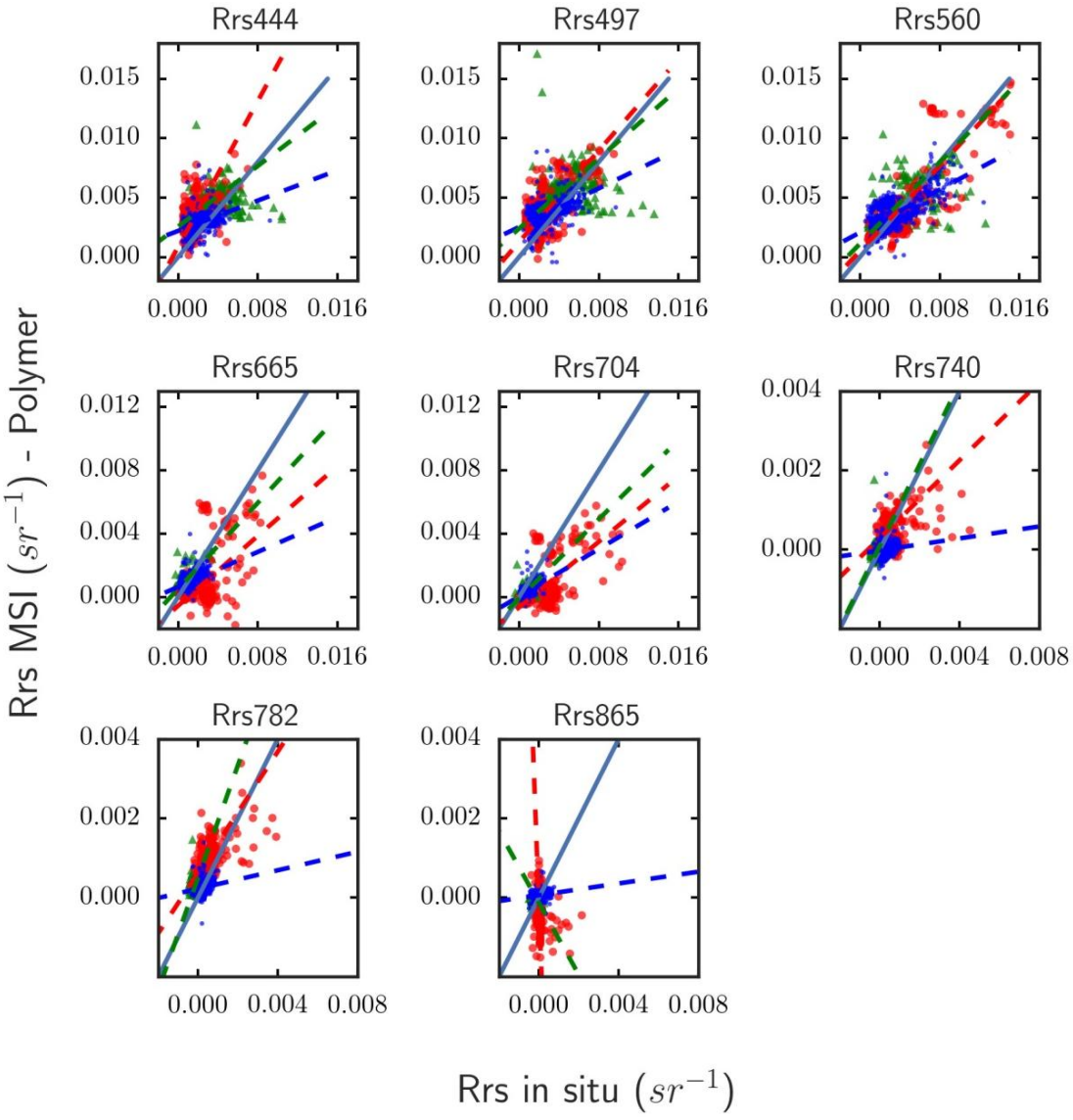


Figure 13: Polymer v4.6 plotted against in situ data for each corrected Sentinel-2 band. All data are plotted as remote-sensing reflectance (Rrs, units sr^{-1}). There are 1735 matchup points. Blue points show Baltic, green triangles show Western Channel and red circles are inland water matchups (Source: Warren et al. 2019)

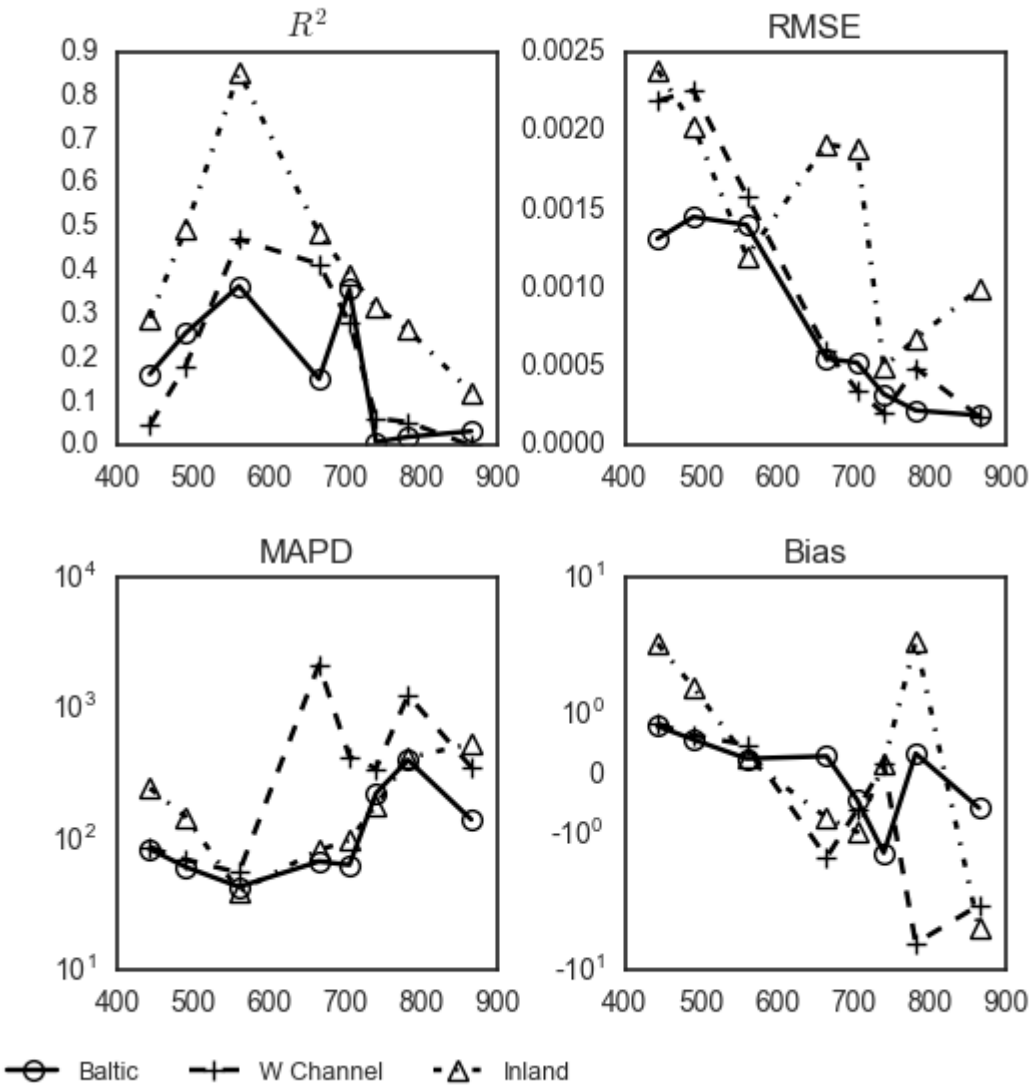


Figure 14: Statistics of Polymer matchups plotted for each region (circles: Baltic Sea, plusses: Western Channel, triangles: inland waters). The metrics are (from top left): coefficient of determination (R2), root mean square error (RMSE), mean absolute percentage difference (MAPD) and bias

Another comparison has been conducted with AERONET-OC stations, which is shown in Figure 15. Here, Polymer v4.10 was used but the analysis is limited to coastal stations, where retrieval of R_w is likely to be less affected by adjacent land compared to inland waters.

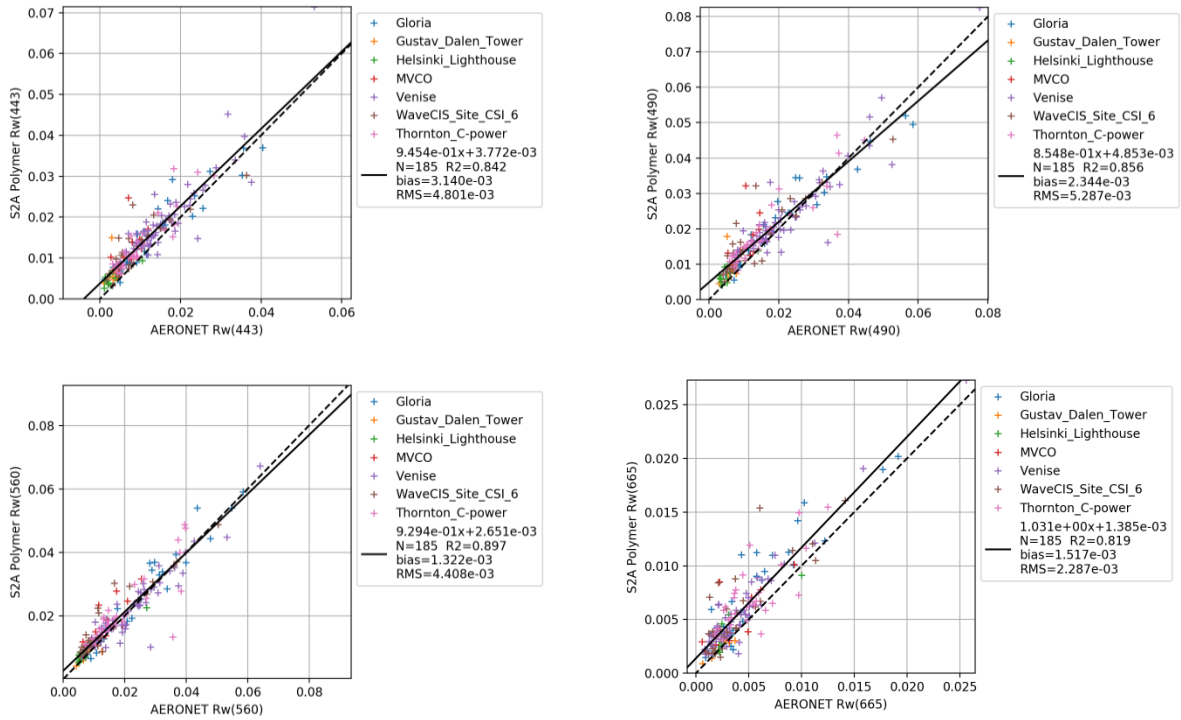


Figure 15: Validation of reflectance (Polymer v4.10) with AERONET-OC stations (coastal waters). (upper left) 443nm; (upper right) 490nm; (lower left) 560nm; (lower right) 665nm

4.2.1 Comparative validation of reflectance with current and candidate atmospheric correction algorithms

Polymer v4.14 was used with the re-introduction of the 100m water quality product into the Service in 2021. Based on results presented here, this version may be upgraded to v4.15 for lake water quality products v3.0. This upgrade introduces a new set of initialization conditions for the underlying bio-optical model, expected to perform better over a wider turbidity range. A similar validation is being carried out for the 300m water quality service component.

The comparison between Polymer v4.14 and v4.15 has been performed using in situ measurements of R_{rs} (With $R_w = R_{rs} \times \pi$). We have used the GLORIA dataset (Lehmann et al, 2023) and observations from automated observation systems, including WISPStations situated at the shore of several lakes in Europe, and So-Rad mobile reflectance platforms installed on ships. Each data source is evaluated independently to allow inspection of dataset-specific uncertainties.

Each matchup with the MSI sensors is constructed from a 3x3 macro-pixel of data centred at the location of the in situ point. There must be at least 5 non-NaN values in the 9 (3x3) pixels and there must be no LAND flagged within the macro-pixel. In addition, invalid pixels were masked out using

a selection of flags from *IdePix*: INVALID, CLOUD, CLOUD_AMBIGUOUS, CLOUD_SURE, CLOUD_BUFFER, CLOUD_SHADOW, CIRRUS_SURE, CIRRUS_AMBIGUOUS, CLUSTERED_CLOUD_SHADOW, and the Polymer OUT_OF_BOUNDS flag. Finally, a homogeneity test was performed on the macro-pixel where the coefficient of variation at 560nm, CV_{560} (standard deviation / mean at $R_w(560)$), must be less than 0.2.

In addition to the above filtering on macro-pixels, the in situ data were also filtered to remove suspect points. The criteria used were to discard any points where the R_{rs} was greater than 0.5 or less than -0.01 sr^{-1} for any equivalent satellite band. Data were also filtered on GLORIA metadata: for GLORIA the "flagged" column was used to remove any points that had been manually flagged as suspect, and for the WISPCloud database the "rrs_quality" was used to remove any points labelled as "suspect". The data were also offset-corrected in the NIR to remove biases, by removing (or adding) signal across the spectrum to make the NIR similarity spectrum conform to the water absorption signature (Ruddick et al. 2006). In situ data were integrated over the sensor response function to match the equivalent satellite bands. This is known to introduce an error in the order of up to 5% (Burggraaff 2020) and can only be resolved by applying the SRF directly to the in situ measured (ir)radiance components prior to deriving R_{rs}, which was not possible for the full dataset provided. If there were multiple in situ observations at the same location on the same day, these were aggregated to use the median spectrum.

When making the comparison between the two Polymer versions, a common set of matchups was used. As not all in situ data contain R_{rs} up to 865nm and higher, these higher wavebands have fewer matchups.

Matchup statistics for the respective datasets follow in the tables and figures below. For each dataset, matchup statistics are provided in tabulated and graphic form, followed by per-band scatter plots, for each version of the atmospheric correction algorithm.

Gloria statistics

Table 4: Statistics for matchups of MSI and GLORIA database

	R_w (n)							
	443 (258)	**490 (258)**	**560 (258)**	**665 (258)**	**705 (258)**	**740 (246)**	**783 (246)**	**865 (136)**
bias_v14	0.0015	0.0000	-0.0017	-0.0011	-0.0014	0.0004	0.0010	0.0001
bias_v15	0.0014	-0.0013	-0.0033	-0.0023	-0.0023	-0.0003	0.0005	-0.0002
rmse_v14	0.0039	0.0051	0.0079	0.0046	0.0043	0.0026	0.0027	0.0016
rmse_v15	0.0103	0.0071	0.0084	0.0046	0.0046	0.0023	0.0022	0.0010
mae_v14	0.0027	0.0029	0.0036	0.0022	0.0020	0.0012	0.0014	0.0007
mae_v15	0.0047	0.0039	0.0045	0.0027	0.0026	0.0012	0.0013	0.0006

r_v14	0.7470	0.7484	0.7999	0.9026	0.9128	0.8031	0.8061	0.8097
r_v15	0.0470	0.4950	0.8156	0.9273	0.9253	0.8642	0.8670	0.8903

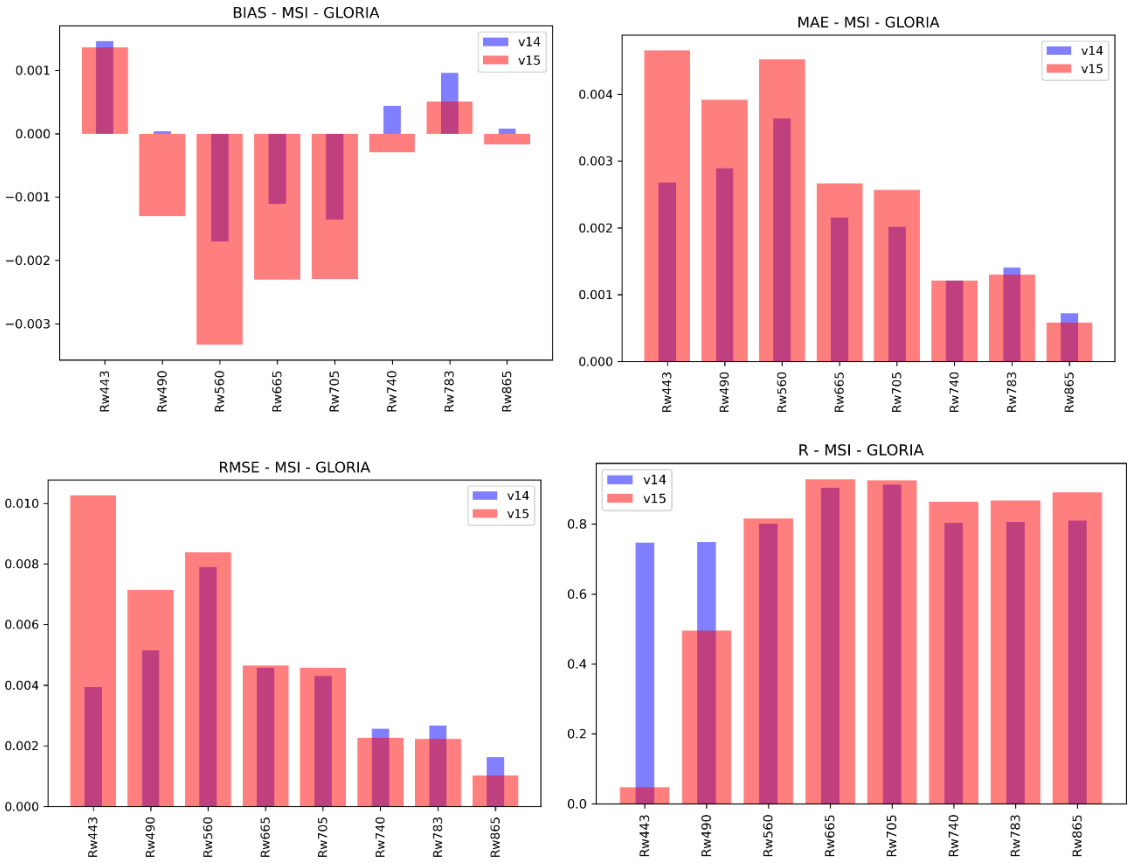


Figure 16: Plots of statistics for Polymer v4.14 and v4.15 for matchups with the GLORIA database. Clockwise from top left: Bias, MAE, R and RMSE

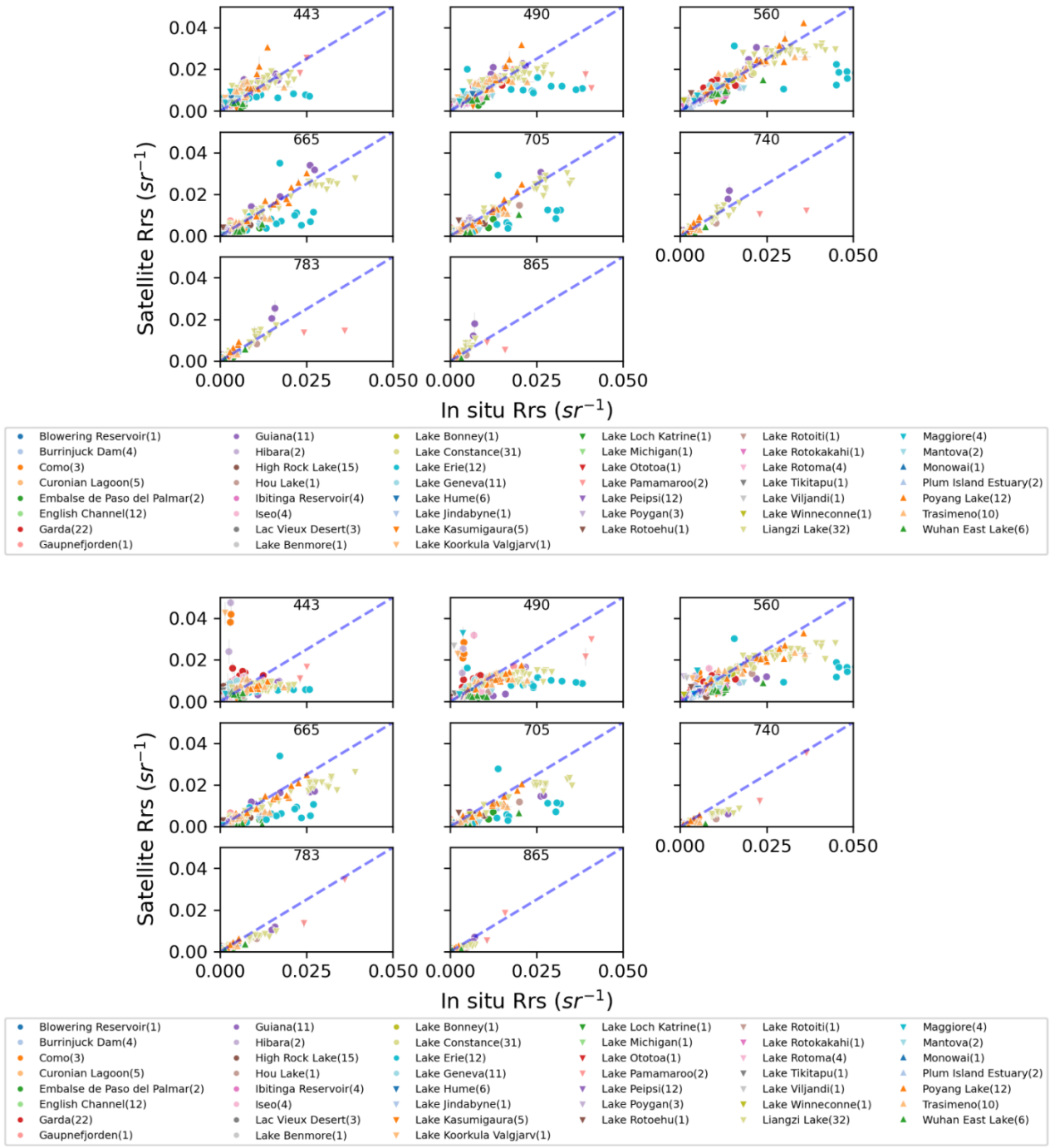


Figure 17: Gloria MSI matchup scatter plots for (top) Polymer v4.14 and (bottom) Polymer v4.15, from the GLORIA database

A comparison between the absolute residual error and distance to land showed that the points with highest residual error are close to land and within the 443nm band. It should be noted that a static

distance to land map was used (Carrea et al, 2015) meaning that the distance to land is referenced against the maximum water extent of the era investigated in the ESA Climate Change Initiative (CCI) Land Cover product, rather than the respective satellite image of the observation in question. Thus, points reported at some distance from the shore may in fact be very close to land at the time of observation.

It can be seen (Figure 18) that Polymer v4.15 band 443nm has a higher error compared to Polymer v4.14. Elsewhere, the residuals imply a similar performance between the two versions.

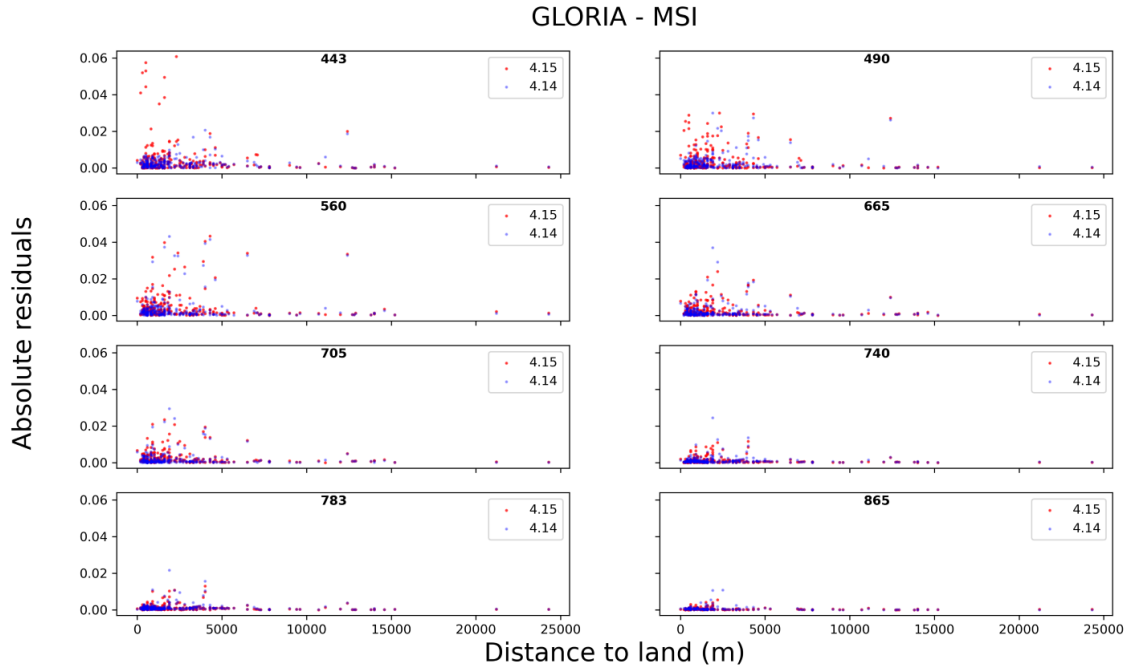


Figure 18: Absolute residual of the MSI retrieved Rrs compared to in situ observations, as a function of the distance to land

WISPCloud statistics

Table 5: Statistics for the matchups of MSI and WISPCloud database

	R_w (n)							
	443 (102)	**490 (102)**	**560 (102)**	**665 (102)**	**705 (102)**	**740 (102)**	**783 (102)**	**865 (102)**
bias_v14	-0.0008	-0.0029	-0.0055	-0.0026	-0.0031	-0.0008	-0.0004	-0.0002
bias_v15	-0.0020	-0.0046	-0.0072	-0.0035	-0.0037	-0.0012	-0.0005	0.0001

rmse_v14	0.0049	0.0058	0.0094	0.0049	0.0054	0.0029	0.0030	0.0031
rmse_v15	0.0062	0.0077	0.0109	0.0054	0.0056	0.0024	0.0021	0.0013
mae_v14	0.0039	0.0040	0.0062	0.0034	0.0038	0.0015	0.0013	0.0009
mae_v15	0.0046	0.0056	0.0077	0.0040	0.0042	0.0016	0.0013	0.0008
r_v14	0.5790	0.8013	0.8784	0.8359	0.7920	0.6153	0.5852	0.4858
r_v15	0.2751	0.7170	0.8720	0.8524	0.8214	0.6843	0.6680	0.5593

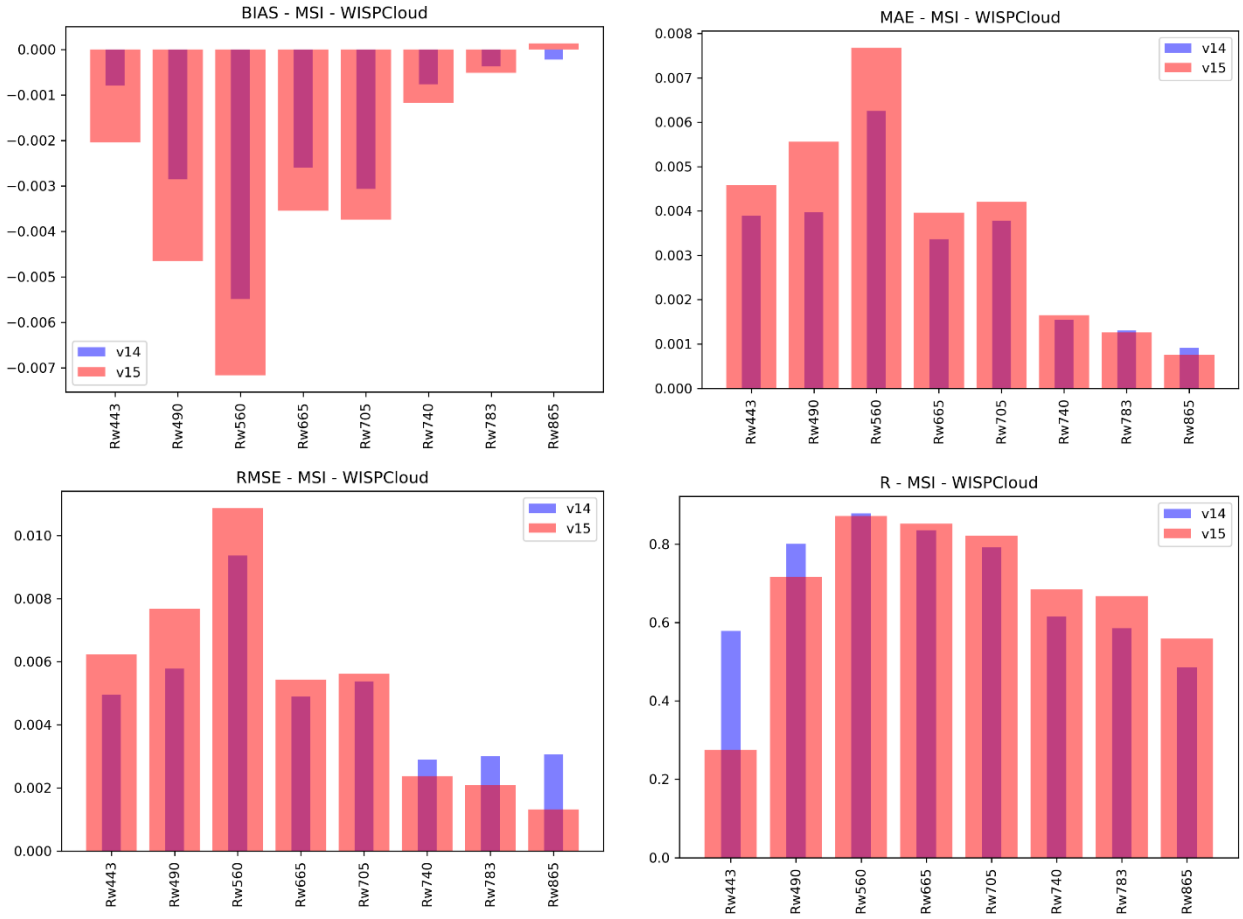


Figure 19: WISPCloud matchups: plots of statistics for Polymer v4.14 and v4.15. Clockwise from top left: Bias, MAE, R and RMSE

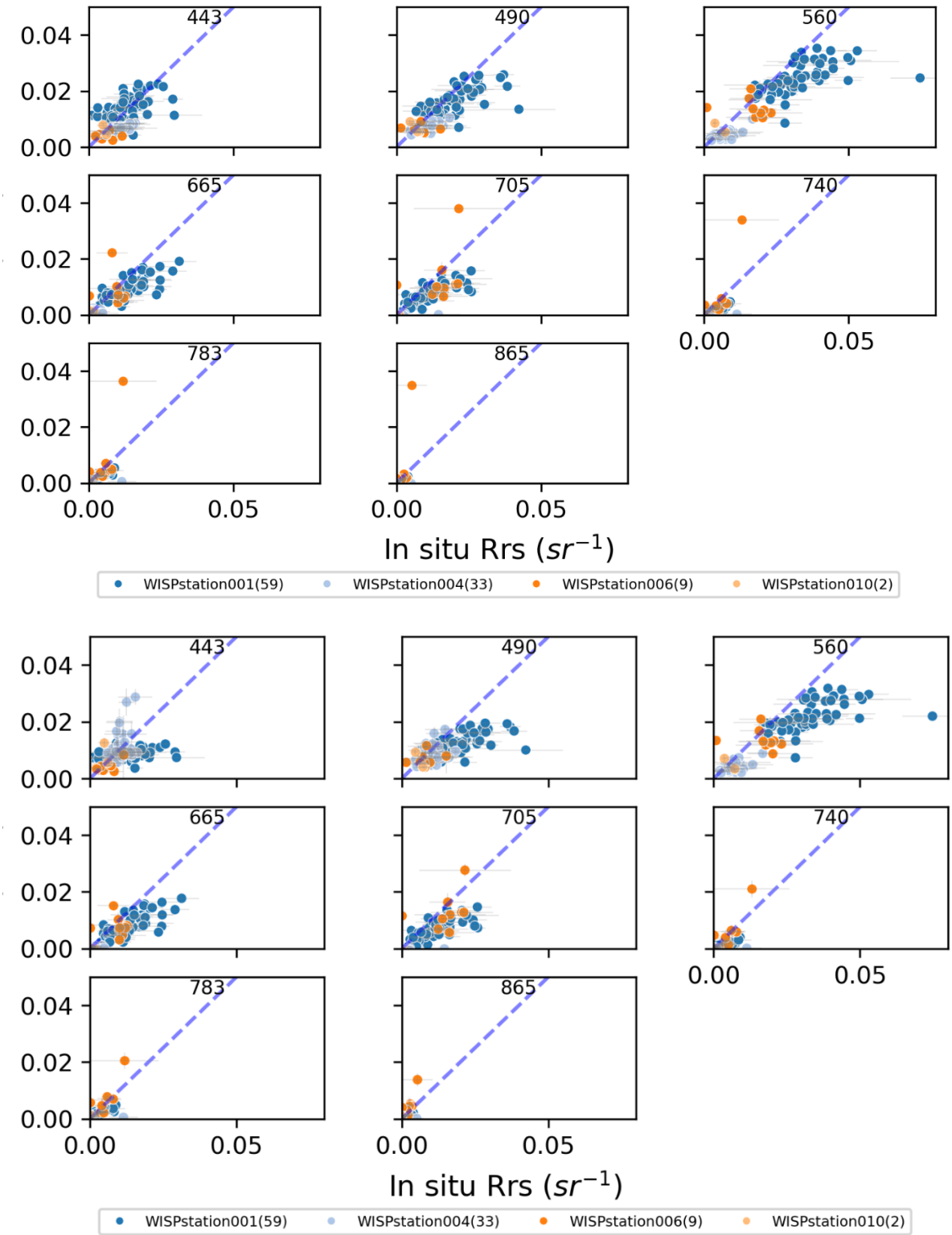


Figure 20: WISPCloud MSI scatter plots for (top) Polymer v4.14 and (bottom) Polymer v4.15

So-Rad Database

Table 6: Statistics for the matchups of MSI and So-Rad database

	R_w (n)							
	443 (7157)	**490 (7157)**	**560 (7157)**	**665 (7157)**	**705 (7157)**	**740 (7157)**	**783 (7157)**	**865 (7157)**
bias_v14	-0.0066	-0.0048	-0.0031	-0.0025	-0.0028	-0.0017	-0.0008	-0.0008
bias_v15	-0.0078	-0.0067	-0.0048	-0.0034	-0.0033	-0.0018	-0.0006	0.0000
rmse_v14	0.0156	0.0104	0.0078	0.0063	0.0057	0.0033	0.0027	0.0017
rmse_v15	0.0162	0.0115	0.0087	0.0068	0.0062	0.0037	0.0032	0.0025
mae_v14	0.0074	0.0057	0.0048	0.0037	0.0035	0.0018	0.0014	0.0009
mae_v15	0.0085	0.0071	0.0058	0.0044	0.0040	0.0024	0.0017	0.0013
r_v14	-0.0507	0.0863	0.2973	0.3703	0.3500	0.2038	0.2001	0.1867
r_v15	0.0114	0.0653	0.2605	0.3147	0.2766	0.0545	0.0638	0.0299

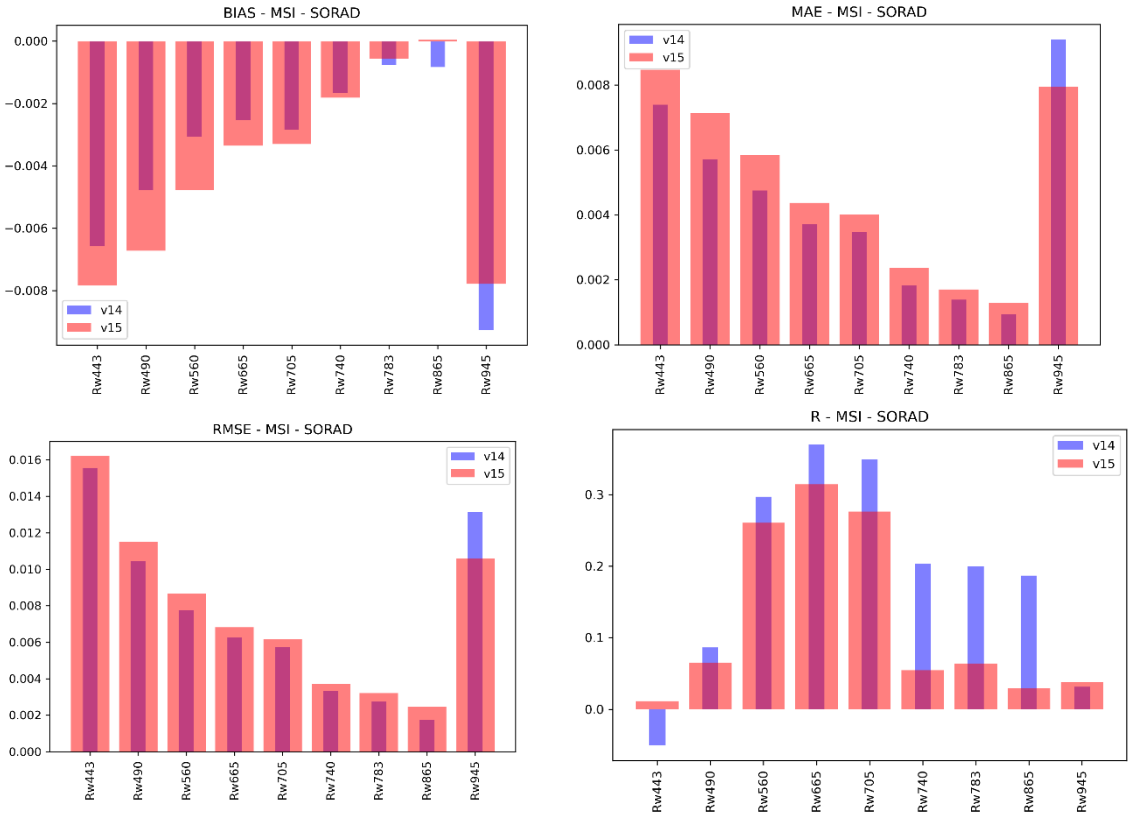
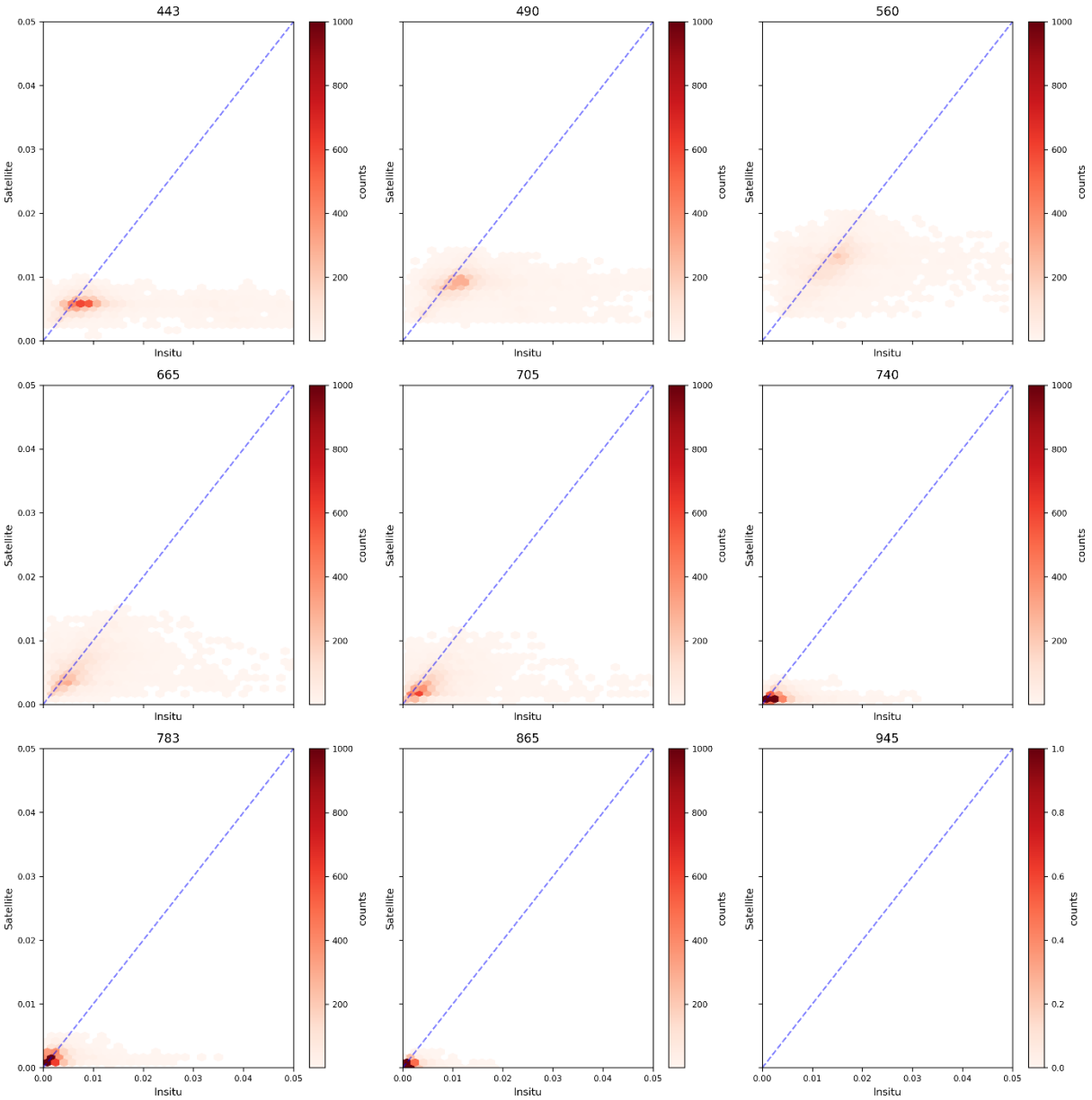


Figure 21: So-Rad matchups: plots of statistics for Polymer v4.14 and v4.15. Clockwise from top left: Bias, MAE, R and RMSE


443
490
560
665
705
740
783
865
945
Satellite
Insitu
counts

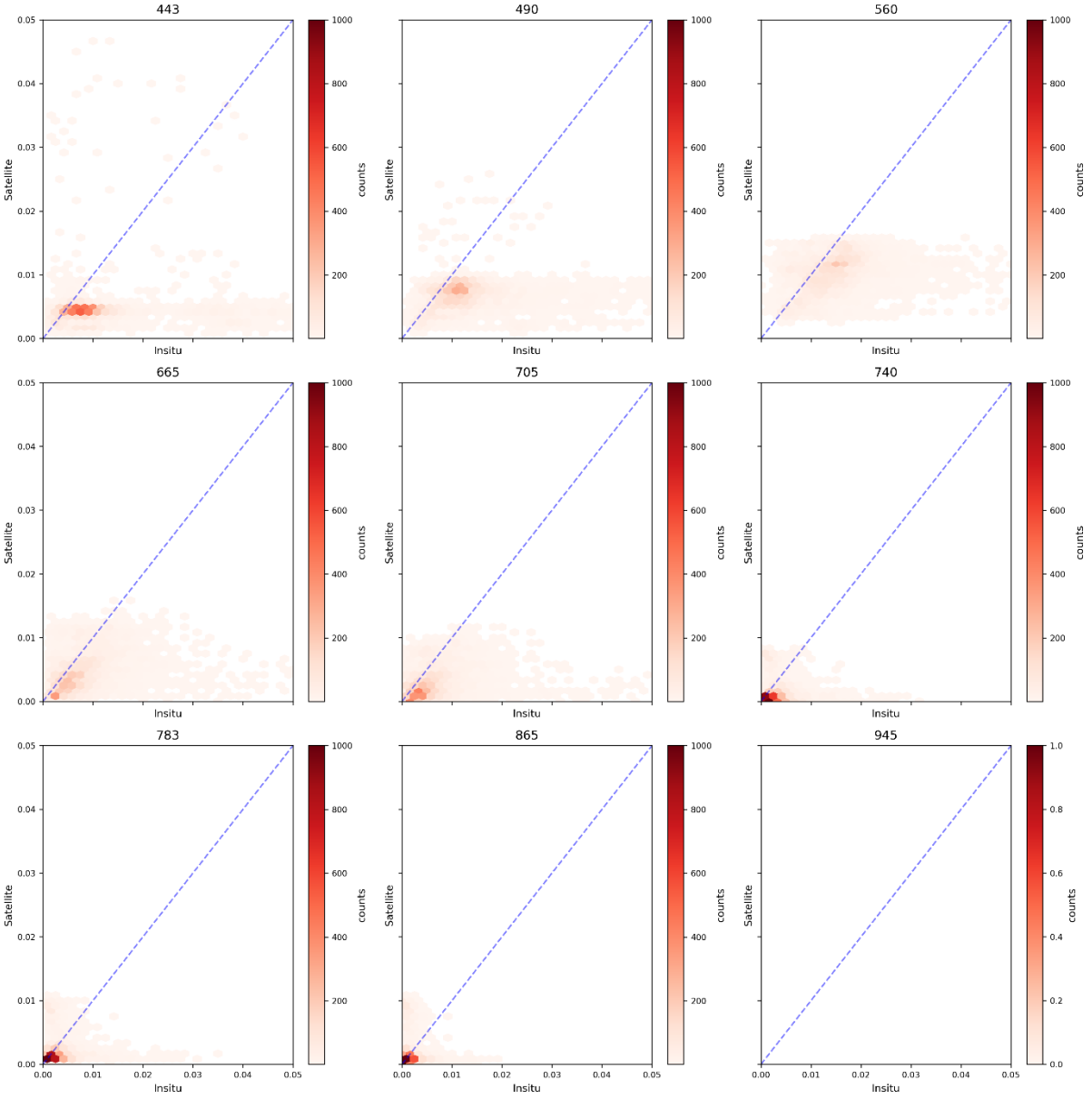


Figure 22: So-Rad matchups: MSI density heatmaps for (top) Polymer v4.14 and (bottom) Polymer v4.15

4.3 CONSISTENCY TESTS BETWEEN SENSORS

4.3.1 Data selection for consistency tests

Four lakes of different size and water type were selected to investigate product consistency between sensors. Time series were extracted at the positions shown in Figure 23. Additionally, transect data were extracted at the indicated locations.

The analysis compares the Sentinel-2 MSI Level 3 product at 100m spatial resolution (v1.5) with the Sentinel-3 OLCI Level 3 product at 300m spatial resolution (v1.4). During extraction, the 100m product is resampled to 300m by nearest neighbour selection. For some investigations on the consistency of the products with respect to the optical water types defined for Sentinel-2 MSI, the transect extractions are carried out on the Level 2 MSI data v1.5. Examples of v1.5 for OLCI are also included.

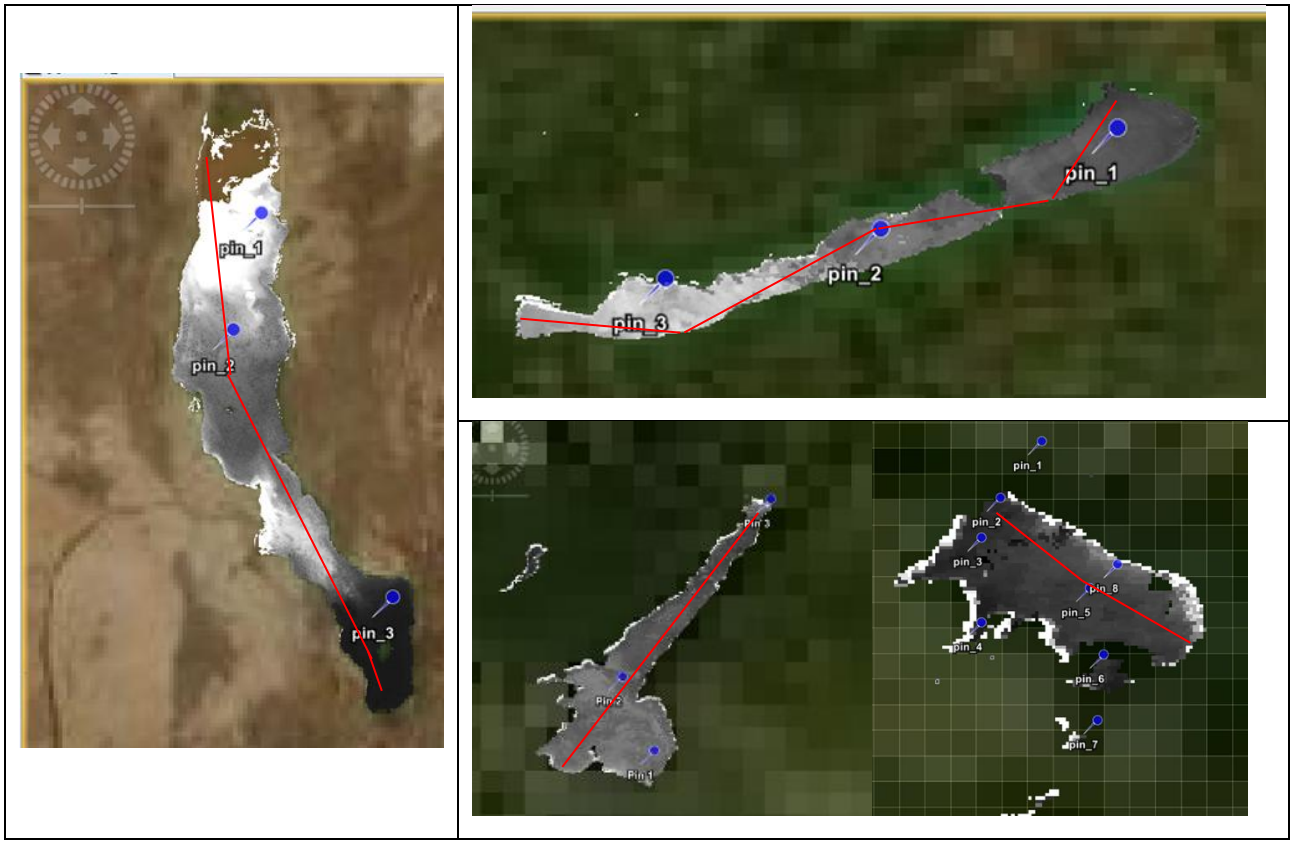


Figure 23: Positions of transects and extraction points for time series in Lake Turkana (left), Lake Balaton (top right), Lake Garda (bottom middle) and Lake Mueritz (bottom right). Spatial scales differ between the selected lakes

4.3.2 Time series Sentinel-2-A MSI versus S3-A OLCI

For the investigation of consistency between sensors, time series plots were generated to compare seasonal cycles of two years of OLCI 300m products v1.4 with MSI 100m products v1.5 (Figure 24 to Figure 31). Each figure shows the time series at one virtual station (for their positions refer to Figure 23) in a lake for each variable (water leaving reflectance, turbidity and TSI). The OLCI 300m time series extends from January 2020 to April 2022 (marked with crosses), while 100m products are available from July 2021 to April 2022 (circles).

4.3.2.1 Lake Mueritz

For the water leaving reflectance spectra (443 to 865nm) the MSI data fits well within the range of the corresponding OLCI data, though R_w665 and R_w705 are a bit higher than OLCI values. Turbidity is in the same range (below 2FNU) in both sensors, but MSI turbidity is substantially lower in most of the cases. Then again, TSI from MSI is in good correspondence with OLCI.

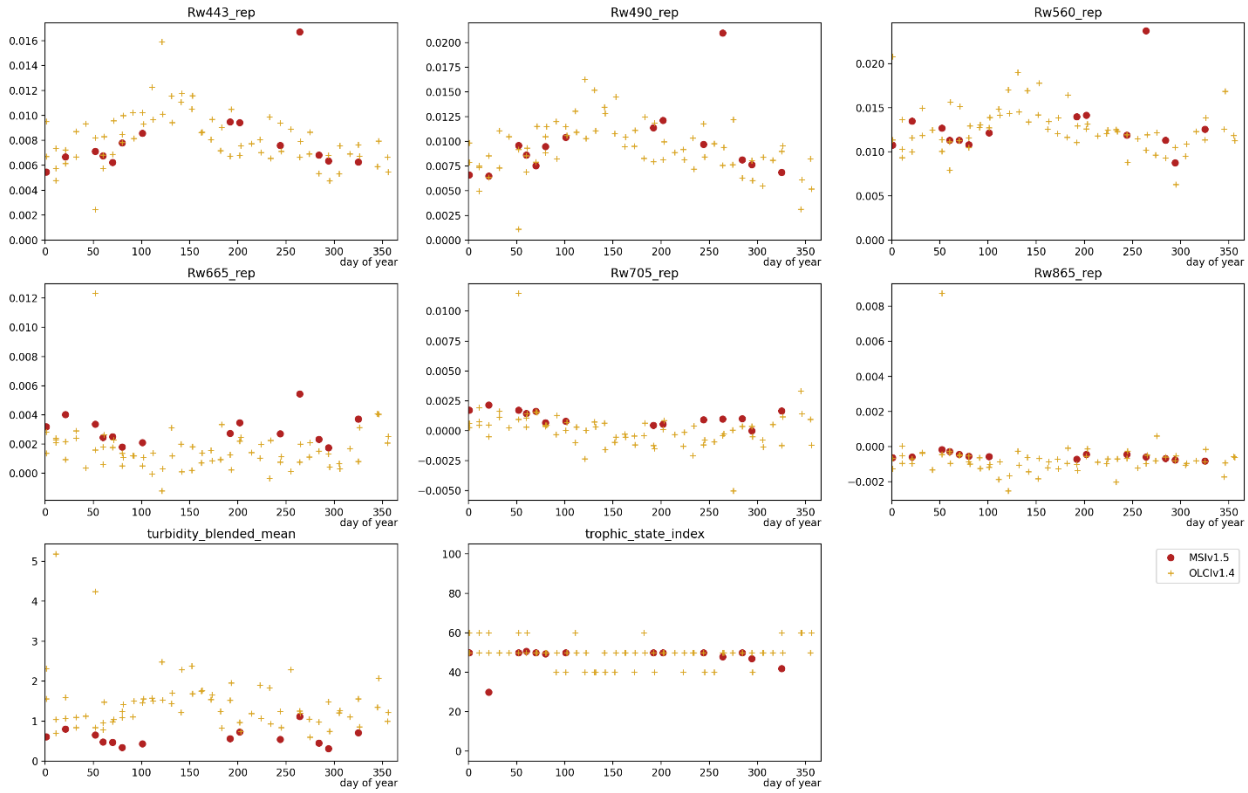


Figure 24: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for central Lake Mueritz at position 5 (close to lake center)

4.3.2.2 Lake Balaton

The waters of Lake Balaton are sometimes not well represented by the underlying bio-optical model in the atmospheric correction, which then occasionally yields erratic water leaving reflectance spectra. In these cases, extremely high turbidity values are obtained (Figure 25, Figure 26). There can be a strong mismatch between MSI and OLCI reflectance especially towards the blue (443 and 490nm bands).

The TSI is derived from both sensors with similar values.

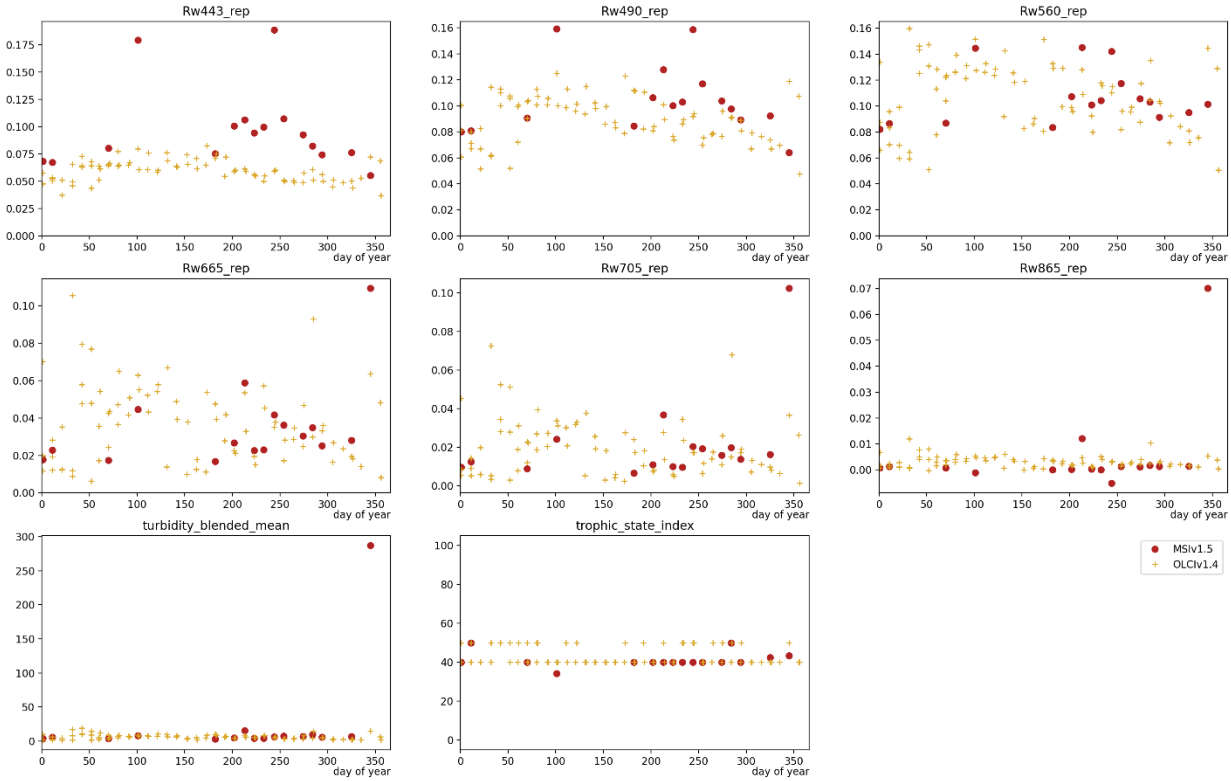


Figure 25: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Balaton (station 1)

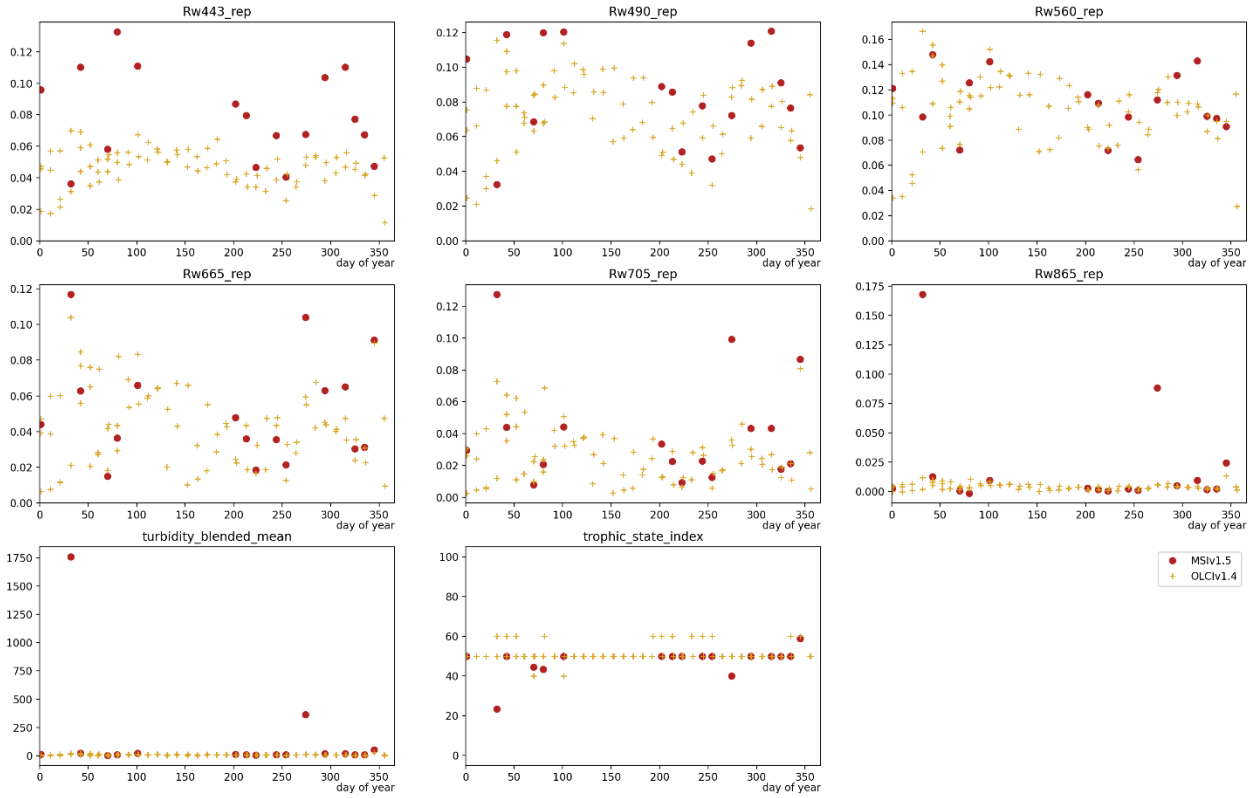


Figure 26: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Balaton (station 2)

4.3.2.3 Lake Turkana

The three stations in Lake Turkana represent very different water types: station 1 lies within the high turbidity zone in the northern part, station 2 is at an intermittent state and station 3 is positioned in a substantially less turbid part in the southern part of the lake (Figure 27 to Figure 29, respectively).

At station 1, the water leaving reflectance for MSI is strongly higher in the blue bands (443 and 490nm) and remain on the upper range of OLCI values for all other bands. Turbidity is higher in MSI L3 including some extreme values above 150NFU. TSI tends to be higher for MSI than for OLCI at station 1 and at the upper range at station 2. At station 3 the agreement in TSI is good.

The seasonal cycle is quite pronounced in the water leaving reflectance from 560 to 865nm at stations 1 and 2 and both sensors show it equally well. For 443 and 490nm, MSI reflectance is much higher than OLCI reflectance, and no clear seasonal cycle is visible at station 1.

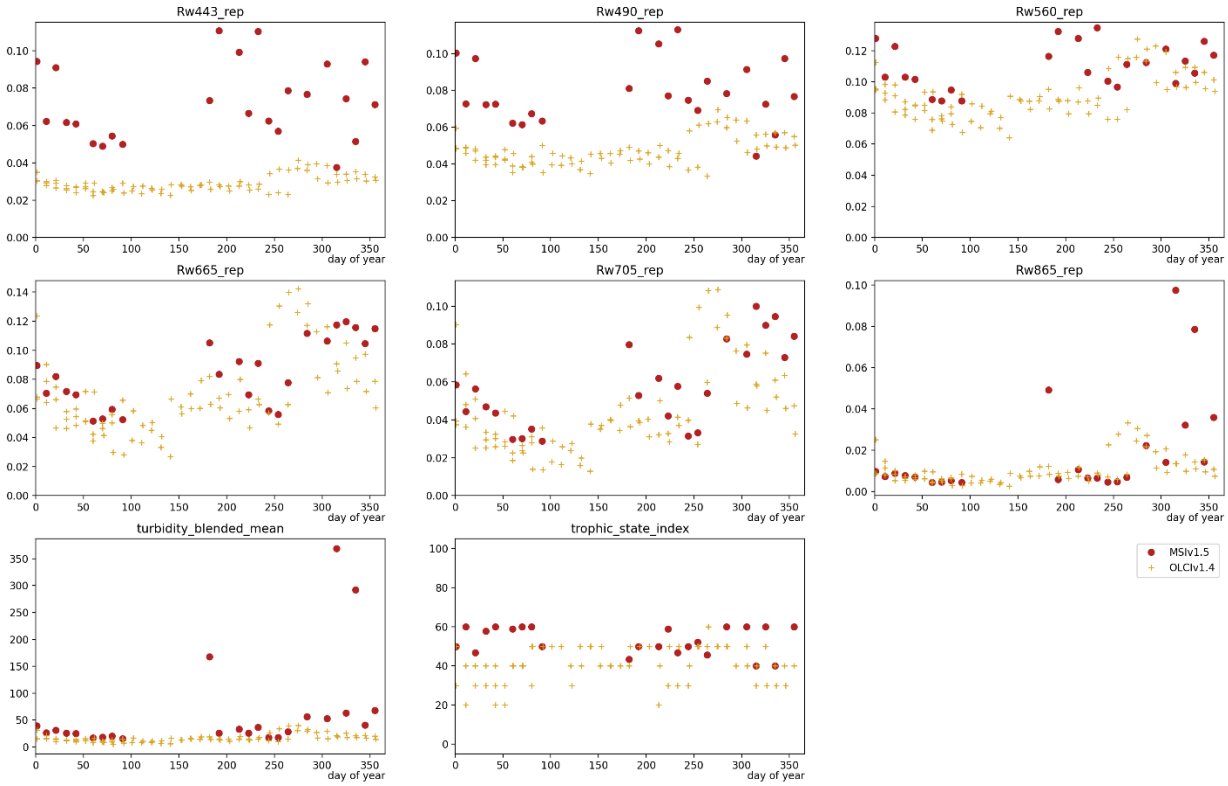


Figure 27: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Turkana (station 1)

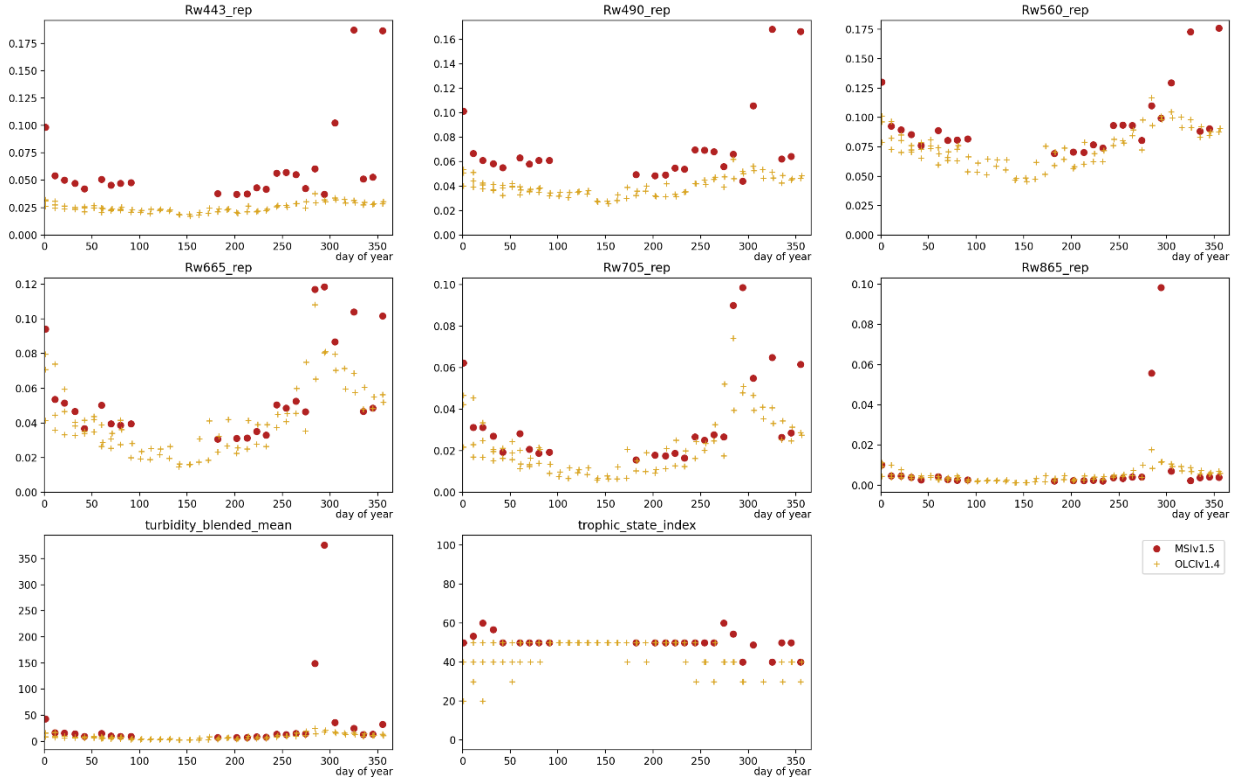


Figure 28: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Turkana (station 2)

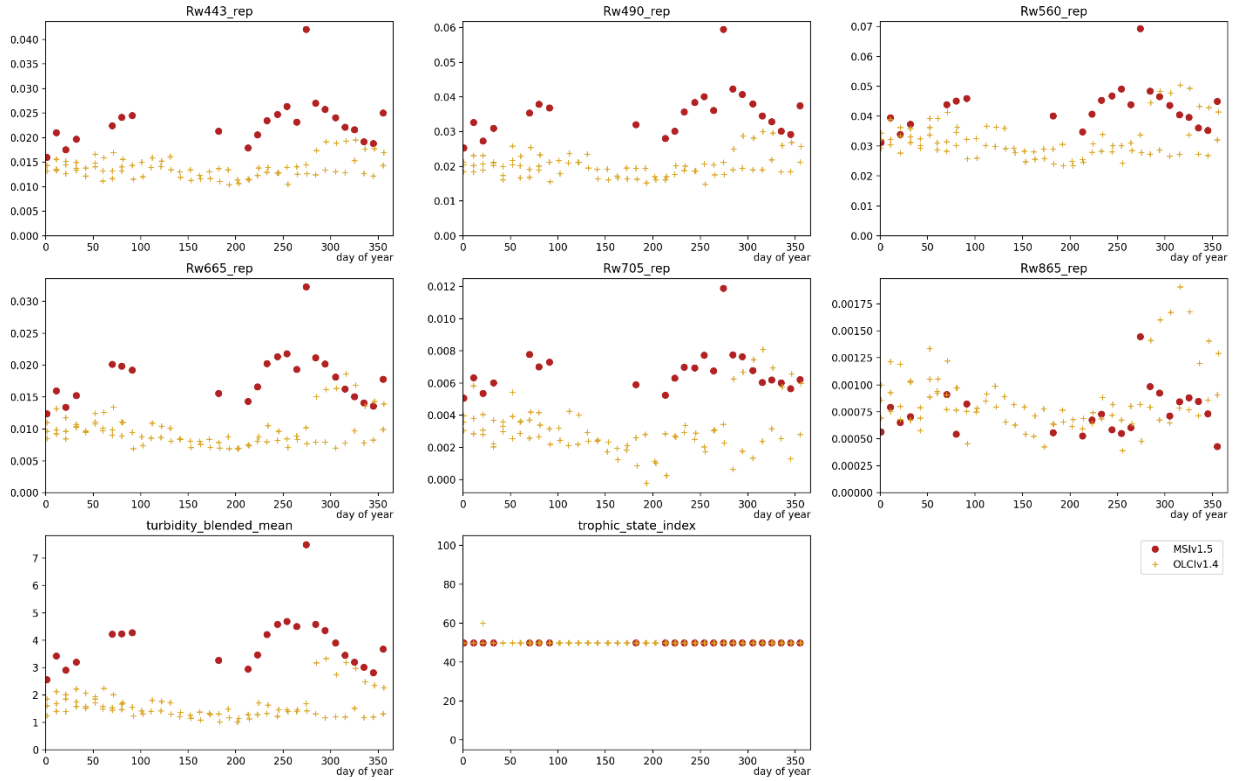


Figure 29: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Turkana (station 3)

4.3.2.4 Lake Garda

The seasonal cycle in water leaving reflectance from Lake Garda is well pronounced. The MSI data show very similar values to the OLCI data at bands 443 to 560 and 865nm (Figure 30 and Figure 31). Only for the 665nm and 705nm bands MSI values are significantly higher than OLCI reflectance. These differences lead to the strong mismatch of MSI turbidity with respect to OLCI turbidity, where MSI turbidity is much lower. The prediction of TSI is similar for both sensors.

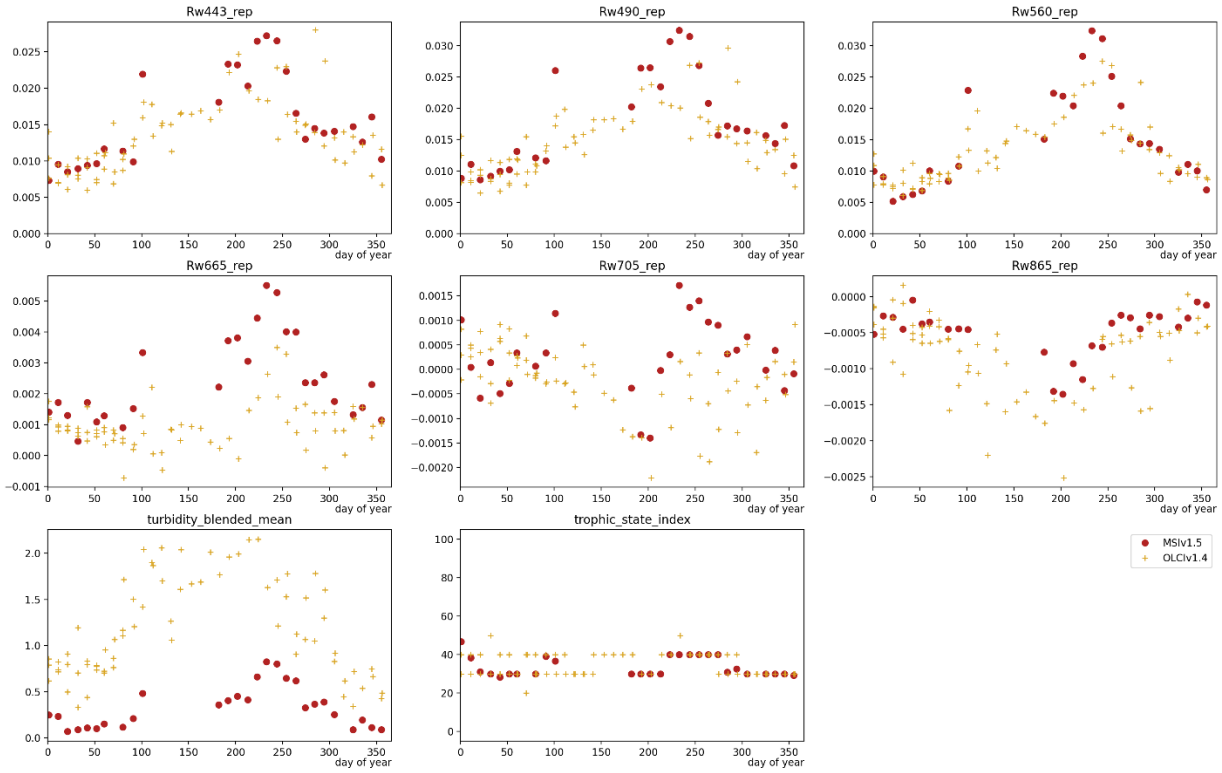


Figure 30: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Garda (station 1)

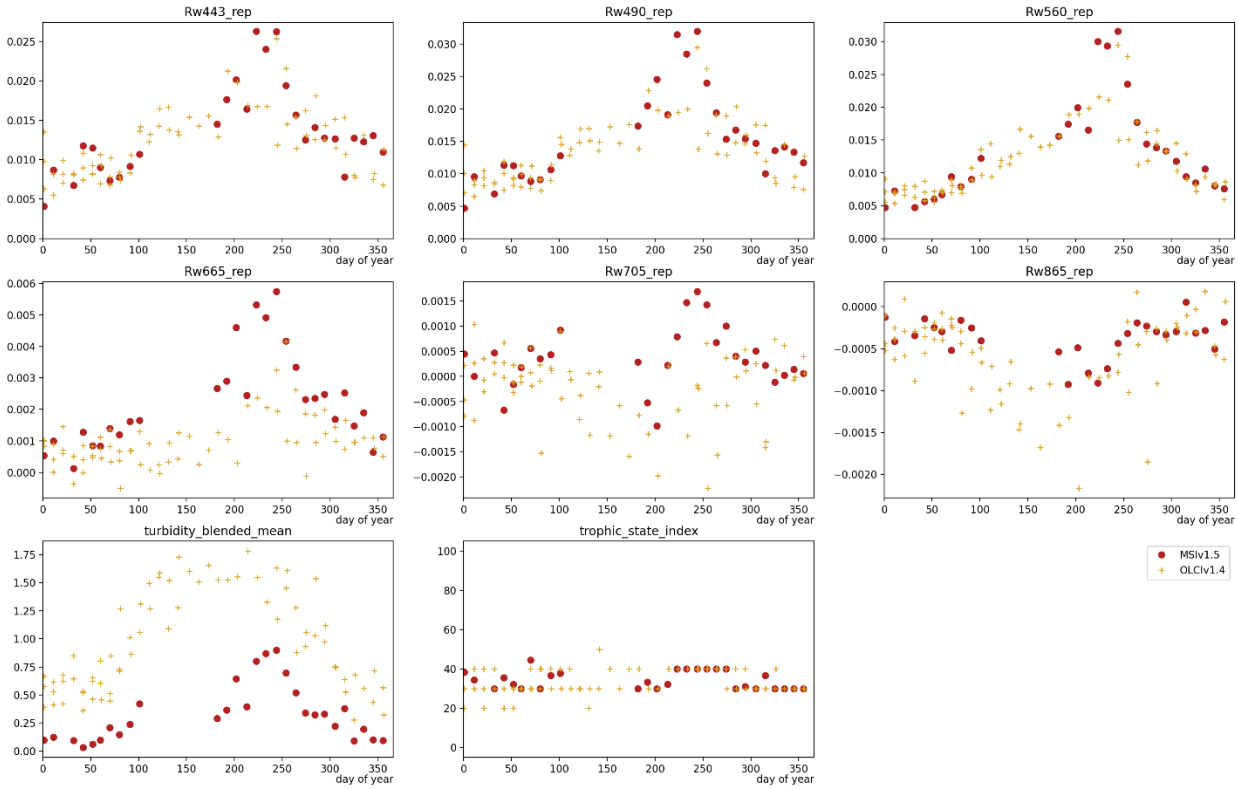


Figure 31: OLCI 300m (v1.4) and MSI 100m (v1.5) seasonal comparison for Lake Garda (station 2)

4.3.3 Spatial consistency using transect extracts on 100m and 300m products

Transects have been defined for the different lakes, for same day acquisitions of MSI and OLCI. Data were extracted for both the 100m and the 300m products along the transects, and 100m products are resampled by nearest neighbour algorithm to the 300m grid. Three water leaving reflectance bands (443, 560 and 665nm) and turbidity have been plotted for the products of both sensors along the transect, so that spatial patterns, the dynamic ranges and the absolute values can be compared.

4.3.3.1 Lake Turkana

Along a transect on 1rst of July 2021 in Lake Turkana, the reflectance spectra differ strongly for 443nm, particularly in the very turbid northern part of the lake. For the most part, reflectance is higher for MSI than for OLCI. The turbidity is substantially lower for the OLCI product and MSI based values can exceed 100FNU easily. Spatial patterns in the MSI data are of higher dynamic range than in the OLCI product. TSI has in large parts a value of 60 for both products.

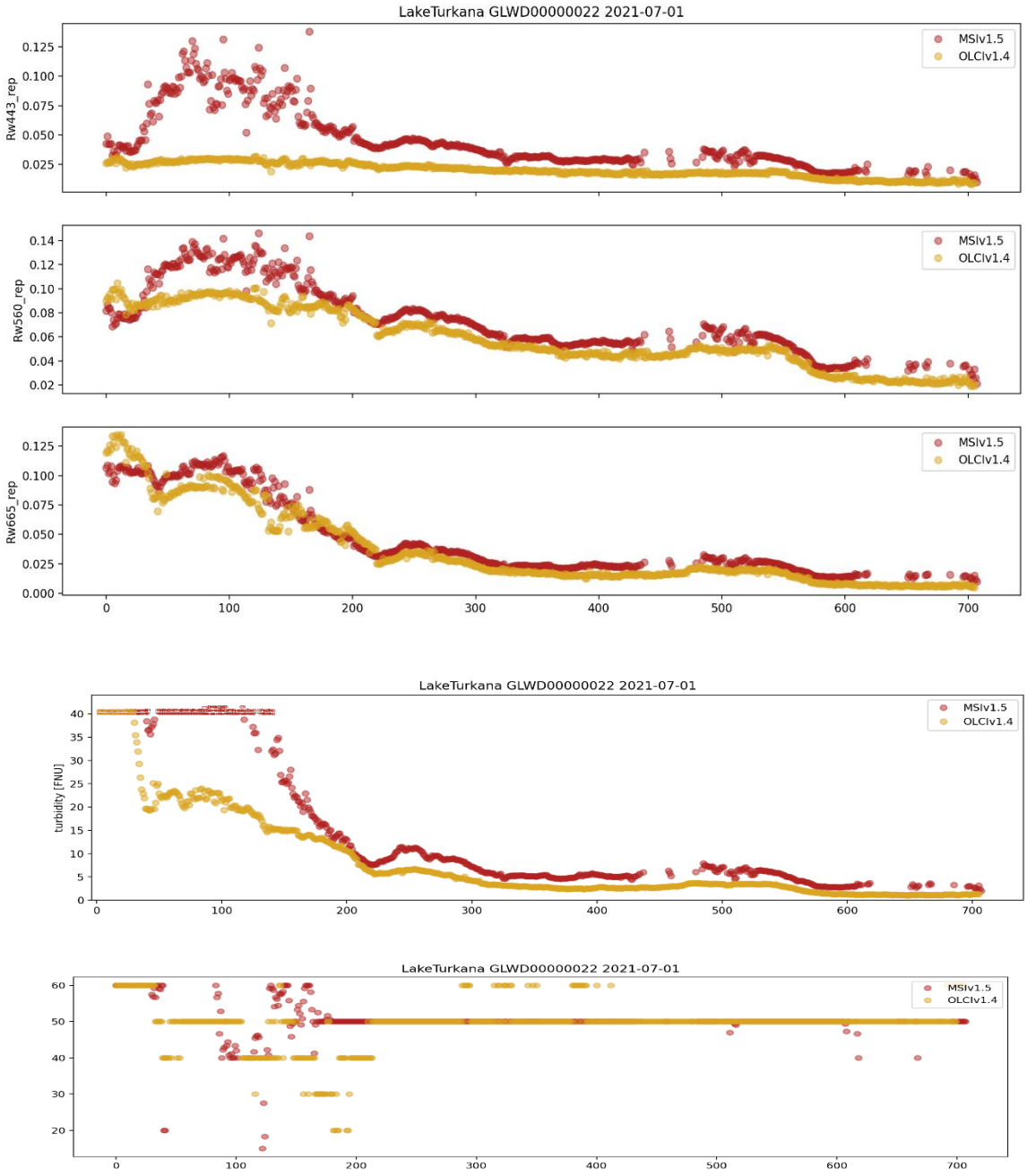


Figure 32: Water leaving reflectance at 443, 560 and 665nm, and turbidity in the MSI Level 3 100m v1.5 product and the OLCI Level 3 300m v1.4 product. Pixels are arranged from North to South. Turbidity is cut-off at 40FNU, to highlight spatial patterns at lower ranges. Maximum values are: OLCI 54.7FNU and MSI 189.6FNU. TSI shown at the bottom graph

4.3.3.2 Lake Garda

Along the transect in Lake Garda the TSI for MSI and OLCI is mostly 40 (Figure 33). Turbidity derived from MSI data is lower than OLCI. Spatial patterns in the MSI data are of lower dynamic range than in the OLCI product for bands 560 and 665nm, which is in contrast with the results seen for Lake Turkana.



Figure 33: Water leaving reflectance at 443, 560 and 665nm, and turbidity and TSI in the MSI Level 3 100m v1.5 product and the OLCI Level 3 300m v1.4 product. Extracts are arranged in the North-South direction. TSI shown at the bottom graph

4.3.3.3 Lake Balaton

Along the transect in Lake Balaton the TSI for MSI is 50 and 40, while for OLCI is mainly 60 and 50. The turbidity for both sensors is similar. High values above 12FNU for MSI products seem too large. Spatial patterns differ for MSI and OLCI, as observation days in the 10-day mean may be different.

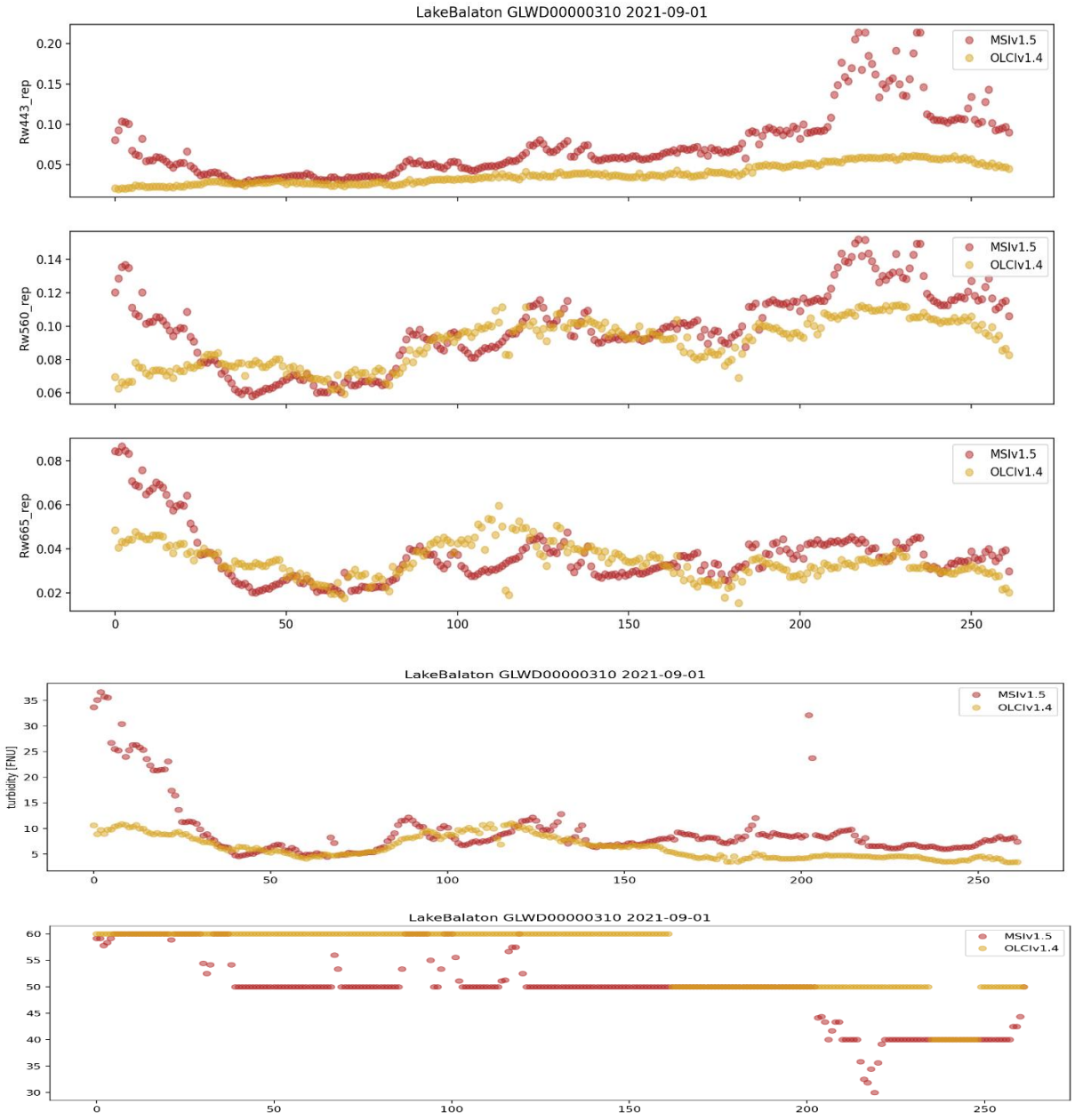


Figure 34: Water leaving reflectance at 443, 560 and 665nm, and turbidity in the MSI Level 3 100m v1.5 product and the OLCI Level 3 300m v1.4 product. Extracts are arranged in the West-East direction. TSI shown at the bottom graph

4.3.3.4 Lake Mueritz

Along the transect in Lake Mueritz TSI for MSI and OLCI is 50. The turbidity values for MSI are significantly lower than for OLCI, but in a low range of turbidity below 4FNU. Reflectance values are similar for R_w443 and R_w560, but MSI shows higher reflectance at 665nm.

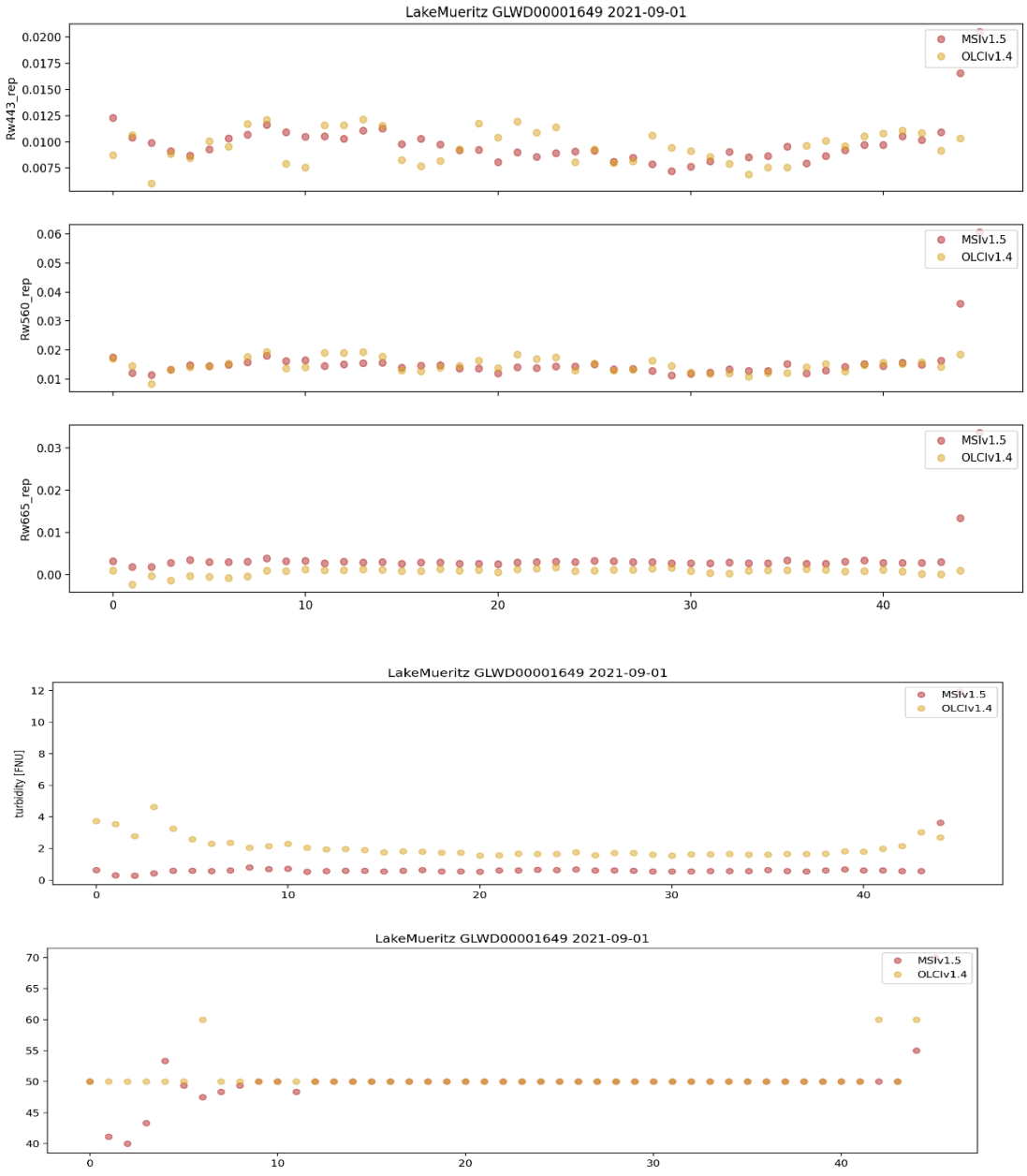


Figure 35: Water leaving reflectance at 443, 560 and 665nm, and turbidity and TSI in the MSI Level 3 100m v1.5 product and the OLCI Level 3 300m v1.4 product. Extracts are arranged in the West-East direction. TSI shown at the bottom graph

4.3.4 Spatial Consistency summary

To each 300m data point along the transect, the nearest 100m Sentinel-2 MSI extraction within the 300m radius is selected. This procedure creates a pseudo-matchup dataset from the transects, which can be represented as scatterplots.

Lake Garda and Lake Mueritz show reasonable spectral consistency between both products, if analysed with the data extracts along the transects (Figure 39, Figure 41). For higher turbidity regions in Lake Turkana and Balaton (Figure 38, Figure 40) reflectance is often strongly overestimated by the 100m product, leading to exceedingly high turbidity values.

Best agreement between both products is achieved in R_w560, R_w665 and R_w709. Lake Turkana and Lake Balaton with the most turbid waters show strong overestimations in R_w443 and R_w490 in 100m products, while for Lake Mueritz the 100m products provide lower values in the blue bands compared to 300m products.

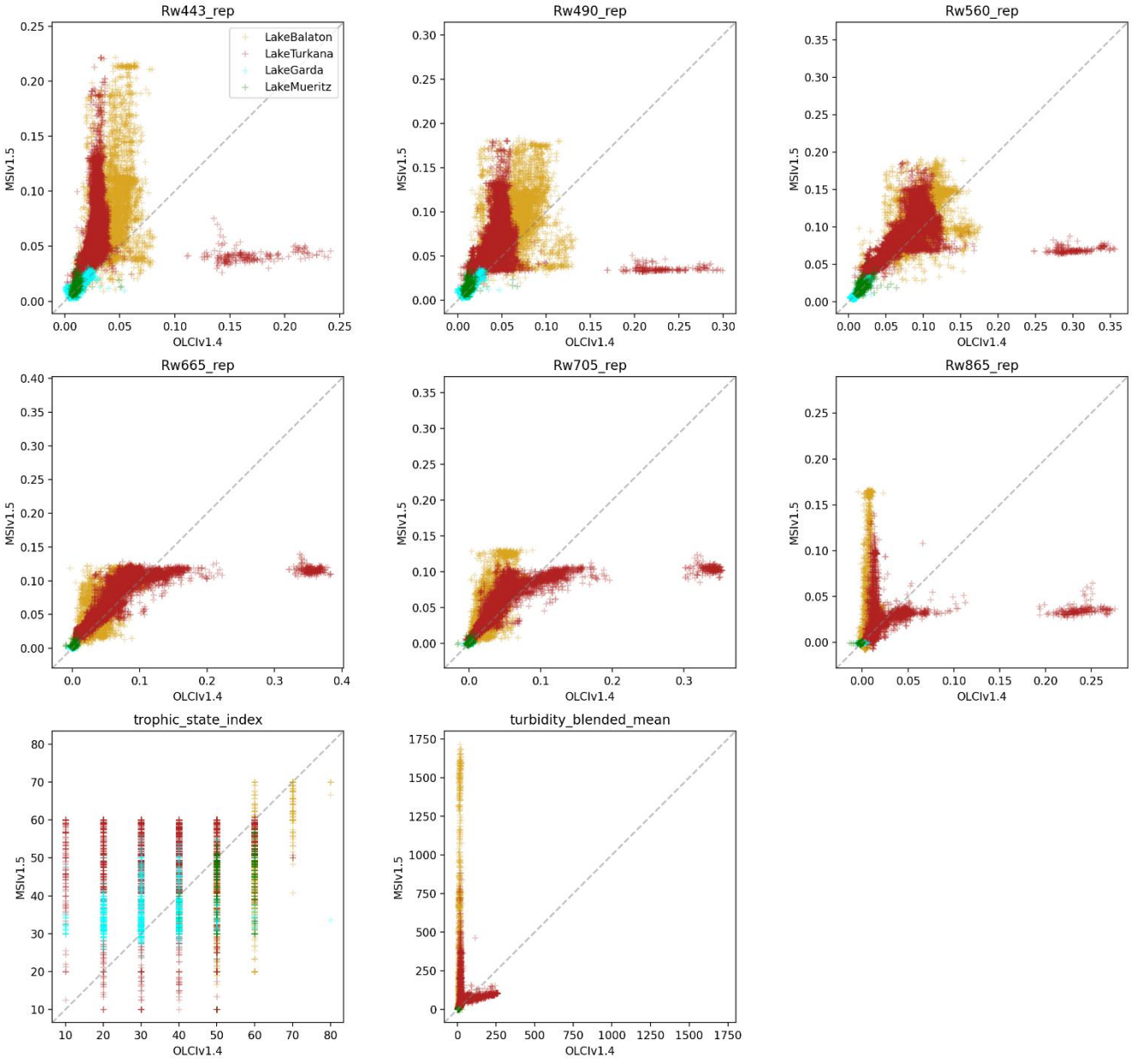


Figure 36: Data points of transect extractions (100m and 300m products, July 2021 to April 2022) for Lake Balaton (yellow), Turkana (red), Garda (light blue), Mueritz (green)

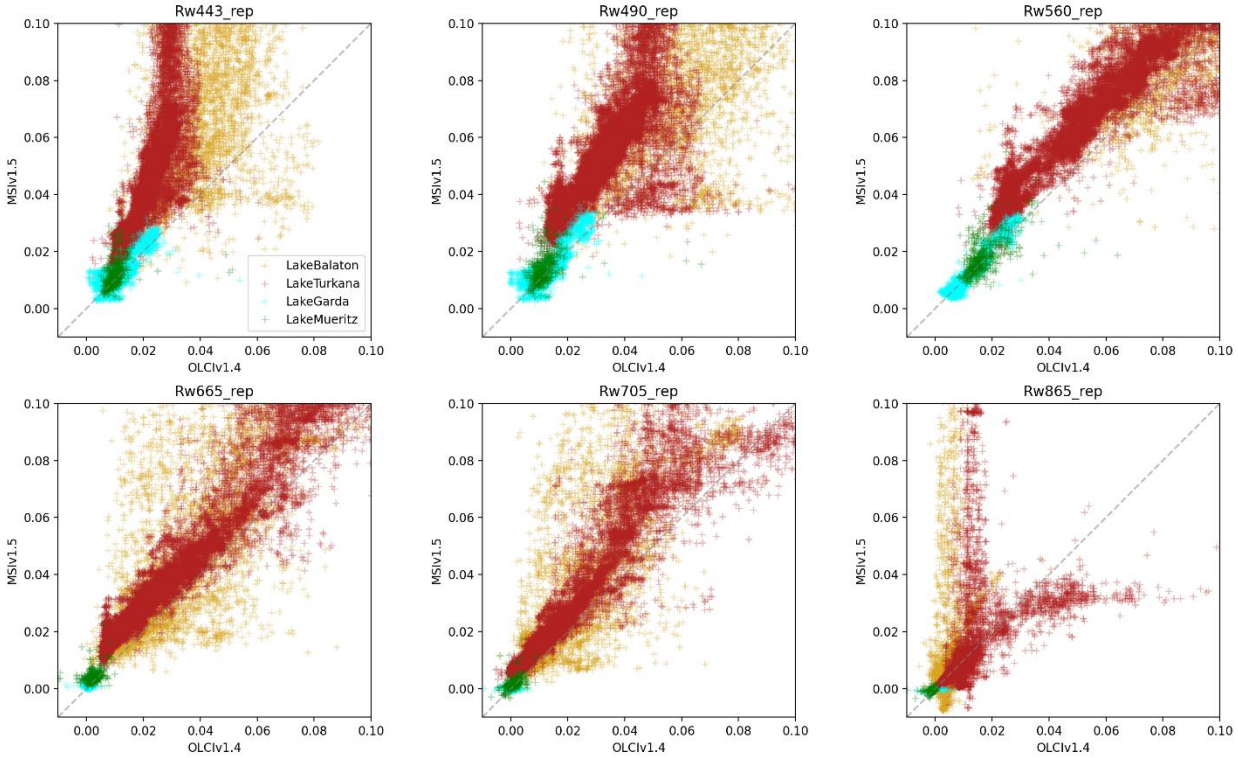


Figure 37: Data points of transect extractions (July 2021 to April 2022) for Lake Balaton (yellow), Turkana (red), Garda (light blue), Mueritz (green). Subset of Figure 36

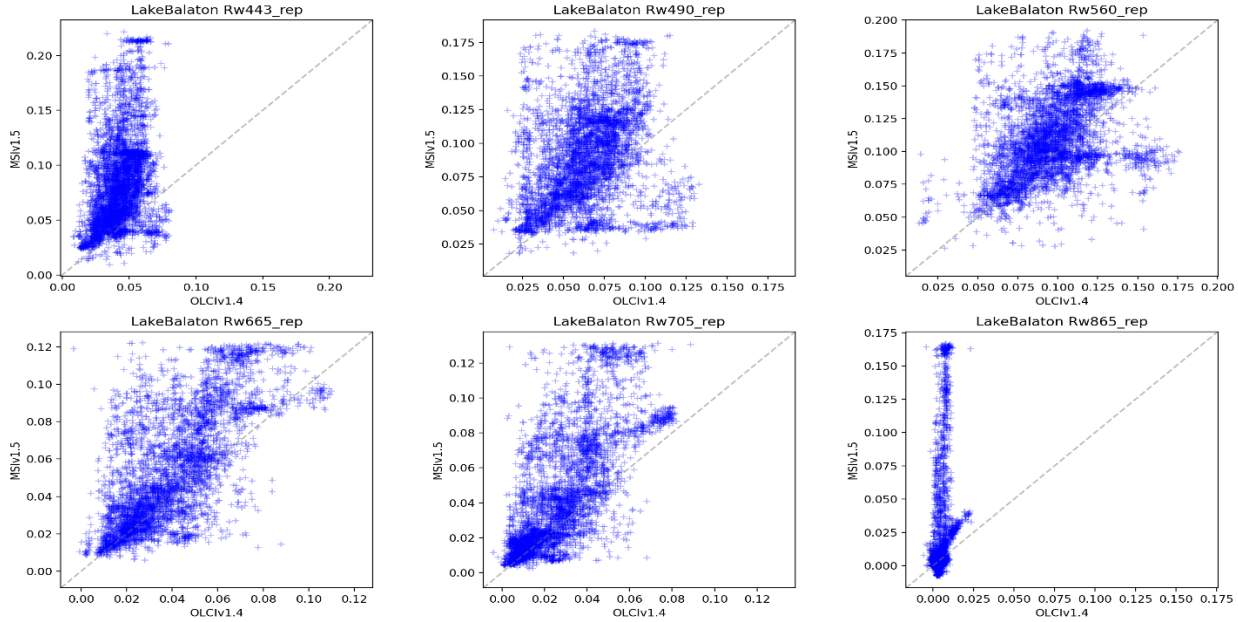


Figure 38: Reflectance at common points in space and time (extracts of transects, July 2021 to April 2022) for Lake Balaton

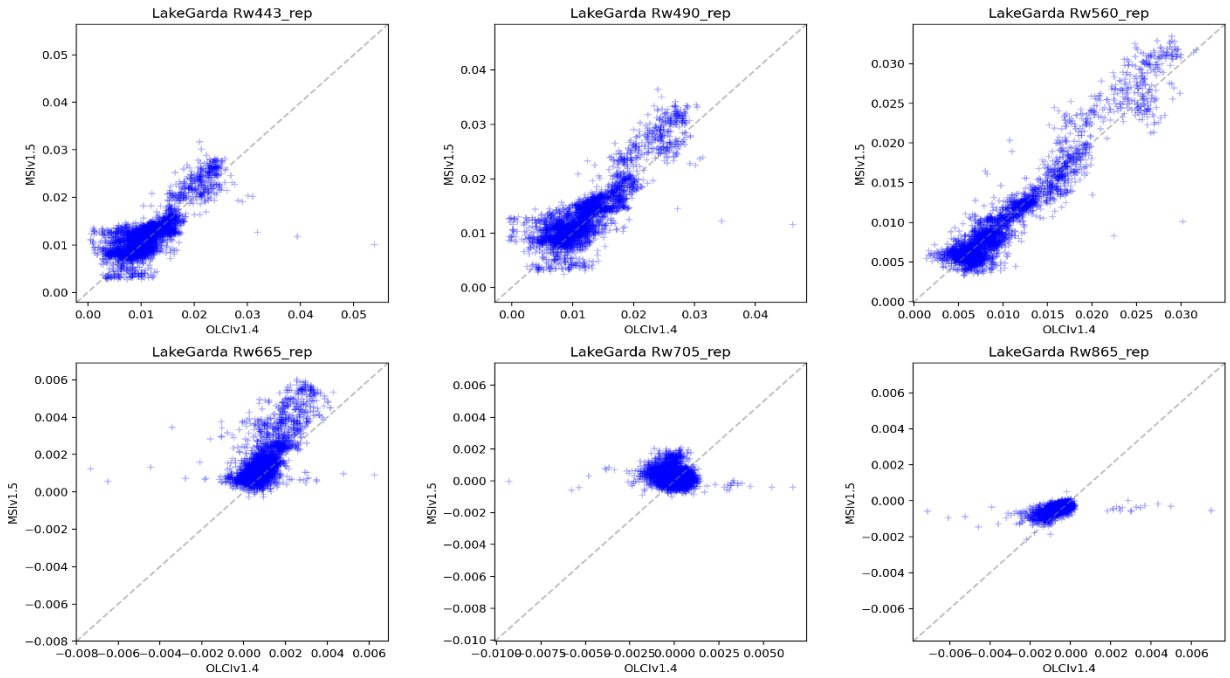


Figure 39: Reflectance at common points in space and time (extracts of transects, July 2021 to April 2022) for Lake Garda

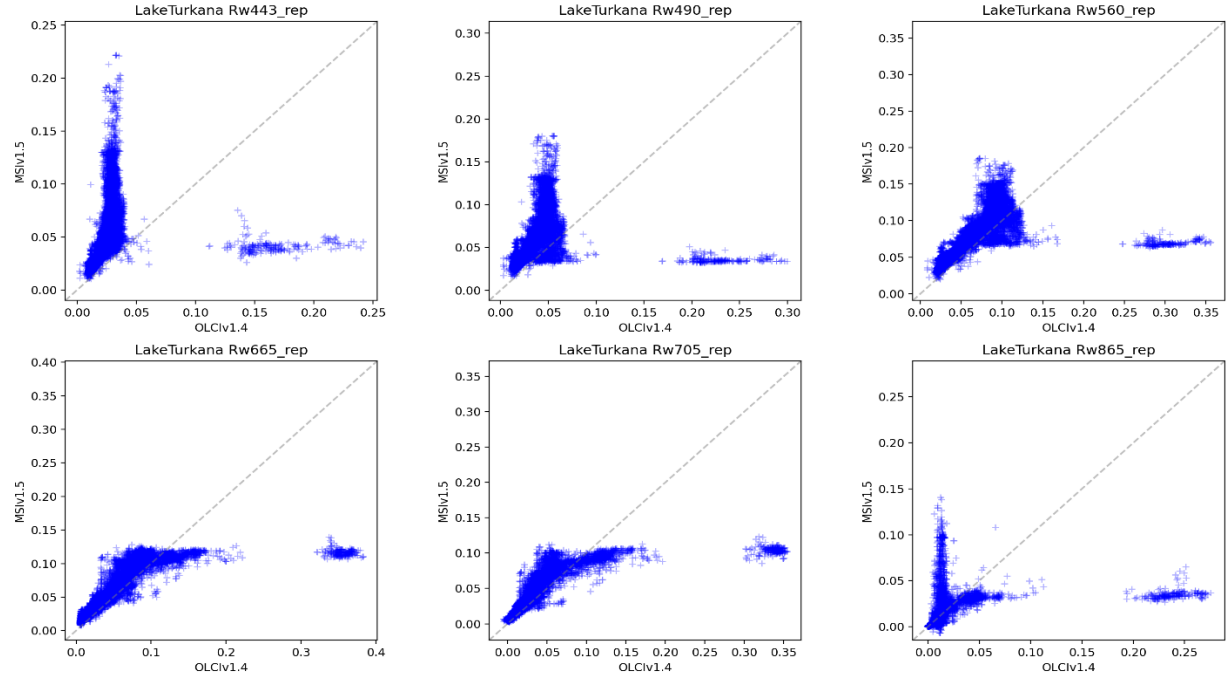


Figure 40: Reflectance at common points in space and time (extracts of transects, July 2021 to April 2022) for Lake Turkana

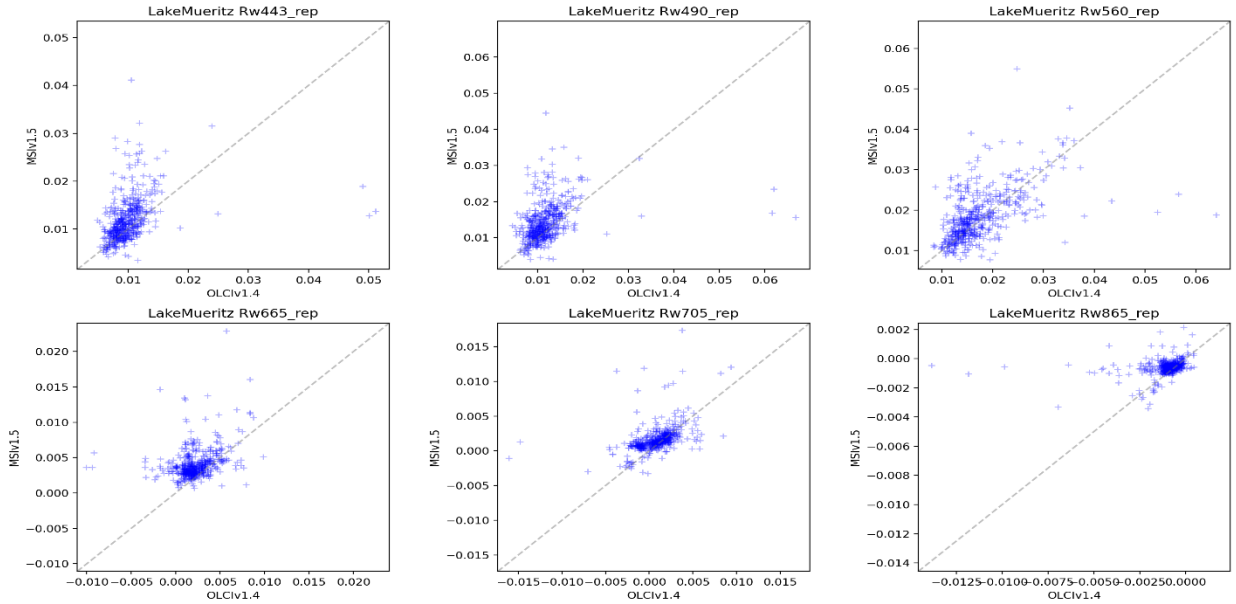


Figure 41: Reflectance at common points in space and time (extracts of transects, July 2021 to April 2022) for Lake Mueritz

4.3.5 Spectral consistency

The spectral consistency of the products can be visualised by extracting coinciding observations and compare their spectral shapes. Here, the analysis was based on Level 2 products for MSI v1.5 and the L3 OLCI product. Data were extracted along the defined transects (see Figure 23) for a selected day per lake (corresponding with the transects shown above, Figure 32 to Figure 35): Lake Garda 2021-09-21, Lake Turkana 2021-07-01, Lake Mueritz 2021-09-01 and Lake Balaton 2021-09-01. Considering that the most representative spectrum is selected for the Level 3 10-day product, this approach also provides insights into the comparability of the 10-day products at 100m and 300m resolution.

The transect extractions are split into groups per lake by the MSI derived optical water type. The dominant OWT is used as classification, membership values are not evaluated. All OWTs which are covered by more than one spectrum per lake are shown here.

For OWT 1 spectral shapes and intensities of R_w are in reasonable agreement for both sensors, if the analysis focuses on the visible part of the spectrum and excludes the bands in the blue (Figure 42). In the blue (443nm), MSI reflectance can be significantly higher in turbid lakes (Balaton, Turkana). OWT 1 comprises a large variety of spectral shapes and intensity ranges.

Some spectra of turbid waters can be classified as OWT 2 (here: Balaton, Turkana, Figure 43). Spectral shapes of R_w of both sensors are similar. Again, MSI derives higher intensities throughout the visible spectrum with particular emphasis on the blue and green bands. In the NIR part, reflectance is often negative for both sensors, with MSI tending towards larger absolute values.

In OWT 5 the deviation in spectral shape reaches its maximum considering this example of one transect in a 10-day period from Lake Balaton (Figure 44).

Spectral shapes in OWT 9 are similar in the visible except the blue, where MSI data yields higher intensities (Figure 45). For spectra with their maximum at 683nm for OLCI or 665nm for MSI, OLCI spectra a brighter than MSI spectra (Turkana). In lake Balaton, all MSI spectra are brighter than OLCI spectra in the visible.

OWT 10 represents very turbid waters (here: Lake Turkana). The spectral shape shows similarities in the visible, but the red and NIR part is a lot higher in intensity for MSI (Figure 46).

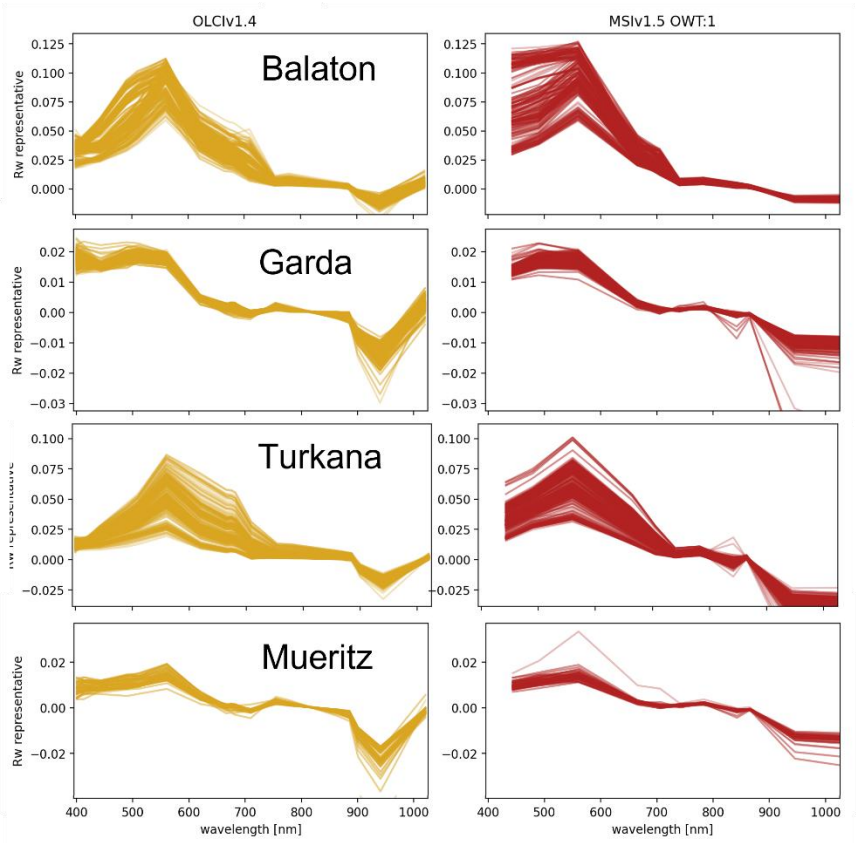


Figure 42: Sentinel-2 MSI L2 v1.5 and Sentinel-3 OLCI L3 v1.4 water leaving reflectance spectra for MSI optical water type 1. The spectra are quite similar in shape and values with exception of the blue bands. Same scale per lake

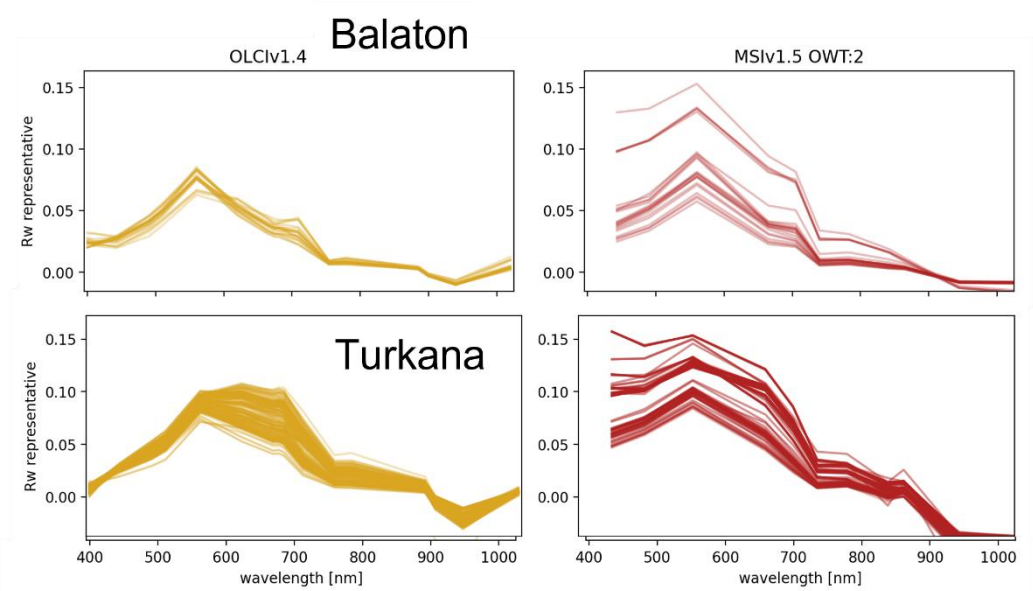


Figure 43: Sentinel-2 MSI L2 v1.5 and Sentinel-3 OLCI L2 v1.4 water leaving reflectance spectra for MSI optical water type 2. Spectral shapes are similar; the values in the blue can divert strongly between OLCI and MSI

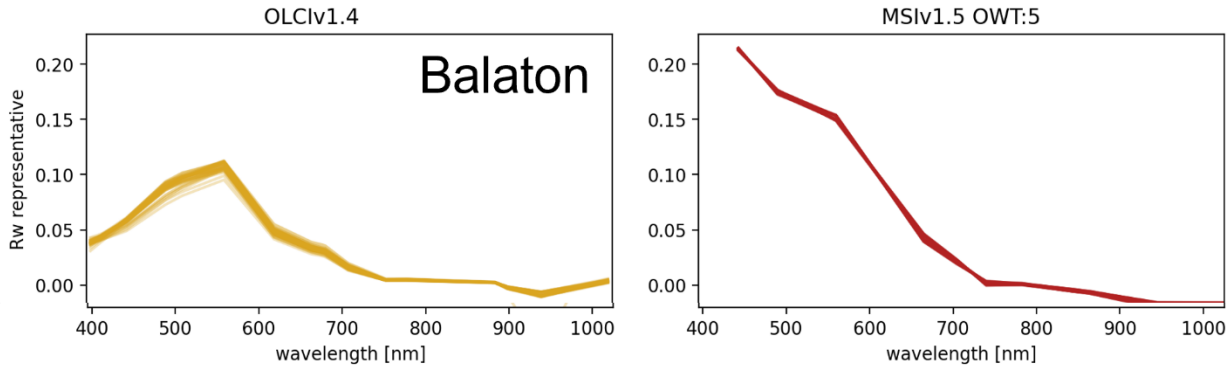


Figure 44: Sentinel-2 MSI L2 v1.5 and Sentinel-3 OLCI L2 v1.4 water leaving reflectance spectra for MSI optical water type 5. The spectral shapes are deviating the most for this water type

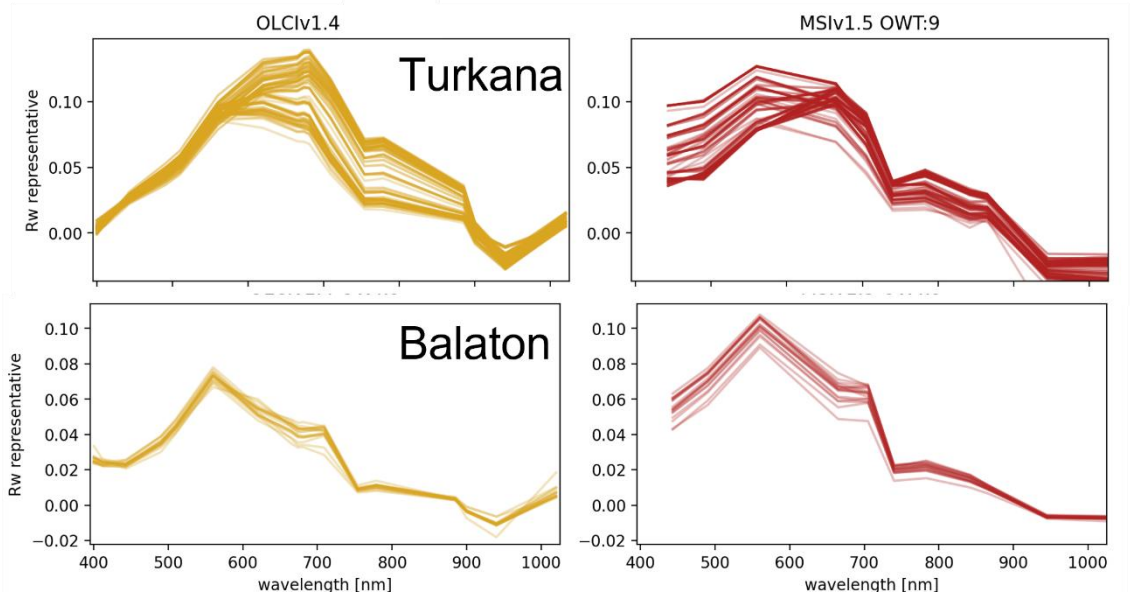


Figure 45: Sentinel-2 MSI L2 v1.5 and Sentinel-3 OLCI L2 v1.4 water leaving reflectance spectra for MSI optical water type 9

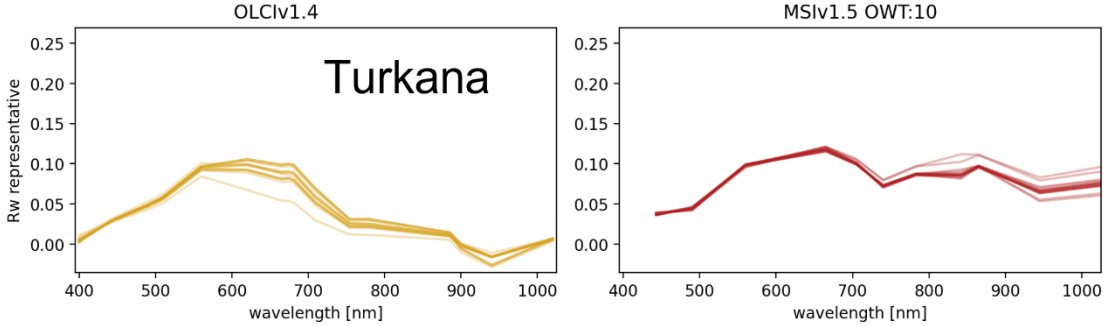


Figure 46: Sentinel-2 MSI L2 v1.5 and Sentinel-3 OLCI L2 v1.4 water leaving reflectance spectra for MSI optical water type 10

Table 7: Overview of data selection per lake and number of spectra per MSI OWT

Lake	OLCI date L3	MSI L2 day(s) of month	OWT 1	OWT 2	OWT 5	OWT 9	OWT 10
Balaton	2021-09-01	A: 4 B: 6	300	23	43	18	-
Turkana	2021-07-01	A: 7, 10 B: 2, 5	401	228	-	115	16
Garda	2021-09-21	A: 23, 30 B: 25, 28	291	-	1	-	-
Mueritz	2021-09-01	A: 3, 6, 10 B: 1,5,8	89	1	-	-	-

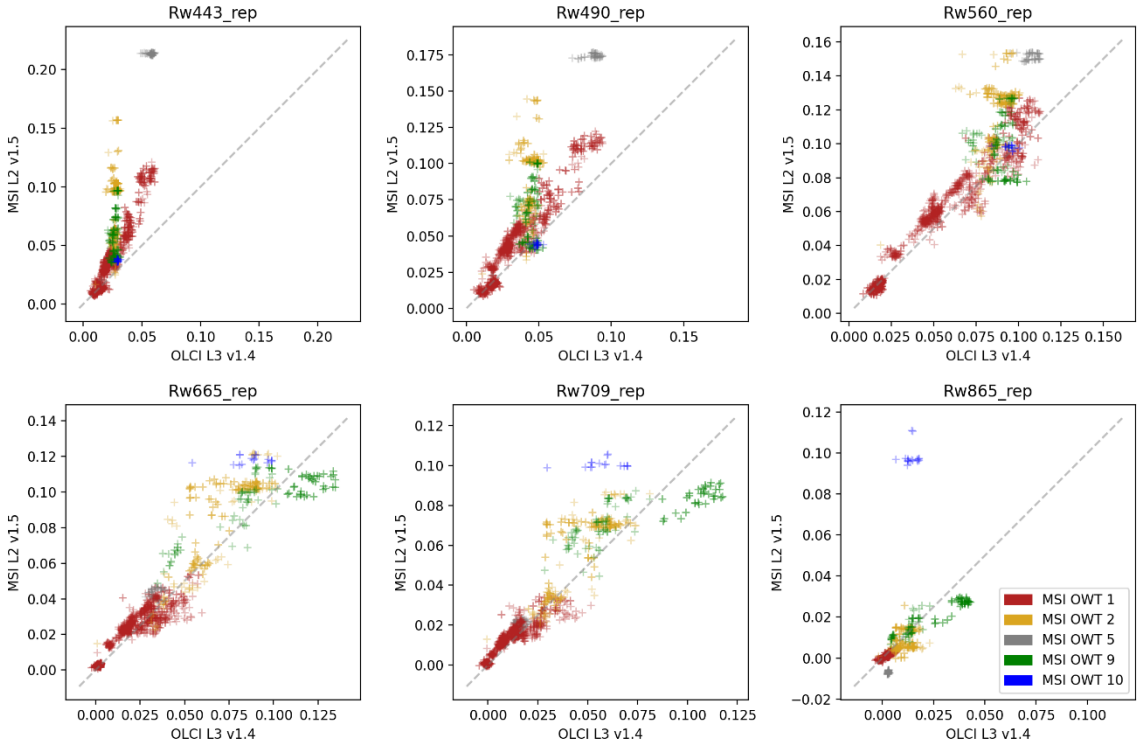


Figure 47: Sentinel-2 MSI L2 reflectance v1.5 versus Sentinel-3 OLCI L3 v1.4 at wavelengths 443, 490, 560, 665, 705 and 865nm. All lakes, four transects from selected dates (see Table 7)

The spectral consistency is dependent on the optical water type (see overview Figure 47: Sentinel-2 MSI L2 reflectance v1.5 versus Sentinel-3 OLCI L3 v1.4 at wavelengths 443, 490, 560, 665, 705 and 865nm. All lakes, four transects from selected dates (see Table 7)Figure 47). In OWT 1 spectra are most likely to be similar in both datasets, though in the blue MSI overestimates OLCI reflectance by a factor of two. The data of the extractions for band 443nm suggests that OLCI reflectance is saturating at about 0.055, while MSI values have a much wider range. Water leaving reflectance of 0.2 in OWT 5 at 443nm seems a strong overestimation. For bands 490 to 709nm reflectance classified as MSI OWT 1 is fairly consistent between the sensors, with a positive bias in bands 490 and 560nm for MSI values. The strongest overestimation for MSI reflectance in the blue (443 and 490nm) is connected to MSI OWT 2 and 9. Again, this observation might be linked to a saturation of

OLCI spectra around 0.025. For OWT 10, consistency is very good in the blue and green, but in the NIR (865nm) MSI shows strong overestimation.

It needs to be checked, which atmospheric correction needs to be improved or which spectra might need a more severe quality flagging: either OLCI with its seeming saturation in certain bands or MSI with its overestimations. Tests against in situ data would be advisable.

4.4 STRIPING IN THE FLAGGING

In regions with sunglint conditions during observations, the MSI product can be flagged in one detector, but not in the next, so that data appears in stripes. As viewing geometry changes strongly from detector to detector, with one facing in flight direction and the other one in the rear direction, geometries of adjacent pixels can be very different at detector borders. This kind of omitted data in stripes cannot be resolved unless the atmospheric correction works in glint conditions perfectly as well.

4.5 EVOLUTION TO VERSION 2.0

Recently we investigated the performance of integrating an additional mask called 'out-of-bounds' (OOB) from Polymer. This mask plays a crucial role in identifying pixels that demonstrate inadequate atmospheric correction results. The OOB flag becomes active when either the 'logchl' or 'f_b' outputs from the default 'PR05' model of Polymer in CGLOPS depart from within the predefined valid range. In the current v2.0 products, an implementation of the OOB flag has taken place which aimed to eliminate pixels affected by imperfect atmospheric correction. This section focuses on evaluations between lake water products v1.4 and v2.0, by assessing the impact of the OOB flag on the final outcomes of specific lakes of different types. Specifically, three lakes are included in this evaluation, including two extremely turbid lakes (Lake Turkana and Lake Tana) and a clear lake (Lake Garda).

Turbid lakes: Lakes Turkana and Tana

Figure 48 and Figure 49 show the maps illustrating the presence of OOB pixels in lakes Turkana and Tana, along with the corresponding R_w665 lake water products from versions 1.4 and 2.0. In the case of lake Turkana, a total of 154,804 OOB pixels were identified, primarily concentrated in the northern part of the lake (Figure 48a). By implementing the OOB flag, pixels with extreme values along the lakeshore and in the central region of the northern area can be effectively eliminated (Figure 48b and c).

Regarding lake Tana, OOB pixels were detected across a significant portion of the lake (Figure 49a). During the evaluation of the satellite scene in lake Tana, a total of 478,577 OOB pixels were identified. Notably, a distinct region displaying visible "artificial boundaries" is observed at the center of Lake Tana. The application of the OOB mask successfully removed this region with visible "artificial boundaries." It is also noteworthy that an updated version of Polymer (v4.15) has been tested successfully to address the issue of "artificial boundaries". This matter is further explored in

section 4.6, and it will be completely resolved in the upcoming v3.0 product through a comprehensive full reprocessing. Although it may not be the optimal solution, those pixels that fail during atmospheric-correction are excluded in v2.0 products by the application of the OOB flag.

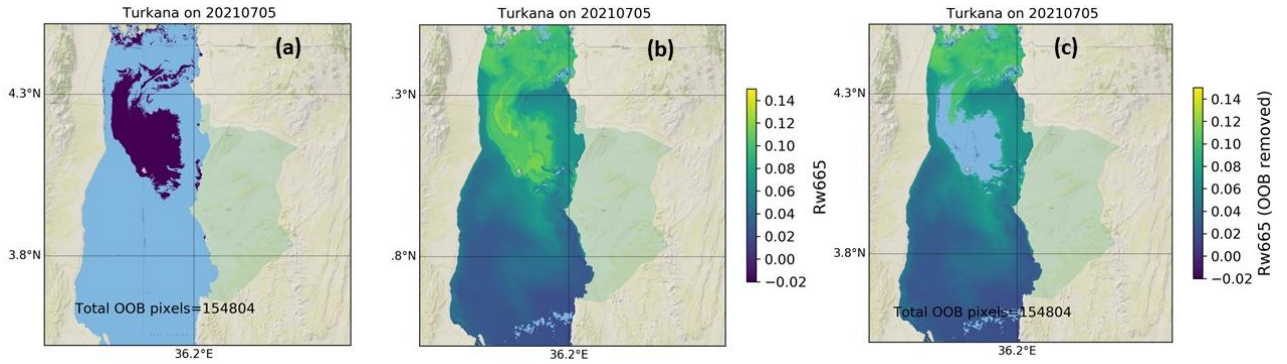


Figure 48: Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water products v1.4, and (c) R_w665 from lake water products v2.0 in Lake Turkana on July 5th, 2021

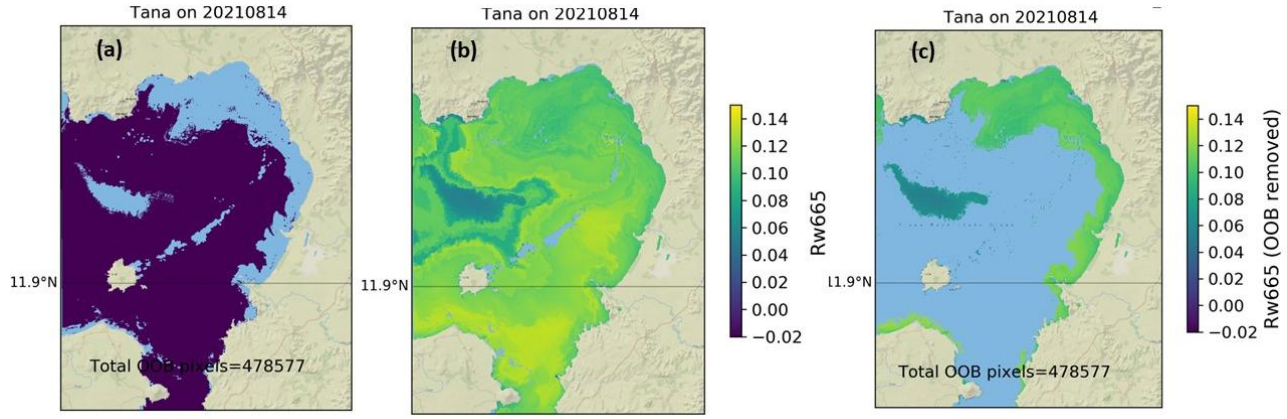


Figure 49: Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water products v1.4, and (c) R_w665 from lake water products v2.0 in Lake Tana on Aug 14th, 2021

Clear lakes: Lakes Garda

During the evaluation of the satellite scene in Lake Garda, a total of 37 OOB pixels were detected on July 1st, 2017 (Figure 50). The distribution of these identified OOB pixels in lake Garda is primarily concentrated along the lakeshore of the southern part of the lake, although they may not be readily visible on the OOB map due to the small number of OOB pixels (Figure 50a).

In comparison to the findings in turbid lakes (Figure 50), it is observed that clear lakes exhibit a lower number of identified OOB pixels. Furthermore, it is worth noting that all the OOB pixels identified in the evaluated lakes are characterised by extreme values and/or abnormal disconnected spatial distributions.

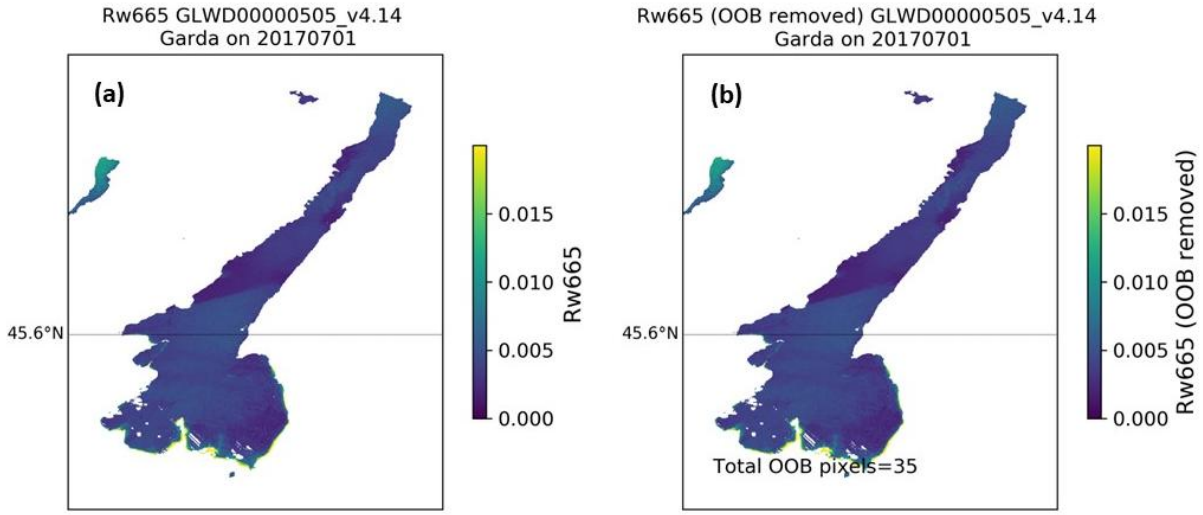


Figure 50: Maps of (a) out-of-bounds from Polymer, and (b) R_w665 from lake water products v1.4, and (c) R_w665 v2.0 in Lake Garda on July 1st, 2017

It is important to acknowledge that the study presented above for the two turbid lakes represents extreme/worst scenarios observed on selected days with abnormal phenomena, which rarely happened even in the lakes studied. Currently, comprehensive assessments of the MSI product on a global scale have not been conducted due to the overwhelming storage and computational requirements involved. As a result, it is not possible to draw a firm conclusion regarding the potential impact of implementing the OOB flag on v2.0 products globally. Nevertheless, as part of the evolution task, a global evaluation was carried out on OLCI (300m resolution), encompassing 4268 lakes on six selected dates in 2021. While the findings may not directly apply to MSI, the global evaluation results from OLCI indicate that, on average, the utilization of the OOB mask could result in the elimination of approximately 1 to 4 pixels out of every 1000 pixels.

4.5.1 Minor update to v2.2

In May 2026, the NASA meteorological data used as input to the processing of the LWQ-100m product were discontinued. This was replaced by permanent ingestion of equivalent ECMWF meteorological data (which is included in the MSI L1C product) following necessary updates to the processing chain, and mainly, but not solely, change of the atmospheric correction processor version. While a newer Polymer version has been employed, its parameterization matches the last used version to result in same behaviour and outputs. Consequently, the integrity and continuity of the product are **maintained** in v2.2 from previous versions 2.0 and v2.1.

Two lakes were selected for comparisons; Étang de Berre (France) and Lake Kariba (Zambia, Zimbabwe). These exhibit spatial gradients in water clarity making them ideal for testing (Figure 51). They are also located in different regions of the world.



Figure 51. True colour S-2 MSI L2A composite in Étang de Berre on 11 May 2026 (left) and in Lake Kariba on 13 May 2026 (right).

Spectral curves derived at different locations in the two lakes from products v2.1 and v2.2 in the same dekad are identical across all wavebands (Figures 52-55).

ECMWF input (LWQ-100m v2.2) **NASA** input (LWQ-100m v2.1)

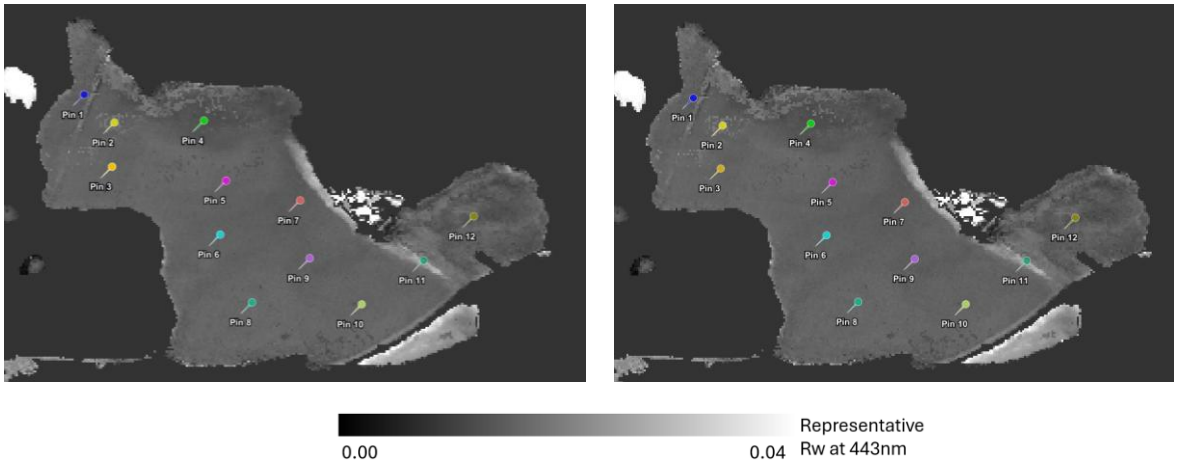


Figure 52. Representative water-leaving reflectance at 443nm from dekadal product LWQ-100m v2 showing the pin locations for which spectrum views were extracted (Figure 53). ECMWF data input (left) and NASA data input (right). Dekad: 1-10 May 2026.

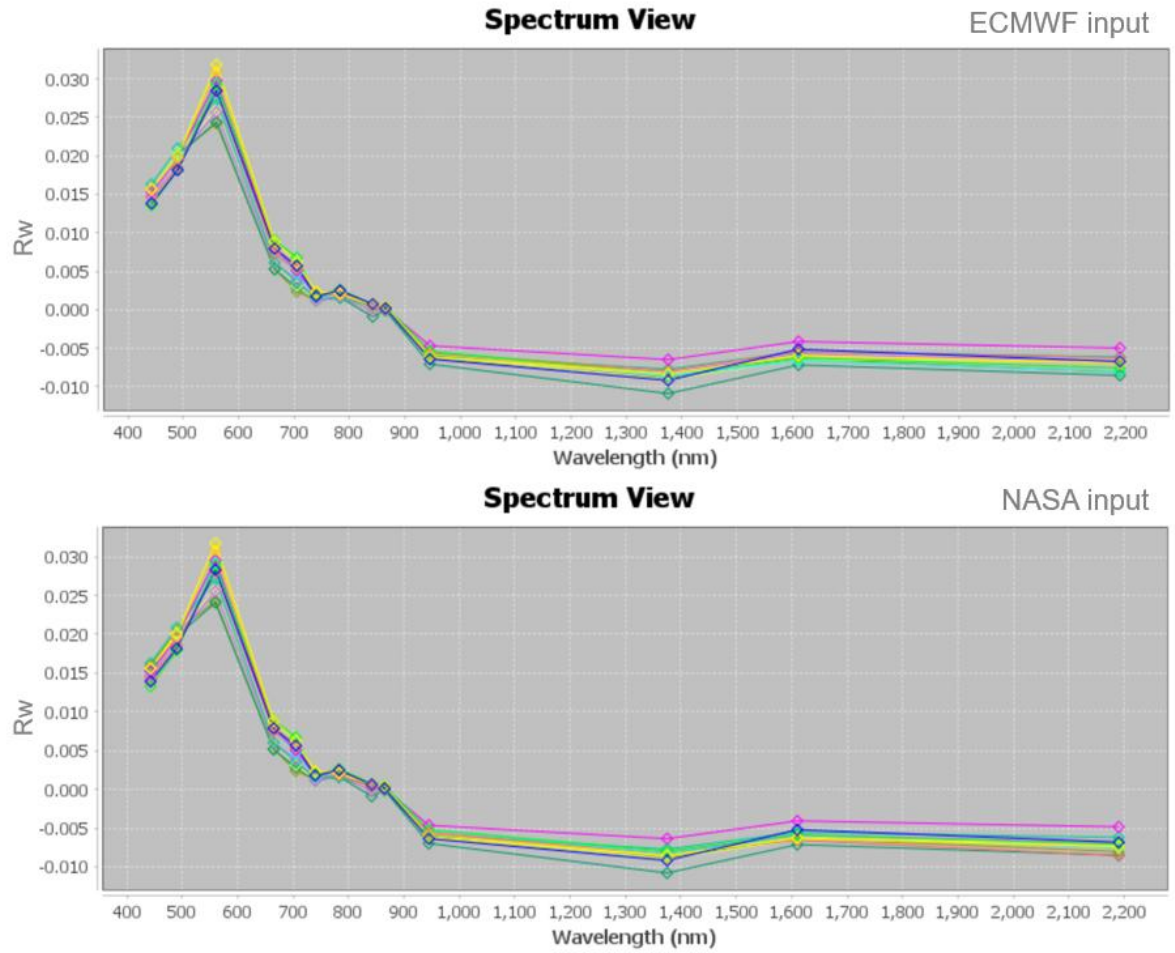
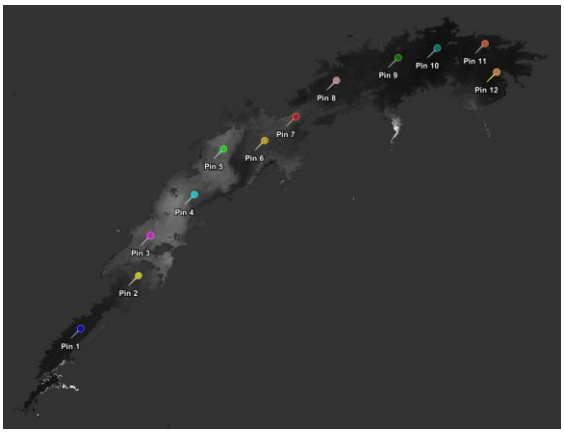


Figure 53. Spectrum view for 12 pin locations in Étang de Berre in dekadal product LWQ-100m v2.2 (above) and LWQ-100m v2.1 (below). Dekad: 1-10 May 2026.

ECMWF input (LWQ-100m v2.2)



NASA input (LWQ-100m v2.1)

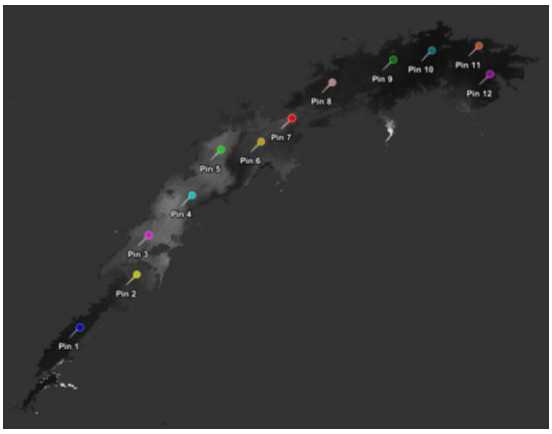




Figure 54. Representative water-leaving reflectance at 443nm from dekadal product LWQ-100m v2 showing the pin locations for which spectra views were extracted (Figure 55). ECMWF data input (left) and NASA data input (right). Dekad: 1-10 May 2026.

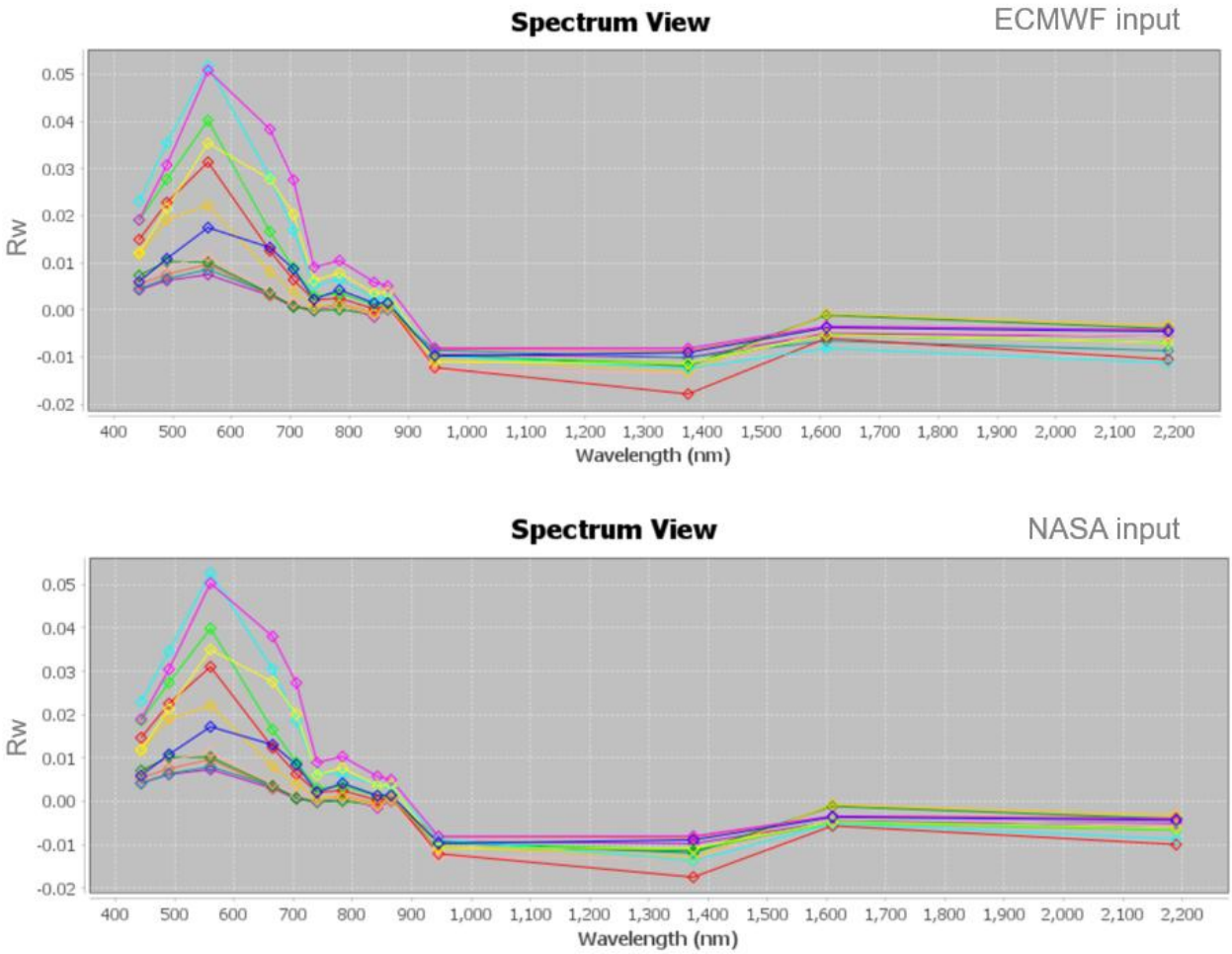


Figure 55. Spectrum view for 12 pin locations in Lake Kariba in dekadal product LWQ-100m v2.2 (above) and LWQ-100m v2.1 (below). Dekad: 1-10 May 2026

Minor discrepancies are observed when water quality bands are compared such as the chlorophyll-a band, and TSI band that is calculated from chl-a ranges. Overall, v2.2 shows minor differences from v2.1 in chl-a and TSI estimates although not systematically (Figures 56-57), i.e., no bias was observed. While the differences in chl-a estimates are relatively small these may lead to different TSI bands in oligotrophic and mesotrophic waters. In chlorophyll-poor waters, small increments of chl-a can classify pixels to a different TSI class, as seen in Figures 56-57. For more information on the chl-a ranges used in each TSI class, please see this product's ATBD document Table 2. The above overall observations are repeated in other dekadal products that were compared, not shown here for the sake of brevity.

Similar results in estimates of TUR and TSM were observed between v2.1 and v2.2 products (Figures 58-59), without noteworthy differences or systematic bias.

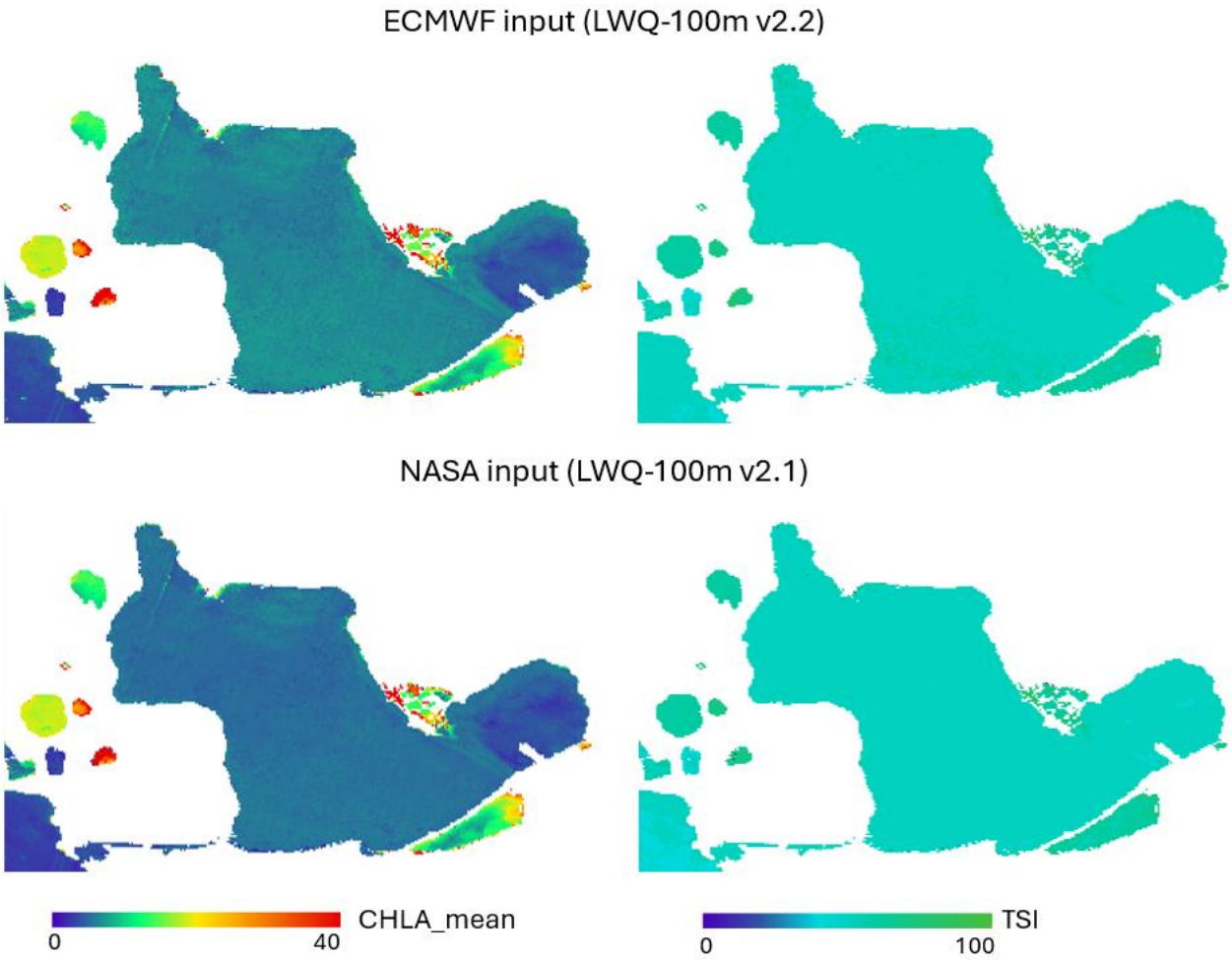


Figure 56. Chlorophyll-a mean (left) and TSI (right) from dekadal LWQ-100m v2 products (dekad: 1-10 May 2026) in Étang de Berre. Top: LWQ-100m v2.2 using ECMWF input. Bottom: LWQ-100m v2.1 using NASA input.

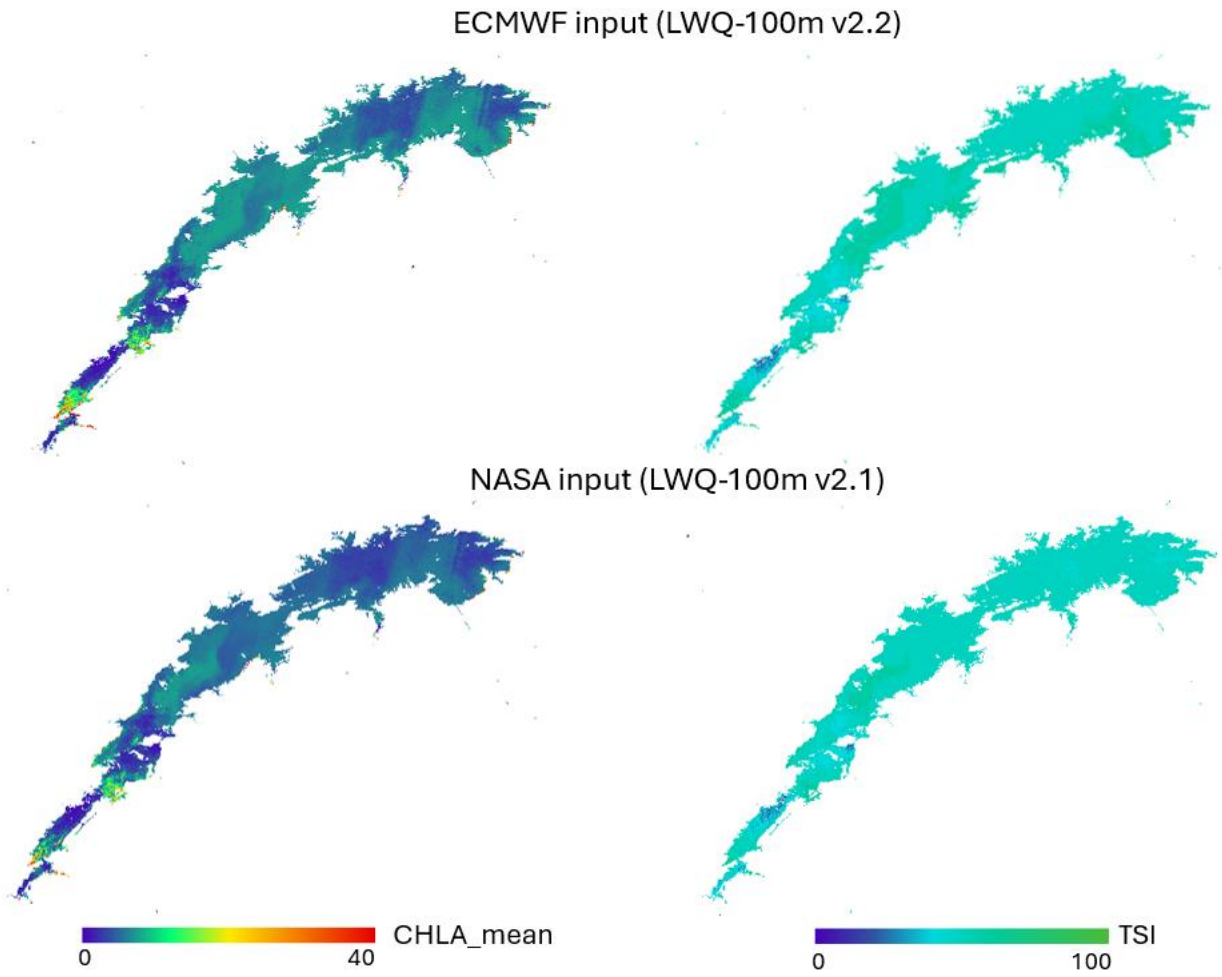


Figure 57. Chlorophyll-a mean (left) and TSI (right) from dekadal LWQ-100m v2 products (dekad: 1-10 May 2026) in Lake Kariba. Top: LWQ-100m v2.2 using ECMWF input. Bottom: LWQ-100m v2.1 using NASA input.

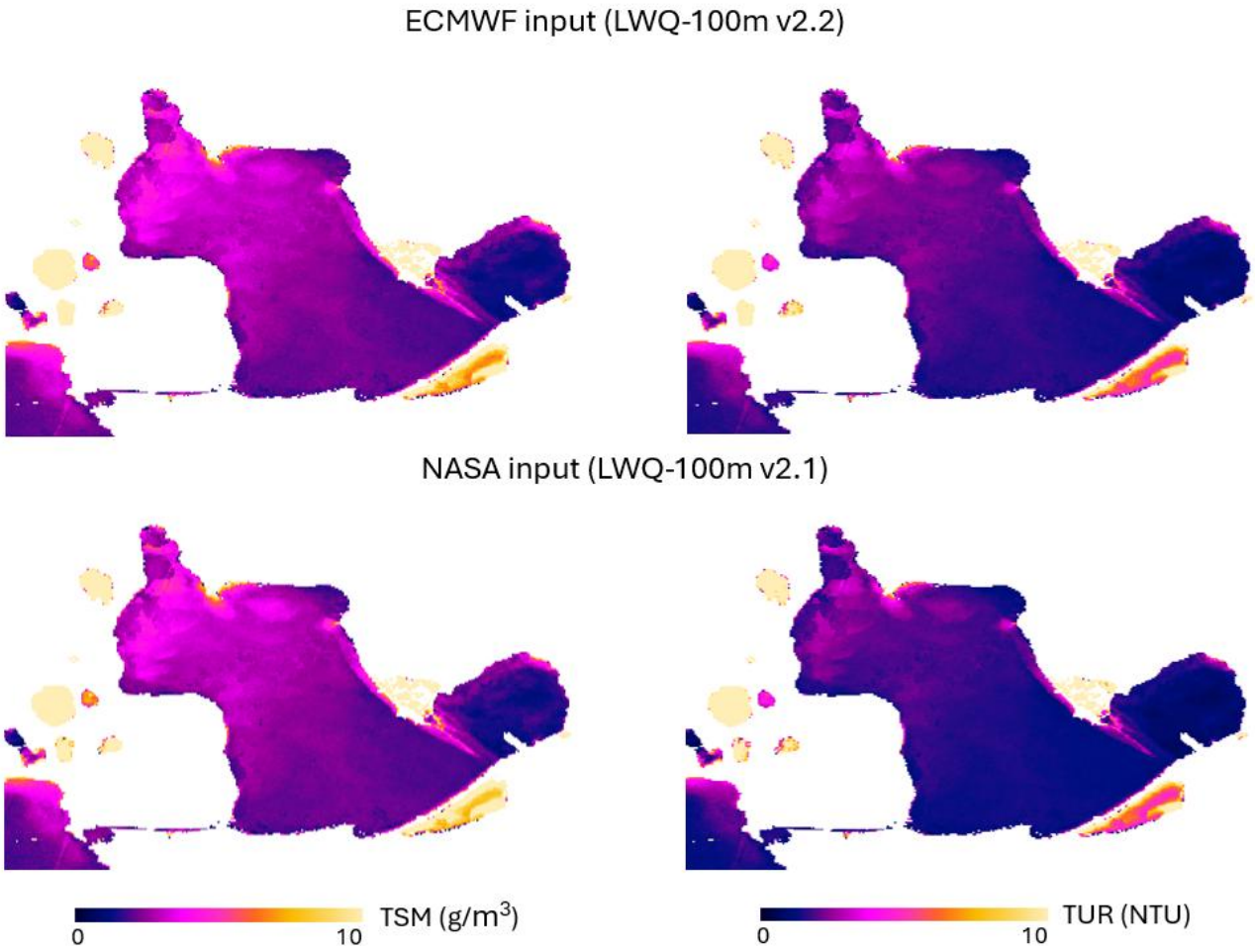


Figure 58. TSM mean (left) and TUR mean (right) from dekadal LWQ-100m v2 products (dekad: 1-10 May 2026) in Étang de Berre. Top: LWQ-100m v2.2 using ECMWF input. Bottom: LWQ-100m v2.1 using NASA input.

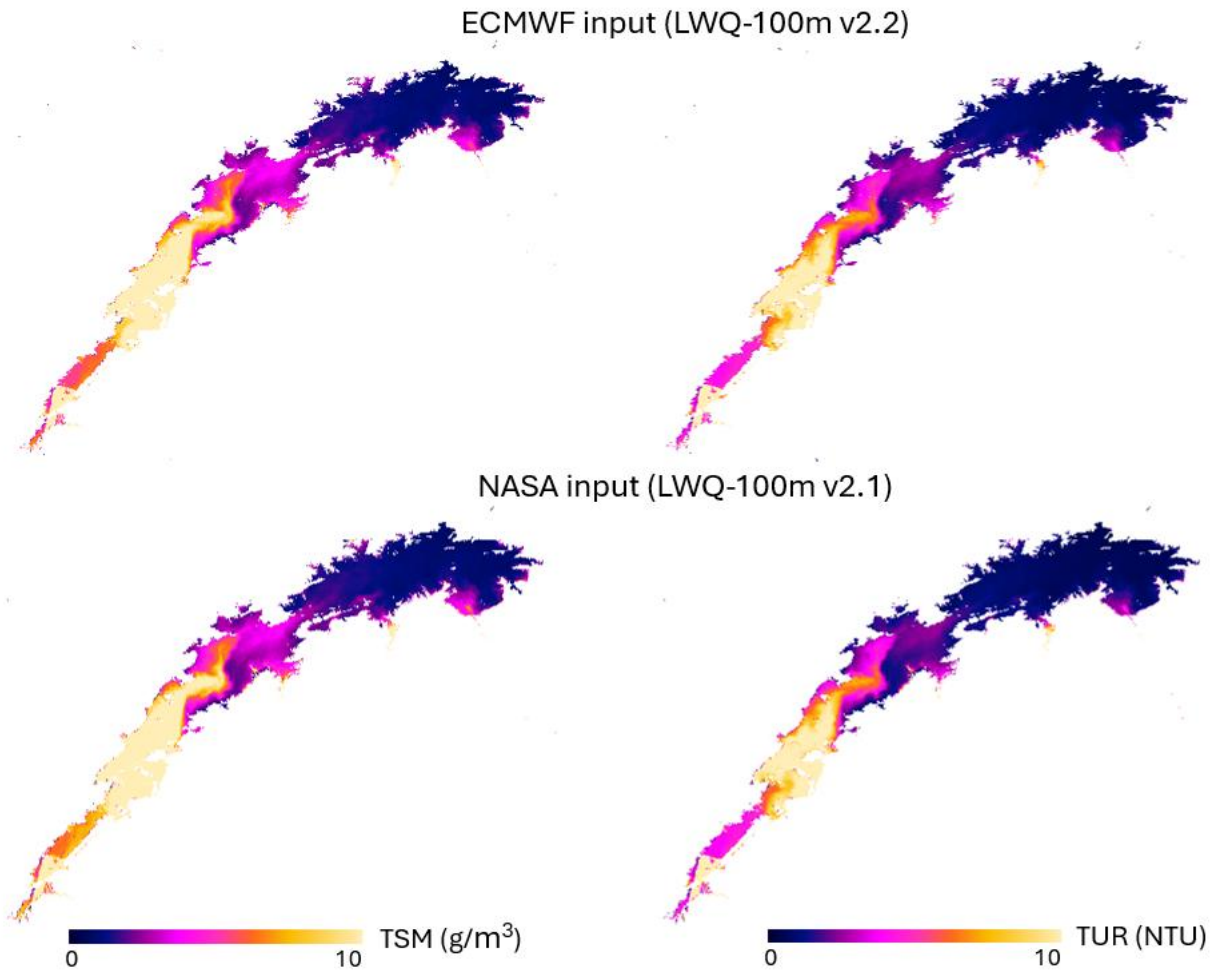


Figure 59. TSM mean (left) and TUR mean (right) from dekadal LWQ-100m v2 products (dekad: 1-10 May 2026) in Lake Kariba. Top: LWQ-100m v2.2 using ECMWF input. Bottom: LWQ-100m v2.1 using NASA input.

4.6 EVOLUTION TO VERSION 3.0

Polymer version v4.15 incorporates experimental processing modes specifically designed for complex coastal and inland waters. Polymer v4.15 includes several key upgraded modes, such as the optional parameterization of mineral absorption switching for Chl>10 (option min_abs=-2), updated first guess to test multiple points across the cost function and not reuse the previous pixel's result (option initial_points), and the inclusion of SWIR bands of 1610 for MSI for bands_corr and bands_oc.

Polymer (v4.15) provides a potential solution to the problem of unrealistic spectral shapes with visible artificial boundaries on maps, which had been a significant issue in extremely turbid regions with

previous versions. In this regard, a comprehensive evaluation is needed between Polymer v4.14 and v4.15, including both satellite-image-based and in situ matchup based analysis (see section 4.2.1).

The objective of this study is to assess the performance of the upgraded version of Polymer (v4.15) on satellite images and to compare it spectrally and spatially with the current version (v4.14). To demonstrate the effectiveness of different Polymer versions in diverse types of lakes, this evaluation includes three lakes - two extremely turbid lakes (Lake Turkana and Lake Tana) and a clear lake (Lake Garda). The purpose of this analysis is to determine if the upgraded version (v4.15) provides improved results compared to the current version (v4.14) and to identify any discrepancies that might require re-tuning of algorithms in certain circumstances.

Turbid lakes: Lakes Turkana and Tana

Figure 60 and Figure 61 present a comparison of the lake water-leaving reflectance obtained from Polymer v1.4 and v1.5 for Lake Turkana and Lake Tana. The comparison was performed using all valid pixels present in the atmospherically corrected images, with MSI bands ranging from 443 to 945nm. The results obtained for Lake Turkana demonstrate good agreement between the two versions, particularly for bands between 560 and 783nm, where scatters closely align with the unity line, and R-values exceed 0.6 (see Figure 60). However, discrepancies were observed for R_w with relatively high values (R_w>0.1). While for Lake Tana, two distinct patterns were observed between the two versions. Specifically, the two versions show good agreement in low-value sessions (e.g., when R_w560<0.1 in v4.14) for all evaluated bands, whereas for high-value sessions of v4.14, v4.15 outputs were significantly elevated.

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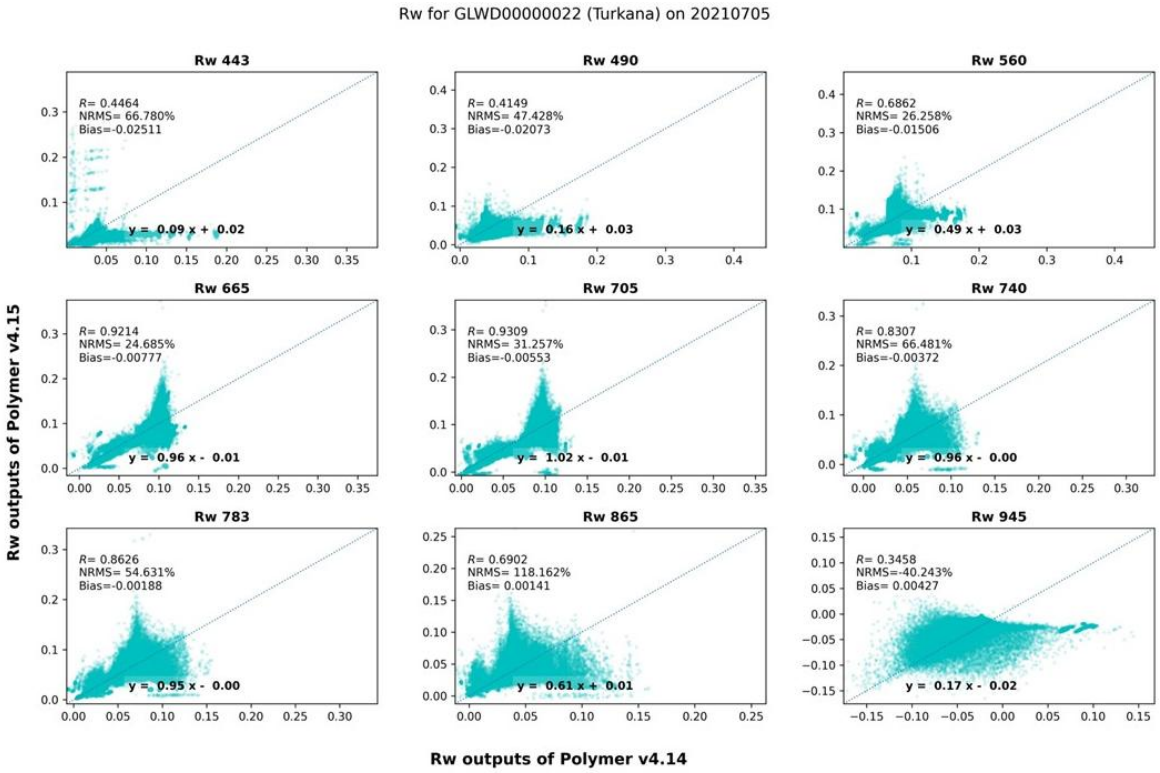


Figure 60: Comparison of lake water-leaving reflectance between Polymer v4.14 and v4.15 in Lake Turkana

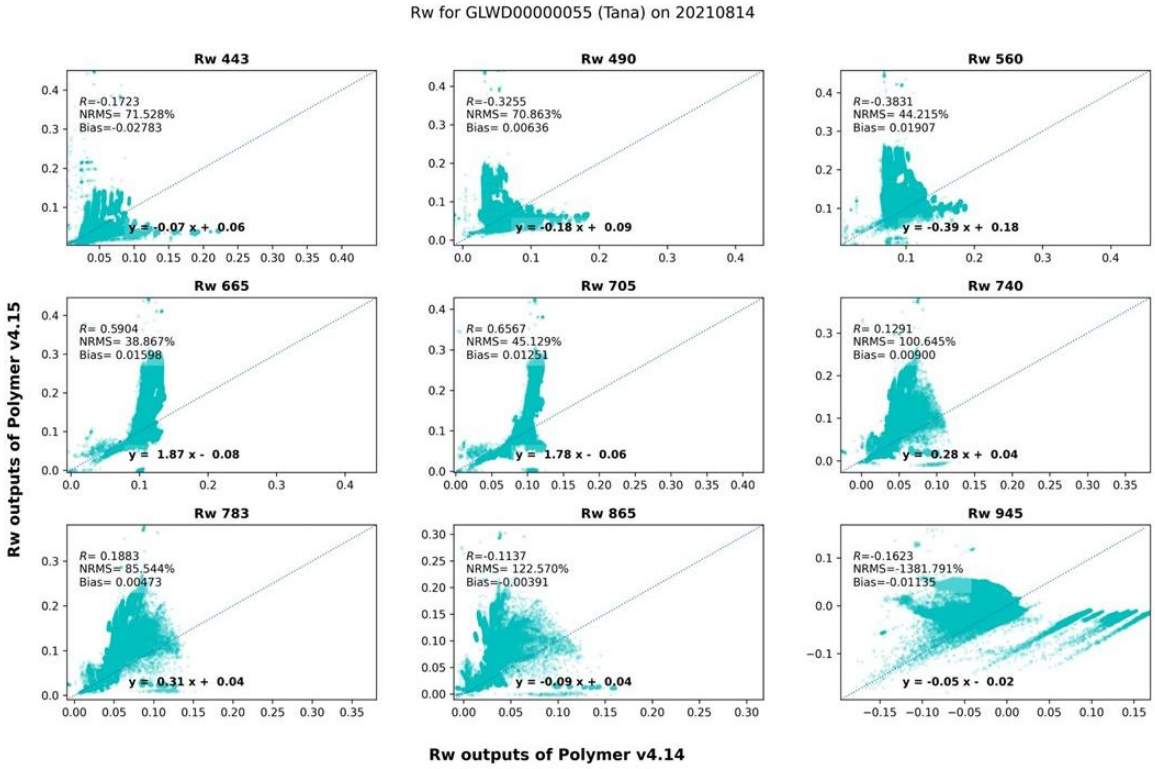


Figure 61: Comparison of lake water-leaving reflectance between Polymer v4.14 and v4.15 in Lake Tana

To further validate Polymer v4.14 and v4.15, Figure 62 displays histograms of the distribution of R_w for the two versions in Lake Turkana (a-c) and Lake Tana (d-f) at bands 443, 560, and 665nm. The median values of the three evaluated bands in Lake Turkana are similar to each other, with v4.14 generally slightly lower than v4.15, consistent with the observations on the scatterplots (Figure 60). Furthermore, the v4.15 bars are slightly shifted towards lower values compared to v4.14. Generally, better agreement was found between the two versions for bands 560 and 665nm, compared to band 443nm. However, in Lake Tana (Figure 62d-f), the differences between the median values of the two versions for each band are more significant than in Lake Turkana (Figure 62a-c). This is particularly evident for band 665nm, where v4.15 is more evenly distributed over a larger range compared to v4.14. This phenomenon is not observed in Lake Turkana, mainly due to the fact that the studied scene in Lake Turkana is less "bright" than that in Lake Tana, where more pixels are distributed in the "high-value" section.

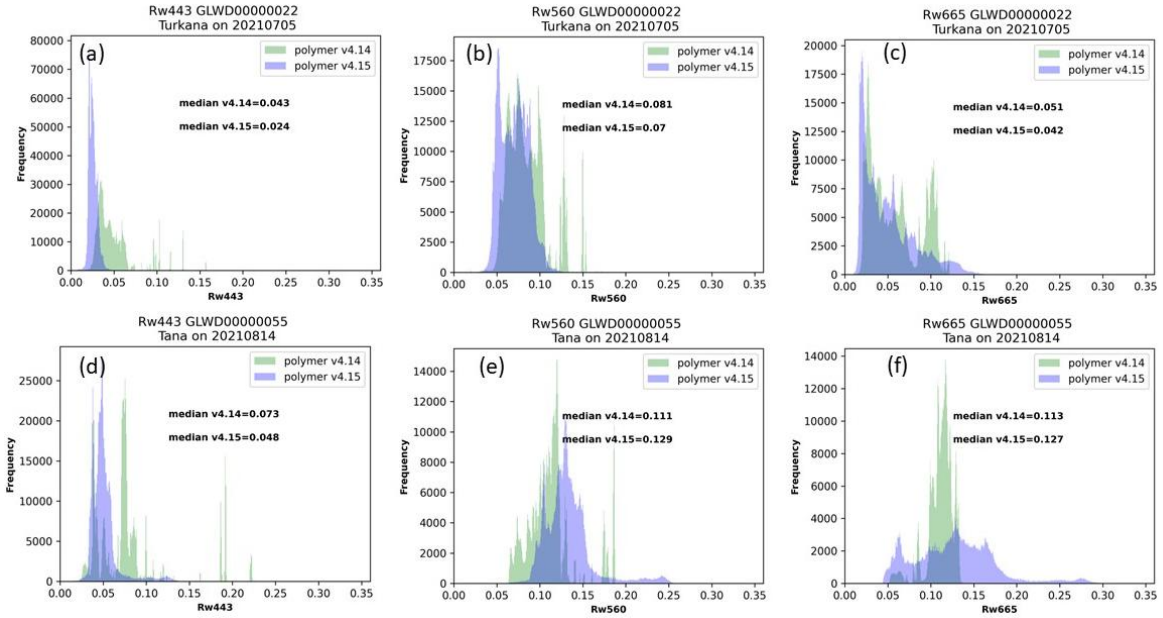


Figure 62: Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for (a-c) Lake Turkana and (d-f) Lake Tana, at bands 443, 560 and 665nm

In terms of spectral characteristics, the median spectra of both Polymer versions in both lakes exhibit similar shapes (see Figure 63). In Lake Turkana, the two versions present almost identical shapes of median spectra and 30-70th percentiles span, however, the median spectra of v4.14 are slightly higher than those of v4.15 for most of the evaluated bands (Figure 63a). Conversely, in Lake Tana, v4.15 shows higher median spectra and span of 30-70th percentiles than v4.14 for bands from 490 to 740 nm (Figure 63b). These findings are consistent with the results presented in Figure 62.

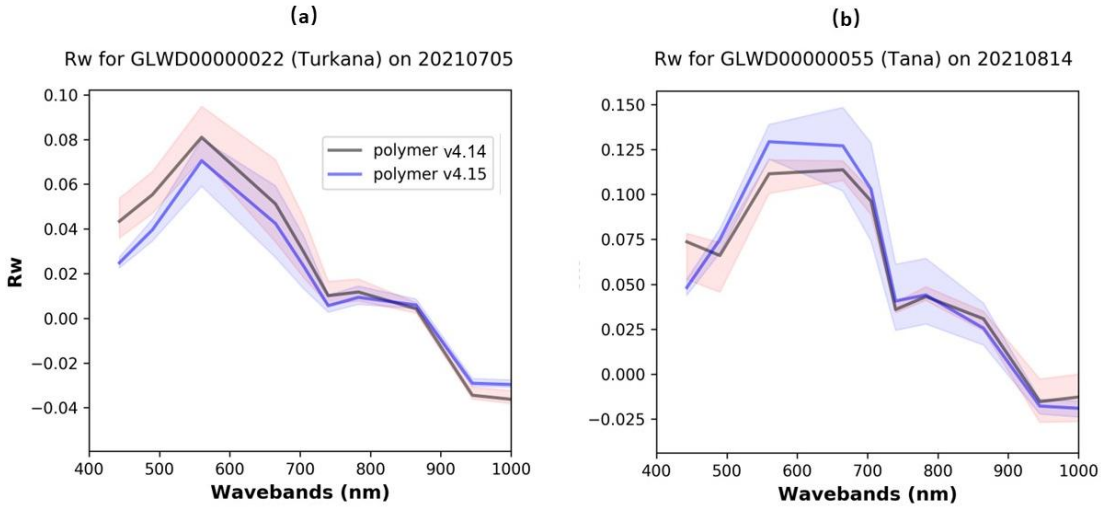


Figure 63: Spectral comparison of Polymer v4.14 and v4.15 in (a) Lake Turkana and (b) Lake Tana. Lines in the center indicate median values, and the filled regions represent 30th to 70th percentiles

To compare the spatial differences in the outputs of Polymer v4.14 and v4.15, Figure 64 displays the maps of the two versions at R_w 665nm for lakes Turkana and Tana. The comparison reveals that in Lake Turkana, both versions exhibit similar distribution patterns, with v4.15 producing higher retrievals than v4.14 in the northern regions with high R_w 665 values. However, a plume shape (with high R_w values) is observed in the northern part of the lake in v4.14, extended from north to south, which is greatly smoothed spatially in v4.15. In contrast, more noticeable differences between the two versions are found in Lake Tana, where the lake center responds differently to the rest of the lake area. Specifically, a region with visible "artificial boundaries" is observed in the lake center of Lake Tana in v4.14, likely due to re-using values from adjacent pixels as initial guesses by default in previous versions of Polymer. However, this phenomenon is not present in v4.15, which has several initial points enabled for processing. While for the vast region along the lake shores, v4.15 generates generally higher retrievals than v4.14.

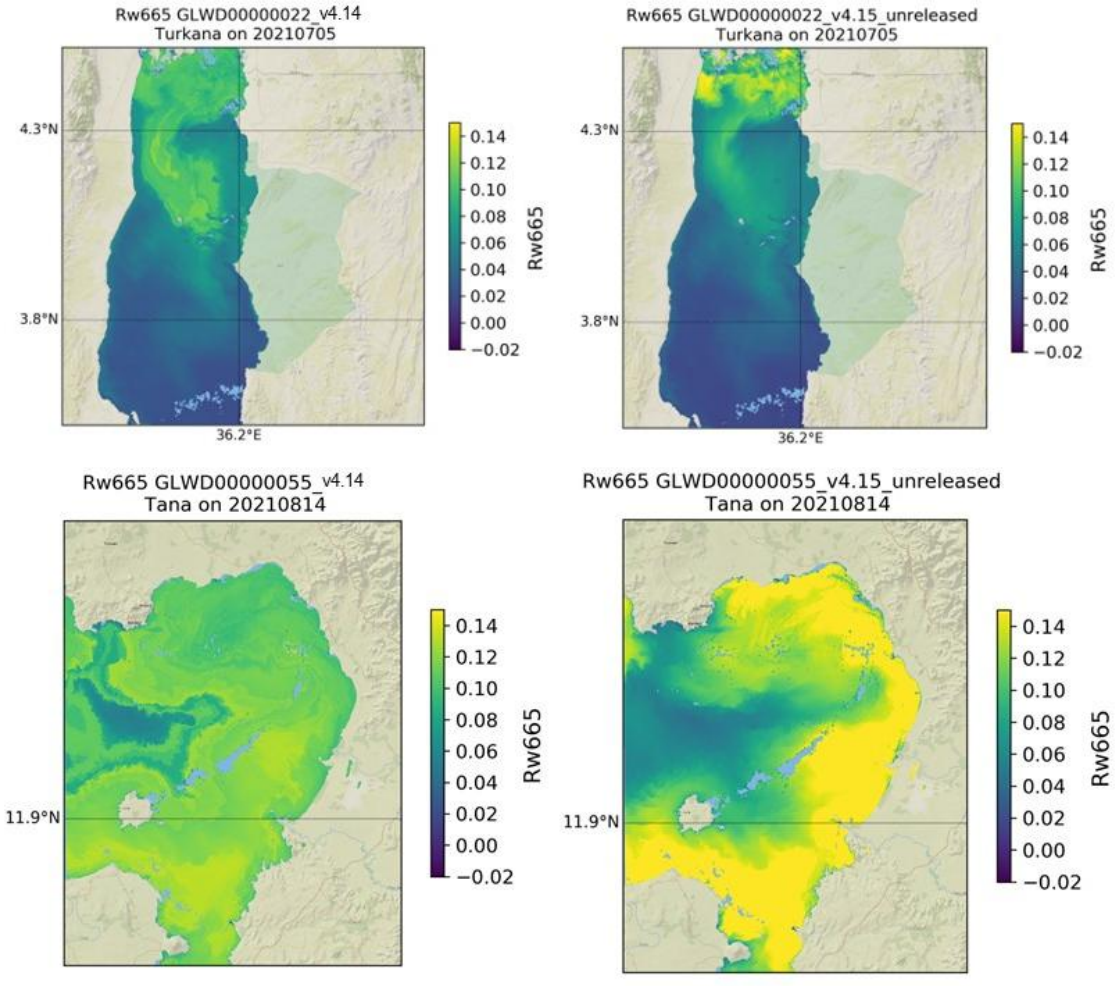


Figure 64: Maps of R_w665 from Polymer v4.14 and v4.15 in Lake Turkana (upper panels) and Lake Tana (lower panels).

Clear lake: Lake Garda

The spatial distribution of R_w 665nm from the two versions of Polymer for a clear lake, Lake Garda, is illustrated in Figure 65. The comparison of the two versions shows good consistency in the majority of the regions in the lake, with v4.15 showing slightly lower values. However, it is worth mentioning that v4.14 and v4.15 behave similarly along the lakeshore for MSI, which is different to what has been observed for OLCI. Specifically, for OLCI, v4.14 has shown extremely low (or even negative) values in the lakeshore, while v4.15 has shown significantly elevated values for the same regions (analysis not shown here).

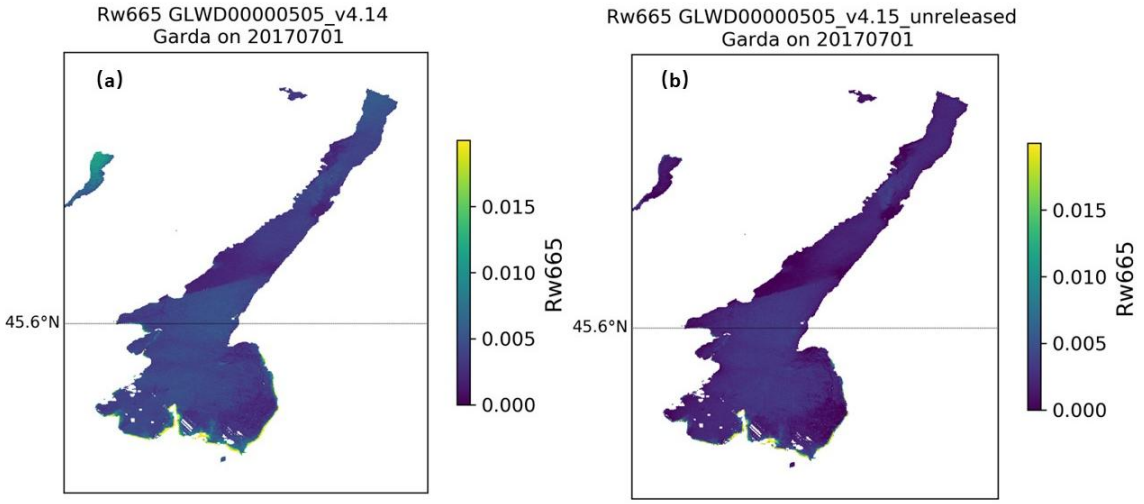


Figure 65: Maps of R_w665 from Polymer v4.14 and v4.15 in Lake Garda

To further analyze the performance of Polymer v4.14 and v4.15 in lake Garda, a comparison of their histograms was conducted for bands 443, 560 and 665nm (Figure 66). The histograms reveal that both versions exhibit similar distribution patterns for all three evaluated bands. However, at band 443nm, v4.15 displays a slight shift towards higher values compared to v4.14. For bands 560 and 665nm, the median values of v4.15 are slightly lower compared to v4.14, which is in line with the findings mentioned earlier (Figure 65). Overall, the histogram comparisons provide further evidence of the similarities and differences in the performance of the two versions of Polymer in Lake Garda.

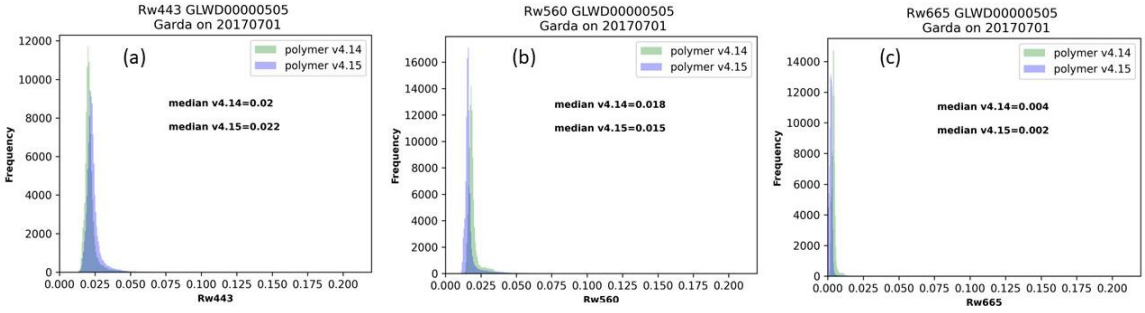


Figure 66: Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) in Lake Garda, at bands (a) 443, (b) 560 and (c) 665nm

Regarding the spectral characteristics, Figure 67 displays the median and 30-70th percentile spectra for Polymer v4.14 and v4.15 in Lake Garda. The two median spectra exhibit perfect agreement between the two versions in the lake (Figure 67). However, it was observed that v4.15 produces slightly higher values of R_w443, while slightly lower values were found for the rest of the evaluated bands compared to v4.14. This finding suggests that a re-tuning of water quality algorithms may be necessary if transitioning from v4.14 to v4.15, particularly for algorithms based on band ratios. Moreover, it was observed that both versions generate negative values of R_w at 945 nm, which is likely due to the over-correction of atmospheric effects.

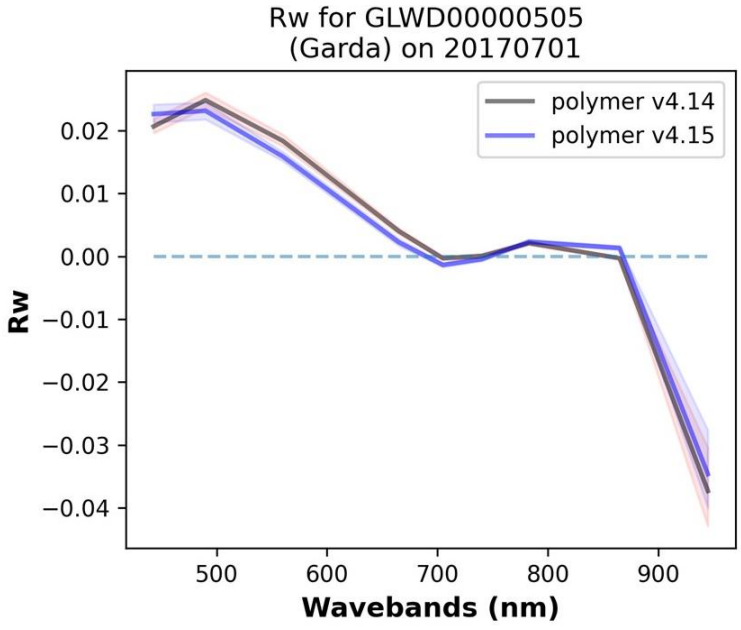


Figure 67: Spectral comparison of Polymer v4.14 and v4.15 in Lake Garda. Lines in the center indicate median values, and the filled regions represent 30th to 70th percentiles

5 CONCLUSIONS

5.1 SUMMARY

Depending on the optical water type, Sentinel-2 MSI derived CGLOPS 100m 10-day products show acceptable agreement with the 300m 10-day Sentinel-3 OLCI products. For example, in MSI OWT 1 the spectral shapes and intensities are quite well matched, with exception of the blue bands, in which overestimation of MSI occurs. For some OWTs, like MSI OWT 5, spectral shapes and intensities are very different between MSI and OLCI, which leads to inconsistent water quality parameters, especially turbidity. MSI products can currently include spatial artefacts of very high turbidity values.

Validation with in situ data remains an important task to do and will be continued, relying on the availability of such measurements.

5.2 LIMITATIONS AND KNOWN ISSUES

MSI, in contrast to ocean colour sensors, has relatively broad and fewer wavebands. It should be expected that, even when the bands are centred on diagnostic absorption features of optically active water constituents, sensitivity of the sensor to quantify these substances will be limited compared to ocean colour sensors.

The major limiting factor in (i) assessing the accuracy of water-leaving reflectance, (ii) the OWT-specific mapping of water constituent concentrations, and (iii) the accuracy of the water-column products is the scarcity of available in situ data to support algorithm tuning and validation. Without dedicated field campaigns, automated monitoring stations, and community data sharing arrangements, this is likely to remain a major source of product uncertainty for some years. Insofar as OWT-specific algorithms have been validated for the MERIS and OLCI sensors, these can be used in the interim to conduct multi-sensor comparisons.

The 100m products include areas that are near-shore and often shallow. Such areas are not yet flagged. One method that was previously investigated (ESA Diversity-II project, 2015) consists of a long-term analysis to identify suspected shallow areas, but this method has not been globally verified. The flagging may also benefit from classifications, which were introduced in ESA Lakes_cci to identify observations influenced by land adjacency. Further research is needed.

Reprocessing is advised once LWQ-100m v3.0 is in place for NRT production to ensure consistency of the time series.

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